

LOCAL EXCLUSION OF RESIDUAL TYPE I NAVIER–STOKES SEREGIN LIMITS

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ABSTRACT. We assemble the refined residual analysis for Type I Navier–Stokes Seregin limits. A finite-energy local pointwise Type I singularity for a suitable weak solution produces, through the endpoint weak Serrin local Type I package, terminal no-escape, and the Seregin blow-up procedure, a nonzero bounded ancient solution of the centered self-similar Navier–Stokes equation carrying positive local velocity L^3 -mass. After the previously treated small, stationary L^3 , uniformly L^3 -tight, and closed structured/decay alternatives have been removed, the remaining residual class is divided into four analytic classes: axisymmetric bounded-circulation profiles, rotational relative equilibria, degenerate stationary-hull profiles with retained local concentration, and the generic terminal residual.

The first class is excluded by a centered angular-circulation equation and an axis-absorption Liouville theorem. The second class is closed by a local Coriolis-flux identity and a local concentration decomposition: a superthreshold rotational flux gives a nontrivial recentered local profile, while a subthreshold flux remains inside the compact local limiting equation. The third class is reduced, by scale reselection for time-translation hulls, to a stationary profile in a degenerate stationary family which occurs as a Seregin limit and still carries local CKN concentration; it is then excluded by the terminal local concentration theorem. The final class is reduced by covariant tail normalization, active-locus extraction, diffuse-defect compactness, critical-tail compactification, active successor path-space recurrence, exclusion of finite separated concentration profiles, indecomposable-profile sequence- L^3 completeness, parasitic-free mildness inheritance, and the Albritton–Barker endpoint Liouville theorem. The paper distinguishes statements inherited from the unified setup package, local compactness and formal reduction statements proved here, and the first-bad-mesoscopic-scale routing theorem which discharges the hidden critical-tail boundary defects. The affine stratum and the coherent critical-tail alternatives are closed here; the log-diffuse and Young alternatives are both reduced to the same first-bad mesoscopic routing theorem, which is then proved by pressure compactness and a bounded ancient variance Liouville argument. No global Coriolis–Leray Liouville theorem for bounded profiles or global critical-norm estimate is used in the residual closure.

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1. INTRODUCTION AND MAIN THEOREM

We write the incompressible Navier–Stokes equations on $\mathbb{R}^3 \times (0, T)$ as

$$(1.1) \quad \partial_t u - \Delta u + (u \cdot \nabla)u + \nabla p = 0, \quad \nabla \cdot u = 0.$$

The residual analysis in this paper begins after the standard Type I blow-up construction has been performed. Thus the basic solution is a smooth bounded ancient solution of the centered equation

$$(1.2) \quad \partial_\tau V - \Delta V + \frac{1}{2}y \cdot \nabla V + \frac{1}{2}V + (V \cdot \nabla)V + \nabla P = 0, \quad \nabla \cdot V = 0,$$

on $\mathbb{R}_y^3 \times \mathbb{R}_\tau$. The equation is obtained from the physical ancient representative (U, Q) by

$$(1.3) \quad y = \frac{x}{\sqrt{-t}}, \quad \tau = -\log(-t), \quad V(y, \tau) = \sqrt{-t}U(x, t), \quad P(y, \tau) = (-t)Q(x, t),$$

where $t < 0$ and pressures are always understood modulo addition of a function of time. Conversely,

$$(1.4) \quad U(x, t) = (-t)^{-1/2}V\left(\frac{x}{\sqrt{-t}}, -\log(-t)\right), \quad t < 0.$$

The purpose of this paper is to present the refined residual decomposition as a single standard PDE argument, using only the reductions already established in the setup package. In this two-paper presentation, the current paper depends only on the unified setup package (`proof_setup.tex`) and on cited peer-reviewed Navier–Stokes regularity and Liouville literature. The rotational-flux estimates, terminal local concentration decompositions, exclusions of local compactness failure and diffuse exterior concentration, and finite separated-profile exclusions are proved in this paper from local estimates rather than imported as global estimates.

Definition 1.1 (Type I convention used in the chain). In this paper and in the final two-paper chain, a terminal singular point $z_* = (x_*, T)$ is called *local pointwise Type I* if there are $\rho > 0$ and $M < \infty$ such that

$$\operatorname{ess\,sup}_{T-\rho^2 < t < T} \sqrt{T-t} \|u(t)\|_{L^\infty(B_\rho(x_*))} \leq M.$$

This is the local pointwise Type I hypothesis used in the setup package and in [Section 1.1](#). For suitable weak solutions this envelope lies in the endpoint weak Serrin class $L_t^{2,\infty} L_x^\infty$. The Albritton–Barker weak-Serrin local Type I estimate, recorded below as [Theorem 1.19](#), upgrades it on smaller cylinders to the scale-invariant local energy package needed for terminal velocity no-escape. Thus the terminal no-escape condition used by the setup package is automatic for the local pointwise Type I singularities considered here. The paper does not use the phrase “Type I” to mean an unrelated global L^∞ , local CKN, weak- L^3 , or Serrin scale condition unless that condition is explicitly stated.

Remark 1.2 (Translation table for standard Type I hypotheses). The global pointwise condition

$$\sup_{t < T} \sqrt{T-t} \|u(t)\|_{L^\infty(\mathbb{R}^3)} < \infty$$

implies [Theorem 1.1](#) at every terminal point. Any stronger local criterion that yields the displayed L^∞ envelope on a neighborhood of x_* enters the same pipeline, provided the solution is suitable in the corresponding cylinder. The endpoint weak Serrin estimate then gives the local scale-invariant energy package and terminal no-escape. Purely CKN-type, weak- L^3 , or other critical criteria are not silently identified with this hypothesis; they require their own implication into the local pointwise envelope, or directly into the local Type I package, before the present exclusion applies.

- (1) Local concentration and terminal no-escape entry: positive local concentration depends on CKN theory and the local setup package; terminal no-escape follows from the endpoint weak Serrin local Type I package in [Theorems 1.19](#) and [1.20](#).
- (2) Seregin extraction and parasitic-free normalization: follows from the setup package and the mildness and compactness machinery in this paper; status: proved/imported.
- (3) Small and stationary L^3 strata: handled in the setup package; status: proved/imported.
- (4) Uniformly tight stratum: handled in the setup package; status: proved/imported.
- (5) Closed structured/decay strata: handled in the setup package; status: proved/imported.
- (6) Axisymmetric and rotational residual strata: proved in this paper.
- (7) Terminal active-locus stratification, active path-space recurrence, and diffuse-defect compactness: proved in this paper.
- (8) Critical-tail compactification and realized critical-tail stratification: proved in this paper.
- (9) Residual indecomposable-profile closure: proved in this paper, with the final sequence- L^3 endpoint supplied by Albritton–Barker.

Theorem 1.3 (Admissible Type I Seregin extraction theorem). *An admissible finite-energy Type I singular point produces a normalized Seregin ancient limit (V, P) satisfying (1.2), with*

$$\|V\|_{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq M < \infty,$$

local suitability and local pressure-gauge compactness, and a compact parabolic cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$ for which

$$(1.5) \quad \iint_K |V|^3 dy d\tau \geq \eta_0 > 0.$$

The collection of all normalized Seregin limits generated by the original solution is denoted by $\mathcal{S}$.

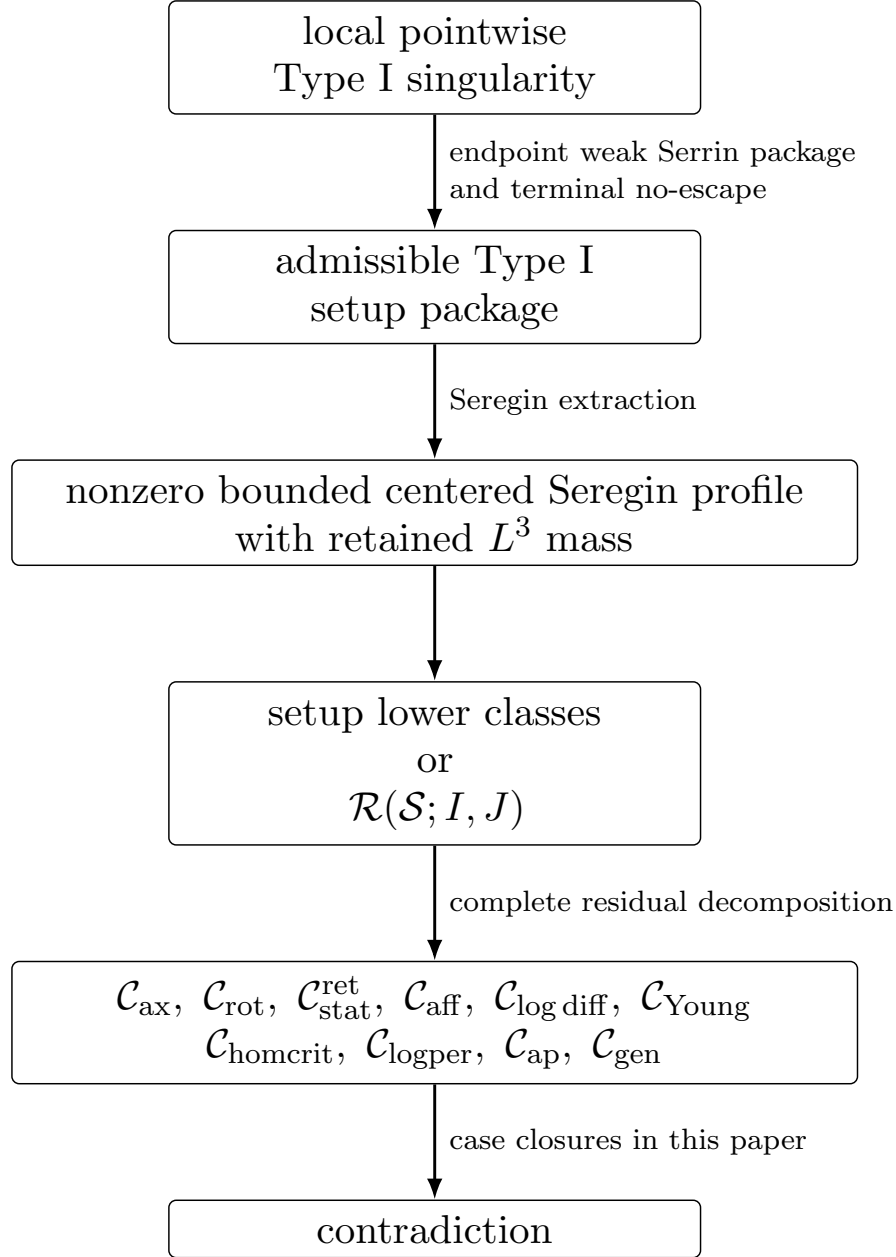


FIGURE 1. Decision diagram for the Type I state-space stratification. The first arrow uses the local pointwise Type I to terminal no-escape bridge supplied by the endpoint weak Serrin local Type I package. The setup lower classes in the split node are imported closed classes. The residual alternatives in the boxed list are discharged by the case-closure theorems cited in the proof of [Theorem 1.9](#).

Proof. This is the base admissible Seregin extraction theorem from `proof_setup.tex`, including velocity nonvanishing, local suitability, and local pressure compactness. The underlying positive local concentration input is also provided there. The present paper uses this as an imported reduction and does not reprove the Type I blow-up compactness theorem. $\square$

Definition 1.4 (Imported earlier alternatives). Within a fixed normalized Seregin collection $\mathcal{S}$, the small-amplitude, stationary L^3 , uniformly L^3 -tight, and closed structured/decay classes are the non-residual alternatives already established in the setup package. The small-amplitude and stationary L^3 alternatives are closed there; the uniformly tight class is excluded there; and the closed structured/decay class is precisely the part proved empty there. Any structured profile not belonging to that closed setup class is residual by definition and is handled by the present paper. The present paper does not reprove those earlier alternatives and does not add any hypothesis to their statements.

Definition 1.5 (Coarse residual class). Let I, J be the index sets used in the base Type I residual criterion. The coarse residual class $\mathcal{R}(\mathcal{S}; I, J)$ is the complement in $\mathcal{S}$ of the already removed alternatives in [Theorem 1.4](#).

Definition 1.6 (Refined residual decomposition). The refined residual decomposition is an ordered definition, not a separate compactness theorem. Starting with the remaining coarse residual class, one removes the following alternatives in order; after the first nine alternatives have been removed, whatever remains is by definition the generic terminal residual $\mathcal{C}_{\text{gen}}$:

$$\begin{array}{ccccc} \mathcal{C}_{\text{ax}}, & \mathcal{C}_{\text{rot}}, & \mathcal{C}_{\text{stat}}^{\text{ret}}, & \mathcal{C}_{\text{aff}}, & \mathcal{C}_{\text{gen}}, \\ \mathcal{C}_{\text{log diff}}, & \mathcal{C}_{\text{Young}}, & \mathcal{C}_{\text{homcrit}}, & \mathcal{C}_{\text{logper}}, & \mathcal{C}_{\text{ap}}. \end{array}$$

Here $\mathcal{C}_{\text{ax}}$ is the axisymmetric bounded-circulation class, $\mathcal{C}_{\text{rot}}$ is the rotational relative-equilibrium class, $\mathcal{C}_{\text{stat}}^{\text{ret}}$ is the degenerate stationary-hull class with retained local concentration, $\mathcal{C}_{\text{aff}}$ is the affine/parasitic lower stratum selected by affine-normalized oscillatory entry, $\mathcal{C}_{\text{log diff}}$ and $\mathcal{C}_{\text{Young}}$ are respectively the log-diffuse and Young critical-tail alternatives, $\mathcal{C}_{\text{homcrit}}, \mathcal{C}_{\text{logper}}, \mathcal{C}_{\text{ap}}$ are the coherent homogeneous, log-periodic, and aperiodic critical-tail alternatives, and $\mathcal{C}_{\text{gen}}$ is the generic terminal residual class. Thus the decomposition inclusion below is a formal consequence of the ordered definition; the non-tautological work is the closure of each alternative, especially $\mathcal{C}_{\text{gen}}$. Equivalently,

$$(1.6) \quad \mathcal{R}(\mathcal{S}; I, J) \subset \mathcal{C}_{\text{ax}} \cup \mathcal{C}_{\text{rot}} \cup \mathcal{C}_{\text{stat}}^{\text{ret}} \cup \mathcal{C}_{\text{aff}} \cup \mathcal{C}_{\text{log diff}} \cup \mathcal{C}_{\text{Young}} \cup \mathcal{C}_{\text{homcrit}} \cup \mathcal{C}_{\text{logper}} \cup \mathcal{C}_{\text{ap}} \cup \mathcal{C}_{\text{gen}}.$$

Definition 1.7 (Ancestor-realized residual objects). Fix a normalized Seregin collection $\mathcal{S}$ and a retained residual candidate $(V, P) \in \mathcal{R}(\mathcal{S}; I, J)$. An auxiliary object used in the residual proof is *ancestor-realized* if it is obtained from (V, P) , or from the terminal sequence which generated (V, P) , by a finite or diagonal sequence of the following operations:

- (i) exact centered symmetries T_a , Θ_s , and the covariant observer maps $\mathcal{R}_z$;
- (ii) raw parabolic recenterings before passing to centered variables, with the local pressure gauges supplied by the Seregin compactness package;
- (iii) retained tail limits or active successors with $\mathcal{E}_1 \geq \varepsilon_*$ after the fixed admissible pressure gauges;
- (iv) normalized diffuse or critical-tail defect limits obtained by restricting the ancestor defect measure to a positive-measure window and renormalizing it;
- (v) first-bad mesoscopic blow-downs selected from such retained or normalized defect windows.

The word “ancestor” always refers to the original singularity-generated candidate (V, P) , not merely to the immediately preceding normalized state.

Theorem 1.8 (Ancestor-realization and discharge principle). *Every descendant, tail limit, active successor, log-window normalization, critical-tail state, and first-bad blow-down used in the proof of [Theorem 1.9](#) is ancestor-realized in the sense of [Theorem 1.7](#). More precisely, each such object satisfies one of the following alternatives.*

- (a) *It is a retained Seregin descendant of the original candidate, with the fixed descendant lower bound $\mathcal{E}_1 \geq \varepsilon_*$.*
- (b) *It is a normalized defect state whose unnormalized ancestor defect measure has positive total mass on the selected window. If every normalized window descendant is discharged, then the original defect is discharged.*
- (c) *It lies in a previously closed lower stratum, and the retained descendant heredity theorem applies.*
- (d) *It generates a retained descendant or a normalized defect descendant satisfying (a), (b), or (c).*

Consequently a contradiction obtained for an ancestor-realized object is a contradiction for the original retained residual candidate, not merely for an auxiliary compactification state.

Proof. The assertion is a formal consequence of the construction of the state spaces and is used below as a logical reduction principle. Exact centered symmetries preserve the centered equation, local suitability, pressure gauges, and retained local concentration; this is recorded in [Theorems 8.2 and 8.6](#). Raw terminal recenterings before centering are the recenterings in the setup package Seregin extraction and are compact after the gauges in [Theorem 1.14](#). Retained tail limits and active successors are ancestor-realized by diagonal lifting: [Theorems 10.10 and 10.11](#) show that the limit remains in the normalized state space and keeps the fixed retained CKN threshold, up to the once-for-all threshold loss in [Theorem 3.2](#).

Normalized defect states are obtained by restricting a positive ancestor defect measure to a window and dividing by its mass. The mass of the chosen window may depend on the index and may tend to zero; this does not detach the normalized state from the ancestor. It records a point in the support of the ancestor defect compactification. The renormalized-window heredity lemma [Theorem 9.24](#) below states explicitly that if all such support descendants are routed into closed lower alternatives, then the ancestor defect measure itself has no surviving mass outside those alternatives.

First-bad blow-downs are selected from retained or normalized defect windows; their centers and scales therefore come with the same ancestor measure or retained CKN lower bound. The pressure/gauge normalization for those blow-downs is treated in [Theorems 9.37, 9.38 and 9.40](#). Thus every contradiction later in the proof is applied either to a retained descendant of the original profile or to a normalized support state whose closure removes the corresponding ancestor defect. This proves the stated principle. $\square$

Theorem 1.9 (Refined residual closure for the complete ordered decomposition). *Relative to the setup alternatives in [Theorems 1.3 and 1.4](#), one has*

$$\begin{aligned} \mathcal{C}_{\text{ax}} &= \mathcal{C}_{\text{rot}} = \mathcal{C}_{\text{stat}}^{\text{ret}} = \mathcal{C}_{\text{aff}} = \mathcal{C}_{\text{gen}} = \emptyset, \\ \mathcal{C}_{\text{log diff}} &= \mathcal{C}_{\text{Young}} = \mathcal{C}_{\text{homcrit}} = \mathcal{C}_{\text{logper}} = \mathcal{C}_{\text{ap}} = \emptyset. \end{aligned}$$

Consequently the coarse residual class $\mathcal{R}(\mathcal{S}; I, J)$ is empty.

Proof. The axisymmetric bounded-circulation class is excluded by [Theorem 4.3](#). The rotational relative-equilibrium class is excluded by [Theorem 5.8](#), which uses the localized Coriolis-flux identity and a terminal local concentration decomposition rather than a whole-space Coriolis–Leray Liouville theorem. The degenerate stationary-hull class with retained local concentration $\mathcal{C}_{\text{stat}}^{\text{ret}}$ is excluded by [Theorem 7.6](#), using the terminal local concentration theorem proved in [Sections 9 and 10](#). The affine/parasitic stratum is excluded by [Theorem 9.53](#). The log-diffuse and Young critical-tail alternatives are excluded by [Theorems 9.26 and 9.54](#). The coherent homogeneous, log-periodic, and aperiodic critical-tail alternatives are excluded by [Theorem 9.59](#). Finally, the generic terminal residual class $\mathcal{C}_{\text{gen}}$ is excluded by [Theorem 11.2](#). These ordered exclusions exhaust the complete decomposition (1.6). $\square$

Corollary 1.10 (Setup residual input). *For every normalized Seregin collection $\mathcal{S}$ generated by a admissible finite-energy Type I singularity in the setup package, the setup remaining class is empty.*

Proof. The setup package defines its remaining class as the complement, inside $\mathcal{S}$, of the small-amplitude, stationary L^3 , uniformly L^3 -tight, and closed structured/decay classes. The notation $\mathcal{R}(\mathcal{S}; I, J)$ used here records the same complement together with the ordered removals used in the residual decomposition; the indices I, J add no analytic assumption. By [Theorem 1.9](#), this class is empty. $\square$

Theorem 1.11 (Local pointwise Type I exclusion after residual closure). *No finite-energy local pointwise Type I singularity can generate a normalized Seregin collection $\mathcal{S}$.*

Proof. Suppose that such a singular point exists. The setup package supplies a positive local velocity concentration sequence. By [Theorems 1.20](#) and [1.22](#), the local pointwise Type I envelope gives the setup terminal velocity no-escape condition for this sequence. Hence [Theorem 1.3](#) applies, and the Seregin blow-up procedure produces a normalized ancient limit (V, P) satisfying (1.2) and the local velocity nonvanishing condition (1.5). By the setup class exhaustion, V is either in one of the non-residual classes recorded in [Theorem 1.4](#) or in $\mathcal{R}(\mathcal{S}; I, J)$. The first alternative is impossible by the corresponding closed inputs: the small and stationary L^3 classes are excluded in the setup package, the uniformly L^3 -tight class is also excluded there, and the closed structured/decay class is excluded there. The second alternative is impossible by [Theorem 1.9](#). This contradiction proves the theorem. $\square$

1.1. Local Type I singularities enter the terminal concentration package. The purpose of this subsection is to make explicit the local reduction that is implicit in the two-paper proof architecture here. There are two logically different issues. The local pointwise Type I envelope gives Seregin compactness on every compact negative-time cylinder

$$B_R \times [-S, -\sigma], \quad \sigma > 0,$$

but this compactness is strictly away from the terminal slice $t = 0$. It does not by itself say that the positive L^3 -mass in $B_1 \times (-1, 0)$ survives away from the terminal layer $(-\sigma, 0)$. The latter is exactly the terminal velocity no-escape condition used in the setup package.

The bridge below first records the admissible form used by the unified setup package. It then uses the endpoint weak Serrin local Type I estimate of Albritton–Barker to derive the required scale-invariant L^2 bound on the terminal interval. Interpolation with the Type I envelope gives terminal velocity no-escape.

For this subsection, if $z_* = (x_*, T)$ and $r_n \downarrow 0$, write

$$u_n(x, t) := r_n u(x_* + r_n x, T + r_n^2 t), \quad p_n(x, t) := r_n^2 p(x_* + r_n x, T + r_n^2 t).$$

Also write $Q_R := B_R(0) \times (-R^2, 0)$ and

$$Q_1(0, 0) = B_1(0) \times (-1, 0).$$

Definition 1.12 (Admissible local Type I concentration sequence). Let $z_* = (x_*, T)$ satisfy the local Type I bound (1.10), and let $r_n \downarrow 0$. The corresponding rescaled velocities u_n form an admissible local Type I concentration sequence if there are $\eta_v > 0$ such that

$$(1.7) \quad \iint_{B_1 \times (-1, 0)} |u_n|^3 dx dt \geq \eta_v \quad \text{for every } n,$$

and the terminal velocity no-escape condition holds:

$$(1.8) \quad \limsup_{\sigma \downarrow 0} \sup_n \iint_{B_1 \times (-\sigma, 0)} |u_n|^3 dx dt = 0.$$

Equivalently, after lowering the mass threshold, there are $\sigma_0 \in (0, 1)$ and $\eta_1 > 0$ such that

$$(1.9) \quad \iint_{B_1 \times (-1, -\sigma_0)} |u_n|^3 dx dt \geq \eta_1 \quad \text{for all } n.$$

Lemma 1.13 (No-escape leaves positive mass away from the terminal layer). *If u_n is an admissible local Type I concentration sequence in the sense of Theorem 1.12, then (1.9) holds for some $\sigma_0 \in (0, 1)$ and some $\eta_1 > 0$.*

Proof. Choose $\sigma_0 \in (0, 1)$ so small that

$$\sup_n \iint_{B_1 \times (-\sigma_0, 0)} |u_n|^3 dx dt \leq \frac{1}{2} \eta_v,$$

which is possible by (1.8). Subtracting this terminal-layer contribution from (1.7) gives

$$\iint_{B_1 \times (-1, -\sigma_0)} |u_n|^3 dx dt \geq \frac{1}{2} \eta_v.$$

Take $\eta_1 = \eta_v/2$. □

Proposition 1.14 (Local Type I rescalings have Seregin compactness). *Let (u, p) be a suitable Leray–Hopf solution on $\mathbb{R}^3 \times (0, T)$, let $z_* = (x_*, T)$ be a singular point, and assume that there exist $\rho > 0$ and $M < \infty$ such that*

$$(1.10) \quad \operatorname{ess\,sup}_{T-\rho^2 < t < T} \sqrt{T-t} \|u(t)\|_{L^\infty(B_\rho(x_*))} \leq M.$$

Let $r_n \downarrow 0$, and let (u_n, p_n) be the rescaled sequence above. Then the following hold.

(i) *For every compact cylinder*

$$K = B_R \times [-S, -\sigma] \Subset \mathbb{R}^3 \times (-\infty, 0), \quad R, S < \infty, \quad \sigma > 0,$$

there exists $n_0(K)$ such that

$$(1.11) \quad \|u_n\|_{L^\infty(K)} \leq M\sigma^{-1/2}, \quad n \geq n_0(K).$$

(ii) *On every compact cylinder $K \Subset \mathbb{R}^3 \times (-\infty, 0)$ there are time-dependent pressure gauges $a_{n,K}(t)$ and a constant $C_K < \infty$ such that, for $n \geq n_0(K)$,*

$$(1.12) \quad \|u_n\|_{L_t^\infty L_x^2(K)} + \|\nabla u_n\|_{L^2(K)} + \|p_n - a_{n,K}(t)\|_{L^{3/2}(K)} \leq C_K.$$

The constant C_K may depend on the fixed Leray energy level of the original solution, but no concentration or no-escape assumption enters it.

(iii) *After subtracting these local pressure gauges, the family (u_n, p_n) is locally precompact in the standard Seregin topology on $\mathbb{R}^3 \times (-\infty, 0)$. Along every subsequence there is a further subsequence and an ancient suitable weak solution (U, Q) such that, on every compact $K \Subset \mathbb{R}^3 \times (-\infty, 0)$,*

$$u_n \rightarrow U \quad \text{strongly in } L^q(K), \quad 1 \leq q < \infty,$$

and

$$p_n - a_{n,K}(t) \rightharpoonup Q \quad \text{weakly in } L^{3/2}(K).$$

The limit is bounded and satisfies

$$|U(x, t)| \leq M(-t)^{-1/2}, \quad t < 0.$$

- (iv) More generally, let $\mathbf{c} = (z_n)$, $z_n = (y_n, \tau_n)$, be a parabolic recentering sequence for (u_n, p_n) with $\tau_n \leq 0$. Assume that $\mathbf{c}$ remains in the local Type I window in the following precise sense: for every $R, S < \infty$, all sufficiently large n satisfy

$$r_n(|y_n| + R) < \rho, \quad T + r_n^2(\tau_n - S) > T - \rho^2.$$

If $\mathbf{c}$ carries retained compact velocity L^3 -mass, then after passing to a subsequence it has no loss of local compactness in the sense of [Theorem 9.3](#), and its recentered fields converge locally to a bounded suitable terminal profile. Recenterings which do not satisfy this local-window condition are, by definition, routed to the exterior or diffuse terminal alternatives rather than to the compact local profile class.

Proof. Fix

$$K = B_R \times [-S, -\sigma] \in \mathbb{R}^3 \times (-\infty, 0).$$

For n sufficiently large,

$$r_n R < \rho/2, \quad r_n^2 S < \rho^2/2.$$

Hence, for a.e. $(x, t) \in K$,

$$x_* + r_n x \in B_\rho(x_*), \quad T + r_n^2 t \in (T - \rho^2, T).$$

The local Type I bound [\(1.10\)](#) gives

$$|u_n(x, t)| = r_n |u(x_* + r_n x, T + r_n^2 t)| \leq r_n M (T - (T + r_n^2 t))^{-1/2} = M(-t)^{-1/2} \leq M\sigma^{-1/2}.$$

This proves [\(1.11\)](#). In particular,

$$\operatorname{ess\,sup}_{-S < t < -\sigma} \int_{B_R} |u_n(x, t)|^2 dx \leq |B_R| M^2 \sigma^{-1},$$

and

$$\iint_K |u_n|^3 dx dt \leq |K| M^3 \sigma^{-3/2}.$$

We next record the pressure estimate because it is the only place where a local Type I proof can hide a nonlocal issue. Use the whole-space Leray pressure representative. Fix a cutoff $\chi_R \in C_c^\infty(B_{8R})$ with $\chi_R \equiv 1$ on B_{4R} , and write in B_{4R}

$$p_n = q_n + h_n, \quad q_n = \mathcal{R}_i \mathcal{R}_j (\chi_R u_{n,i} u_{n,j}).$$

The local term satisfies

$$\|q_n\|_{L^{3/2}(B_{4R} \times [-S, -\sigma])} \leq C_{R,S,\sigma,M}.$$

For the harmonic part choose the gauge $a_{n,K}(t) = h_n(0, t)$. The standard kernel difference estimate for the Riesz pressure kernel gives, for $x \in B_{2R}$,

$$|h_n(x, t) - a_{n,K}(t)| \leq C_R \sum_{m \geq 2} 2^{-4m} \int_{2^m R < |z| < 2^{m+1} R} |u_n(z, t)|^2 dz.$$

For annuli whose physical image remains in $B_\rho(x_*)$, the local Type I bound gives

$$\int_{2^m R < |z| < 2^{m+1} R} |u_n(z, t)|^2 dz \leq C M^2 \sigma^{-1} (2^m R)^3.$$

The corresponding part of the series is bounded by a geometric sum. For the remaining outer annuli one uses only the finite Leray energy:

$$\int_{\mathbb{R}^3} |u_n(z, t)|^2 dz = r_n^{-1} \int_{\mathbb{R}^3} |u(y, T + r_n^2 t)|^2 dy.$$

If m_n is the first index for which $2^{m_n} R r_n \gtrsim \rho$, then

$$\sum_{m \geq m_n} 2^{-4m} \int_{\mathbb{R}^3} |u_n(z, t)|^2 dz \leq C_{\rho, R} r_n^3 \|u\|_{L^\infty(0, T; L^2)}^2 \leq C_{\rho, R, u}.$$

Thus $h_n - a_{n, K}$ is uniformly bounded on $B_{2R} \times [-S, -\sigma]$, and

$$\|p_n - a_{n, K}(t)\|_{L^{3/2}(B_{2R} \times [-S, -\sigma])} \leq C_K.$$

Apply the local energy inequality for (u_n, p_n) with a cutoff equal to one on K and supported in a slightly larger compact cylinder $K' \Subset \mathbb{R}^3 \times (-\infty, 0)$. The already obtained local L^2 , L^3 , and pressure-gauge bounds control every term on the right-hand side, and therefore

$$\iint_K |\nabla u_n|^2 dx dt \leq C_K.$$

This proves (1.12).

The Navier–Stokes equation now bounds $\partial_t u_n$ locally in a negative Sobolev space, for instance in $L_t^{3/2} H_x^{-2}(K)$. Aubin–Lions compactness and the uniform L^∞ bound give strong local convergence of the velocity in every finite L^q , after passing to a subsequence. The pressure bound gives weak $L^{3/2}$ convergence after gauges. Passing to the limit in the distributional equations and in the local energy inequality gives an ancient suitable weak solution (U, Q) . The pointwise Type I bound passes to the limit, so U is bounded on every compact negative-time cylinder. This proves (iii).

Finally, let $\mathbf{c} = (y_n, \tau_n)$ be as in (iv), and set

$$u_n^{\mathbf{c}}(y, s) = u_n(y + y_n, s + \tau_n), \quad p_n^{\mathbf{c}}(y, s) = p_n(y + y_n, s + \tau_n).$$

The local-window condition, together with $\tau_n \leq 0$, is exactly what is needed to repeat the preceding estimates on each fixed backward compact cylinder in the (y, s) -variables: the physical cylinders remain in $B_\rho(x_*) \times (T - \rho^2, T)$, the local Type I envelope gives the same L^∞ bound on compact windows, and the pressure decomposition has the same local and finite-energy tail estimates. Thus the two failure modes in Theorem 9.3, blow-up of the local dissipation measure or failure of local pressure-gauge control, cannot occur for such a retained local recentering sequence after subsequence extraction. The recentered fields therefore converge locally to a bounded suitable terminal profile. If the local-window condition fails, then the recentering leaves every bounded terminal window attached to the singular point, or its critical mass persists only through expanding regions; those are precisely the exterior and diffuse terminal alternatives rather than loss of local compactness inside the local Type I bridge. $\square$

1.2. Pressure-projected reduction of the terminal no-escape gap. The preceding compactness result is intentionally stated away from the terminal slice. We next isolate the exact additional estimate needed to cross the terminal layer. The local pressure projection removes the nonlocal pressure flux from the local energy inequality; what remains is an interior shielding problem, namely preventing the Caccioppoli estimate on a fixed interior ball from cascading through larger and larger balls until it reaches the edge of the local Type I window.

Let $v = u^{(r)}$, $\pi = p^{(r)}$, and $\Lambda = \rho/r$. After translating z_* to $(0, 0)$, the local Type I bound gives

$$|v(x, s)| \leq M(-s)^{-1/2} \quad (x, s) \in B_\Lambda \times (-4, 0),$$

for all sufficiently small r . For $a \in \mathbb{R}^3$, $L \geq 1$, and $-4 < s < t < 0$, write

$$A_L(a, s) := \int_{B_L(a)} |v(x, s)|^2 dx, \quad D_L(a; s, t) := \int_s^t \int_{B_L(a)} |\nabla v|^2 dx d\sigma.$$

Lemma 1.15 (Local pressure-projected Caccioppoli inequality). *There is a universal constant C with the following property. Let (v, π) be a suitable weak solution in $B_{8L}(a) \times (s_-, 0)$, $L \geq 1$, and assume*

$$|v(x, s)| \leq M(-s)^{-1/2} \quad \text{on } B_{8L}(a) \times (s_-, 0).$$

If $s_0 \in (s_-, 0)$ is a Lebesgue time for v and ∇v , then for a.e. $t \in (s_0, 0)$,

$$(1.13) \quad \int_{B_L(a)} |w(x, t)|^2 dx + D_L(a; s_0, t) \leq CA_{4L}(a, s_0) + C \int_{s_0}^t \left(L^{-2} + ML^{-1}(-s)^{-1/2} \right) A_{4L}(a, s) ds.$$

The estimate is entirely local on $B_{8L}(a)$; no pressure tail from outside $B_{8L}(a)$ appears. Here $w = v + \nabla p_h$ is the local projected velocity defined in the proof below.

Proof. We recall the standard local pressure-projection argument in the scale L normalization. Let $B = B_{8L}(a)$. Denote by E_B^* the Stokes pressure projection on B : for $F \in W^{-1,q}(B)$, $1 < q < \infty$, $\nabla E_B^*(F)$ is the gradient part of the stationary Stokes solution in B with zero boundary velocity. It satisfies the scale-invariant estimates

$$\|\nabla E_B^*(\nabla \cdot F)\|_{L^q(B)} \leq C_q \|F\|_{L^q(B)}, \quad \|\nabla E_B^*(\Delta w)\|_{L^2(B)} \leq C \|\nabla w\|_{L^2(B)}.$$

Set

$$\nabla p_h = -\nabla E_B^*(v), \quad w := v + \nabla p_h.$$

The harmonic pressure p_h is harmonic in B , and the projection estimates give

$$\|w\|_{L^2(B)} + \|\nabla p_h\|_{L^2(B)} \leq C \|v\|_{L^2(B)}, \quad \|\nabla w\|_{L^2(B_{6L}(a))} \leq C \|\nabla v\|_{L^2(B)}.$$

The projected local energy inequality for w , obtained by applying the Stokes projection to the distributional Navier–Stokes equations and testing with $\zeta^2 w$, reads

$$(1.14) \quad \begin{aligned} & \int |w(t)|^2 \zeta^2 + \int_{s_0}^t \int |\nabla v|^2 \zeta^2 \\ & \leq C \int |w(s_0)|^2 \zeta^2 + C \int_{s_0}^t \int |w|^2 (|\partial_s \zeta| + |\Delta \zeta| + |\nabla \zeta|^2) \\ & \quad + C \int_{s_0}^t \int |v| |w|^2 |\nabla \zeta| + C \int_{s_0}^t \int |v| |\nabla p_h|^2 |\nabla \zeta|. \end{aligned}$$

Here $\zeta \in C_c^\infty(B_{4L}(a))$, $\zeta \equiv 1$ on $B_L(a)$, and

$$|\nabla \zeta| \leq CL^{-1}, \quad |\Delta \zeta| + |\partial_s \zeta| \leq CL^{-2}.$$

The pressure-projection estimates allow w and ∇p_h to be bounded by v in $L^2(B_{4L}(a))$. Therefore the cutoff terms in (1.14) are bounded by

$$CL^{-2} \int_{s_0}^t A_{4L}(a, s) ds.$$

The two convective terms are bounded, using the Type I envelope, by

$$CL^{-1} \int_{s_0}^t \|v(s)\|_{L^\infty(B_{4L}(a))} A_{4L}(a, s) ds \leq CML^{-1} \int_{s_0}^t (-s)^{-1/2} A_{4L}(a, s) ds.$$

The initial projected energy is controlled by

$$\int |w(s_0)|^2 \zeta^2 dx \leq CA_{4L}(a, s_0).$$

Combining the preceding bounds gives (1.13). $\square$

Definition 1.16 (Interior shielding estimate). The local pointwise Type I window at z_* is said to satisfy the interior shielding estimate if, for every fixed $R < \infty$, there are

$$r_R > 0, \quad B_R < \infty,$$

such that every rescaling $v = u^{(r)}$, $0 < r < r_R$, with $\Lambda = \rho/r$, satisfies

$$(1.15) \quad \operatorname{ess\,sup}_{-1 < s < 0} \int_{B_{8R}} |v(x, s)|^2 dx + \int_{-1}^0 \int_{B_{8R}} |\nabla v|^2 dx ds \leq B_R.$$

Equivalently, the dependence on A_{4L} in [Theorem 1.15](#) can be stopped inside a fixed multiple of the target ball, uniformly as the edge $\partial B_{\rho/r}$ of the local Type I window recedes to infinity.

Proposition 1.17 (Pressure projection reduces the entry gap to interior shielding). *Assume the local pointwise Type I bound (1.10) at z_* . If the interior shielding estimate (1.15) holds with $R = 1$, then the velocity no-escape condition (1.8) holds for every positive local Type I concentration sequence at z_* . Consequently such a sequence is admissible in the sense of [Theorem 1.12](#).*

Proof. Let $u_n = u^{(r_n)}$ be any positive local Type I concentration sequence. By (1.15) with $R = 1$, after discarding finitely many indices,

$$\operatorname{ess\,sup}_{-1 < s < 0} \int_{B_1} |u_n(x, s)|^2 dx \leq B_1.$$

The local Type I envelope gives

$$\|u_n(s)\|_{L^\infty(B_1)} \leq M(-s)^{-1/2}.$$

Therefore, for $0 < \sigma < 1$,

$$\begin{aligned} \int_{-\sigma}^0 \int_{B_1} |u_n|^3 dx ds &\leq \int_{-\sigma}^0 \|u_n(s)\|_{L^\infty(B_1)} \int_{B_1} |u_n(x, s)|^2 dx ds \\ &\leq 2MB_1\sigma^{1/2}. \end{aligned}$$

The finitely many discarded indices have absolutely continuous $|u_n|^3$ -integrals on $B_1 \times (-1, 0)$, so taking the supremum over the whole sequence and then letting $\sigma \downarrow 0$ gives (1.8). The positive Q_1 mass hypothesis is the other part of [Theorem 1.12](#). $\square$

Remark 1.18 (The narrowed final bridge). [Theorem 1.15](#) shows, independently of the Albritton–Barker shortcut used below, how the terminal no-escape issue reduces to an interior shielding estimate: after local pressure projection, the local energy inequality controls the projected velocity and the dissipation up to the larger-scale quantity A_{4L} . A proof of (1.15) by this route would prevent an edge-fed terminal cascade directly. In the present paper this shielding is supplied instead by the endpoint weak Serrin local Type I package in [Theorem 1.19](#).

Theorem 1.19 (Albritton–Barker weak Serrin local Type I package). *Let (v, q) be a suitable weak solution in a parabolic cylinder*

$$Q_\rho(z_0) = B_\rho(x_0) \times (t_0 - \rho^2, t_0).$$

Assume

$$v \in L_t^{q, \infty} L_x^{p, \infty}(Q_\rho(z_0)), \quad 3 \leq p \leq \infty, \quad 2 \leq q \leq \infty, \quad \frac{3}{p} + \frac{2}{q} = 1.$$

Then for every $0 < \theta < 1$, the scale-invariant local Type I package is finite on $Q_{\theta\rho}(z_0)$: there is

$$K_\theta < \infty$$

such that every cylinder $Q_\lambda(y, \tau) \subset Q_{\theta\rho}(z_0)$ satisfies

$$A(Q_\lambda(y, \tau)) + C(Q_\lambda(y, \tau)) + D(Q_\lambda(y, \tau)) + E(Q_\lambda(y, \tau)) \leq K_\theta.$$

Here

$$A(Q_\lambda(y, \tau)) := \lambda^{-1} \operatorname{ess\,sup}_{\tau - \lambda^2 < s < \tau} \int_{B_\lambda(y)} |v(x, s)|^2 dx,$$

and C, D, E denote the usual scale-invariant L^3 -velocity, $L^{3/2}$ -pressure, and dissipation quantities of Albritton–Barker. In the endpoint $p = \infty, q = 2$, the constant is allowed to depend on the local suitable-solution background quantities on $Q_\rho(z_0)$, including the local C - and D -bounds.

Proof. After translating and scaling to the unit cylinder, this is Lemma 2.5 of Albritton–Barker [1]. Their proof combines the weak Serrin bound with critical Morrey estimates for suitable weak solutions. The endpoint $p = \infty, q = 2$ is included in the weak Serrin lemma; as emphasized there, the resulting local Type I bound is a suitable-solution estimate and its constant may depend on the background local energy quantities on the larger cylinder. $\square$

Theorem 1.20 (Local pointwise Type I implies terminal no-escape). *Let (u, p) be a suitable Leray–Hopf solution on $\mathbb{R}^3 \times (0, T)$, let $z_* = (x_*, T)$, and assume that*

$$u \in L^\infty(0, T; L^2(\mathbb{R}^3)) \cap L^2(0, T; \dot{H}^1(\mathbb{R}^3)).$$

Assume that z_ satisfies the local pointwise Type I bound: there are $\rho > 0$ and $M < \infty$ such that*

$$(1.16) \quad \operatorname{ess\,sup}_{T - \rho^2 < t < T} \sqrt{T - t} \|u(t)\|_{L^\infty(B_\rho(x_*))} \leq M.$$

For $0 < r < \rho$, set

$$u^{(r)}(x, s) = r u(x_* + rx, T + r^2 s), \quad p^{(r)}(x, s) = r^2 p(x_* + rx, T + r^2 s).$$

Then for every $0 < R < \infty$ there are constants

$$r_R \in (0, \rho/R), \quad B_R < \infty,$$

and time-dependent pressure gauges $a_{r,R}(s)$, depending on R, M, ρ, T and the local suitable-solution background quantities of (u, p) in $Q_\rho(z_)$, such that the following terminal local energy package holds for every $0 < r < r_R$:*

$$(1.17) \quad \begin{aligned} & \operatorname{ess\,sup}_{-1 < s < 0} \int_{B_R} |u^{(r)}(x, s)|^2 dx + \int_{-1}^0 \int_{B_R} |\nabla u^{(r)}|^2 dx ds \\ & + \int_{-1}^0 \int_{B_R} |p^{(r)}(x, s) - a_{r,R}(s)|^{3/2} dx ds \leq B_R. \end{aligned}$$

Consequently, for every $0 < \sigma < 1$ and every $0 < r < r_R$,

$$(1.18) \quad \int_{-\sigma}^0 \int_{B_R} |u^{(r)}(x, s)|^3 dx ds \leq 2MB_R \sigma^{1/2},$$

and in particular

$$(1.19) \quad \lim_{\sigma \downarrow 0} \sup_{0 < r < r_R} \int_{-\sigma}^0 \int_{B_R} |u^{(r)}(x, s)|^3 dx ds = 0.$$

Proof. Fix once and for all $0 < \theta < 1$, for instance $\theta = 1/2$, and put $L_R := \max\{R, 1\}$. By decreasing r_R , we may assume

$$B_{L_R r}(x_*) \times (T - L_R^2 r^2, T) \subset Q_{\theta \rho}(z_*) \quad (0 < r < r_R).$$

The pointwise Type I bound implies the endpoint weak Serrin estimate on $Q_\rho(z_*)$. Indeed, after writing

$$f(t) = \|u(t)\|_{L^\infty(B_\rho(x_*))},$$

one has

$$|\{t \in (T - \rho^2, T) : f(t) > \alpha\}| \leq M^2 \alpha^{-2}, \quad \alpha > 0,$$

and therefore $u \in L_t^{2,\infty} L_x^\infty(Q_\rho(z_*))$. Applying [Theorem 1.19](#) gives a finite constant K_θ such that $A, C, D, E \leq K_\theta$ on all cylinders contained in $Q_{\theta\rho}(z_*)$.

Taking the cylinder $Q_{L_R r}(z_*)$ gives

$$(L_R r)^{-1} \operatorname{ess\,sup}_{T-L_R^2 r^2 < t < T} \int_{B_{L_R r}(x_*)} |u(x, t)|^2 dx \leq K_\theta,$$

and similarly the D - and E -bounds give

$$(L_R r)^{-1} \int_{T-L_R^2 r^2}^T \int_{B_{L_R r}(x_*)} |\nabla u|^2 dx dt \leq K_\theta,$$

$$(L_R r)^{-2} \int_{T-L_R^2 r^2}^T \int_{B_{L_R r}(x_*)} |p - a_{r,R}^{\text{phys}}(t)|^{3/2} dx dt \leq K_\theta$$

for suitable time-dependent pressure gauges $a_{r,R}^{\text{phys}}(t)$. Scaling these three estimates and restricting to

$$B_R \times (-1, 0) \subset B_{L_R} \times (-L_R^2, 0)$$

gives [\(1.17\)](#) with

$$a_{r,R}(s) := r^2 a_{r,R}^{\text{phys}}(T + r^2 s),$$

after enlarging the constant B_R by a harmless factor depending on R and K_θ .

After translating (x_*, T) to $(0, 0)$, the rescaled fields also satisfy

$$|u^{(r)}(x, s)| \leq M(-s)^{-1/2} \quad (x, s) \in B_R \times (-1, 0)$$

for all $r < r_R$, after decreasing r_R if necessary. The terminal local energy package just proved gives

$$\operatorname{ess\,sup}_{-1 < s < 0} \int_{B_R} |u^{(r)}(x, s)|^2 dx \leq B_R.$$

For $0 < \sigma < 1$, the local Type I bound above and the kinetic bound give

$$\begin{aligned} \int_{-\sigma}^0 \int_{B_R} |u^{(r)}|^3 dx ds &\leq \int_{-\sigma}^0 \|u^{(r)}(s)\|_{L^\infty(B_R)} \int_{B_R} |u^{(r)}(x, s)|^2 dx ds \\ &\leq M B_R \int_{-\sigma}^0 (-s)^{-1/2} ds = 2M B_R \sigma^{1/2}. \end{aligned}$$

This proves [\(1.18\)](#) and [\(1.19\)](#). □

Corollary 1.21 (No diffuse terminal-layer mass). *Under the hypotheses of [Theorem 1.20](#), for every fixed $R < \infty$ there do not exist $r_n \downarrow 0$, $\sigma_n \downarrow 0$, and $\varepsilon_0 > 0$ such that*

$$r_n^{-2} \int_{T-\sigma_n r_n^2}^T \int_{B_{R r_n}(x_*)} |u(x, t)|^3 dx dt \geq \varepsilon_0 \quad \text{for all } n.$$

Proof. For $u_n = u^{(r_n)}$, the left hand side is

$$\int_{-\sigma_n}^0 \int_{B_R} |u_n(x, s)|^3 dx ds.$$

By [\(1.18\)](#), this is bounded by

$$2M B_R \sigma_n^{1/2} \rightarrow 0,$$

which contradicts any fixed positive lower bound. □

Corollary 1.22 (Local pointwise Type I concentration sequences are admissible). *Let (u, p) and $z_* = (x_*, T)$ satisfy the hypotheses of [Theorem 1.20](#). Let $r_n \downarrow 0$ be any velocity concentration sequence satisfying*

$$\iint_{B_1 \times (-1, 0)} |u_n|^3 dx dt \geq \eta_v > 0, \quad u_n(x, t) = r_n u(x_* + r_n x, T + r_n^2 t).$$

Then u_n is an admissible local Type I concentration sequence in the sense of [Theorem 1.12](#).

Proof. Apply [Theorem 1.20](#) with $R = 1$. After discarding finitely many n , all $r_n < r_1$, and

$$\limsup_{\sigma \downarrow 0} \sup_n \int_{-\sigma}^0 \int_{B_1} |u_n|^3 dx dt = 0.$$

The finitely many discarded indices do not affect the limit as $\sigma \downarrow 0$, because each fixed suitable weak solution has $|u_n|^3 \in L^1(B_1 \times (-1, 0))$. The positive mass hypothesis is the other condition in [Theorem 1.12](#). $\square$

Lemma 1.23 (Local Type I concentration enters the terminal dichotomy). *Under the hypotheses of [Theorem 1.14](#), let $r_n \downarrow 0$ be a velocity concentration sequence at z_* satisfying*

$$\iint_{B_1 \times (-1, 0)} |u_n|^3 dx dt \geq \eta_v > 0.$$

Then r_n is admissible by [Theorem 1.22](#). Let σ_0, η_1 be given by [Theorem 1.13](#). For the associated terminal extraction sequence (V_N, P_N) generated by a Seregin limit of this sequence, choose the small-data threshold $\varepsilon_{\text{sd}} \leq \eta_1/2$. Then the origin extraction produces a bounded ancient Seregin profile with retained compact local velocity mass away from the terminal layer. In particular the retained compact concentration generated from the origin recentering has no loss of local compactness. Any retained local recentering which stays in the local Type I window has the same compactness property by [Theorem 1.14](#); any concentration leaving that window is routed to the exterior or diffuse terminal remainder. The paired terminal decomposition of [Theorem 9.9](#), together with the mixed-state exclusions and active descendant-chain exclusions [Theorems 9.61](#) and [10.20](#), reduces the retained possibilities to a single retained concentrating terminal profile or a finite separated family of retained concentrating terminal profiles.

Proof. By [Theorem 1.13](#),

$$\iint_{B_1 \times (-1, -\sigma_0)} |u_n|^3 dx dt \geq \eta_1 \quad \text{for all } n.$$

The cylinder $B_1 \times [-1, -\sigma_0]$ is compactly contained in $\mathbb{R}^3 \times (-\infty, 0)$. Therefore [Theorem 1.14](#) gives, after passing to a subsequence, strong local L^3 convergence on this cylinder:

$$u_n \rightarrow U \quad \text{in } L^3(B_1 \times (-1, -\sigma_0)).$$

Consequently

$$\iint_{B_1 \times (-1, -\sigma_0)} |U|^3 dx dt \geq \eta_1.$$

Cover $B_1 \times (-1, -\sigma_0)$ by finitely many unit terminal cylinders strictly away from $t = 0$. At least one such cylinder carries a fixed positive velocity lower bound, and since the local CKN measure dominates the velocity part, choosing $\varepsilon_{\text{sd}} \leq \eta_1/2$ after adjusting by this finite cover makes the corresponding origin-frame recentering ε_{sd} -concentrating. This argument is precisely where no-escape is used: without [\(1.8\)](#), the lower bound could remain entirely in $B_1 \times (-\sigma, 0)$ for every fixed $\sigma > 0$.

The distinguished recentering satisfies the local-window condition in [Theorem 1.14](#), so it is not in the local-compactness-failure class. Other retained local recenterings inside the same

Type I window have the same compactness. Recenterings leaving the window are exterior or diffuse terminal remainders. The paired decomposition [Theorem 9.9](#) records these remainders alongside the active profile locus rather than treating them as mutually exclusive alternatives. The sole-carrier radiative cases are excluded by [Theorem 9.12](#), mixed compact–diffuse or compact–noncompact coexistence is excluded by [Theorem 9.61](#), and infinite active descendant chains are excluded by [Theorem 10.20](#). Hence [Theorem 9.10](#) leaves only a single retained concentrating terminal profile or a finite separated family of retained concentrating terminal profiles. $\square$

Theorem 1.24 (Local Type I singularities enter the full exclusion pipeline). *Let (u, p) be a suitable Leray–Hopf solution on $\mathbb{R}^3 \times (0, T)$, let $z_* = (x_*, T)$ be a singular point, and assume that z_* satisfies the local Type I bound [\(1.10\)](#). Then there exists a bounded centered Type I Seregin profile with retained local concentration generated from (u, p, z_*) . Consequently the local terminal concentration package of [Theorem 10.24](#) applies. In particular, no finite-energy local pointwise Type I singular point exists.*

Proof. By the local velocity concentration theorem in the setup package, choose $r_n \downarrow 0$ and $\eta_v > 0$ with

$$\iint_{B_1 \times (-1, 0)} |u_n|^3 dx dt \geq \eta_v, \quad u_n(x, t) = r_n u(x_* + r_n x, T + r_n^2 t).$$

[Theorems 1.20](#) and [1.22](#) show that this concentration sequence satisfies the terminal velocity no-escape condition required by the setup package. Apply [Theorem 1.23](#) to this sequence. The local Type I rescalings therefore produce at least one bounded terminal profile carrying retained local concentration. After passing to the centered variables, the bound $|U(x, t)| \leq M(-t)^{-1/2}$ becomes the global centered Type I bound

$$|V(y, \tau)| \leq M,$$

so each retained profile generated by the bridge is a bounded centered Type I Seregin profile.

If every such retained profile belongs to one of the previously closed lower classes, then the contradiction is supplied by the setup package: small-amplitude and stationary L^3 profiles are excluded there, the uniformly tight class is excluded there, and the closed structured/decay class is excluded there. Otherwise choose a retained profile outside those lower classes. It is then a refined residual candidate, so [Theorem 10.24](#) applies. In that residual setting, a finite separated family of retained concentrating terminal profiles is excluded by [Theorem 10.23](#), infinite active descendant chains are excluded by [Theorem 10.20](#), and mixed compact–diffuse tails are excluded by [Theorem 9.61](#). The remaining single retained profile is terminally indecomposable, has the backward finite-critical-norm sequence, has parasitic-free finite-shift mildness, and is then excluded by the Albritton–Barker endpoint Liouville input used in this paper. Both routes are impossible. Hence the assumed finite-energy local Type I singular point does not exist. $\square$

Corollary 1.25 (Unconditional local pointwise Type I exclusion). *Let (u, p) be a suitable Leray–Hopf solution on $\mathbb{R}^3 \times (0, T)$. If $z_* = (x_*, T)$ is a singular point satisfying the local Type I bound [\(1.10\)](#), then the completed case closures give a contradiction. Equivalently, in the local Type I/local Type II dichotomy of the setup package, every finite-energy singular point belongs to the local Type II alternative.*

Proof. The first assertion is exactly [Theorem 1.24](#). The equivalence is the local Type I/local Type II dichotomy of the setup package. $\square$

2. RENDERED DECISION TREE FOR THE STATE-SPACE PARTITION

The following rendered decision tree records the ordered partition used in the paper. Each diamond is a stratification criterion. Each terminal node is a possible singularity or candidate

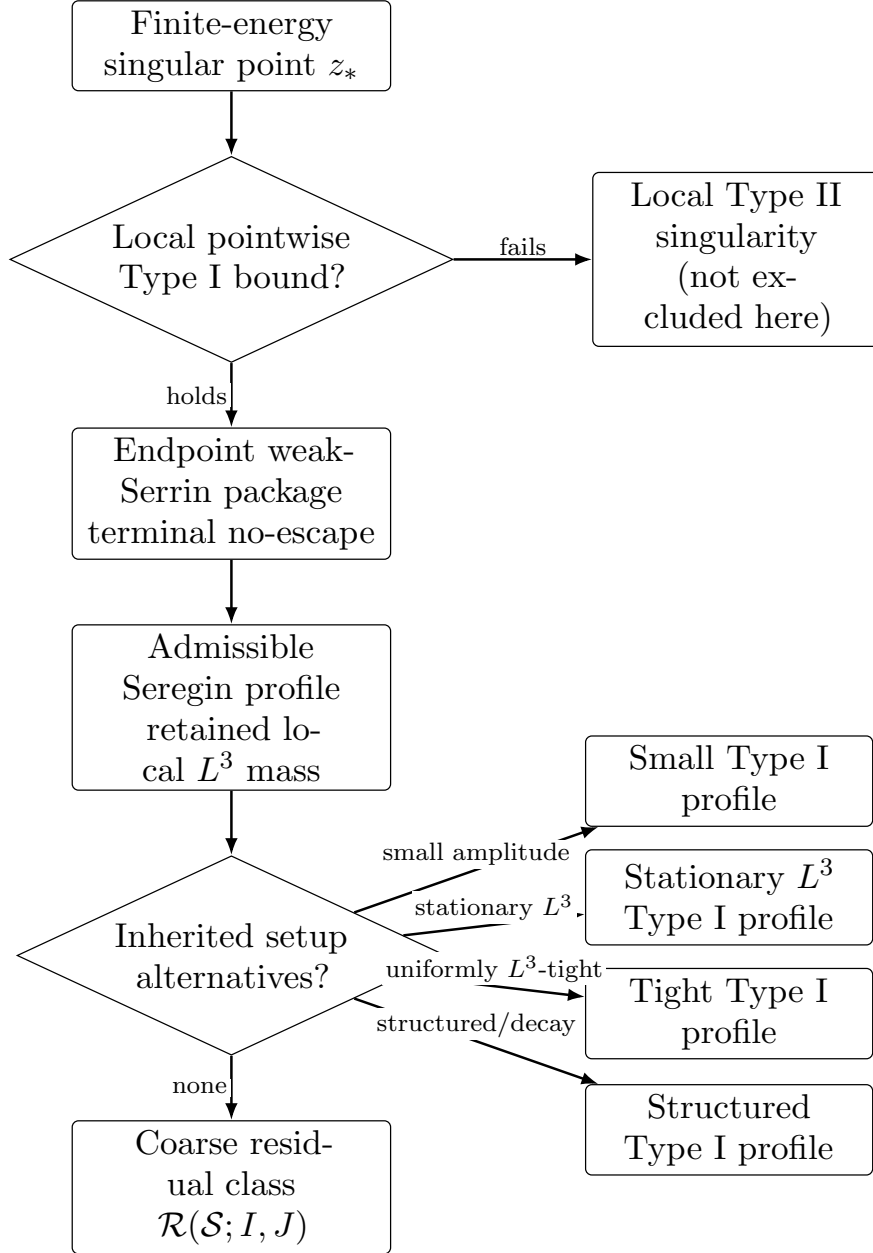


FIGURE 2. Rendered entry tree. Local pointwise Type I enters the setup admissible Seregin package through endpoint weak Serrin estimates and terminal no-escape. If the bound fails, the point belongs to the local Type II alternative, which is not excluded here.

blow-up profile type selected by the criteria. The local Type II terminal alternative is the remaining regime after the local pointwise Type I case is excluded; every Type I terminal alternative is discharged either by the inherited setup alternatives or by the case-closure theorems proved below.

The loop from the last criterion back to the residual class is only a graphical way to indicate the ordered nature of the decomposition: if a lower stratum has not actually been removed, it is routed to its corresponding terminal alternative rather than to $\mathcal{C}_{\text{gen}}$. After the ordered removals

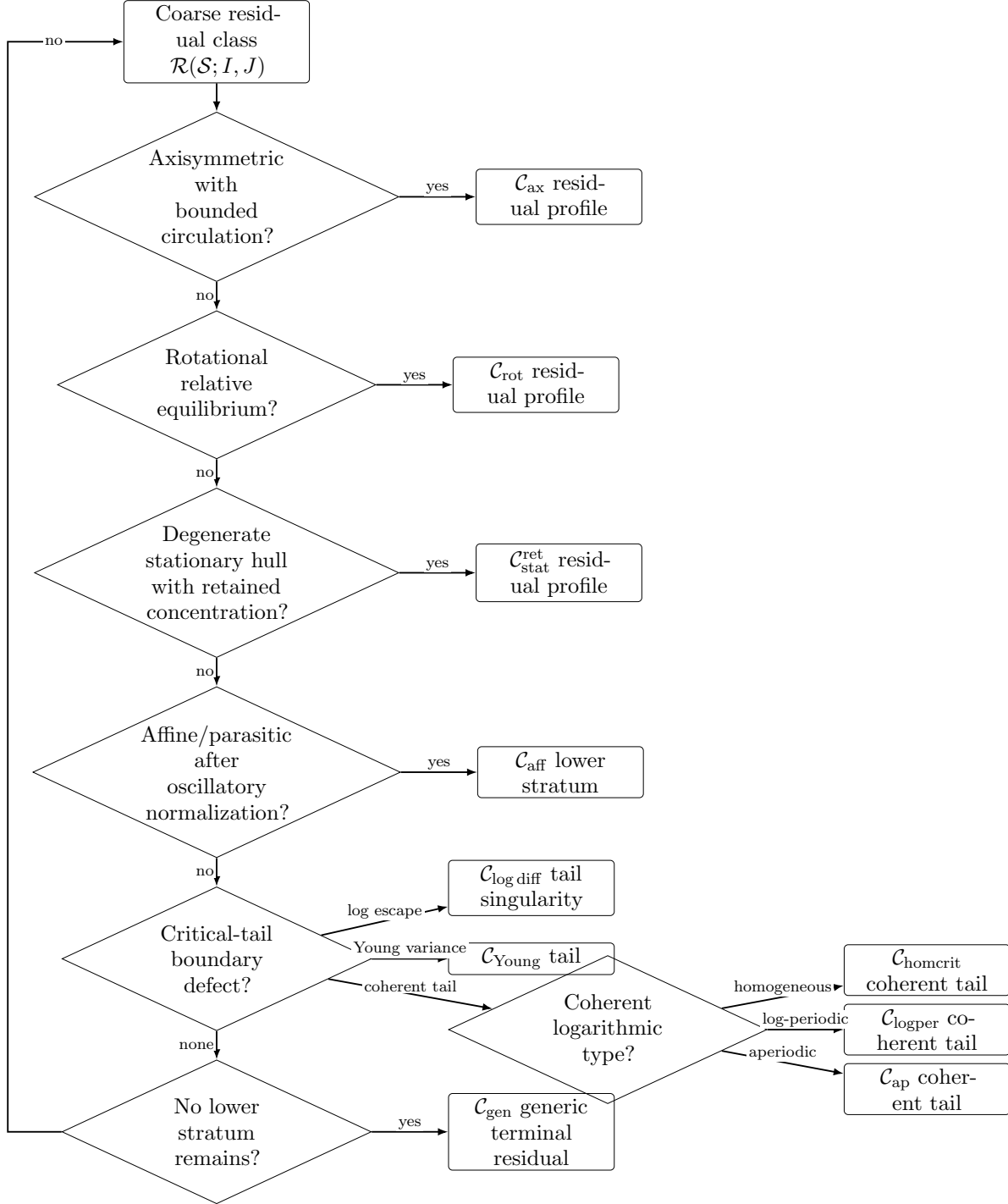


FIGURE 3. Rendered residual stratification tree. The final return arrow only records the ordered nature of the decomposition: if a lower stratum has not been removed, the profile is routed to that terminal alternative; after all lower alternatives are removed, the remaining residual class is $\mathcal{C}_{\text{gen}}$.

are performed, $\mathcal{C}_{\text{gen}}$ is exactly the remaining residual class by definition. For the table, write

$$\mathfrak{T}_\rho(u; z_*) := \operatorname{ess\,sup}_{T-\rho^2 < t < T} \sqrt{T-t} \|u(t)\|_{L^\infty(B_\rho(x_*))}.$$

Table 1: Classification and closure table for the decision tree. Each row gives the criterion selecting the terminal alternative, the mechanism that discharges it when it is a Type I alternative, and the statement used at that point. For the inherited setup alternatives, the displayed condition is a representative form of the inherited definition; the precise definitions and closures are those of `proof_setup.tex`.

Terminal alternative	Criterion, discharge mechanism, and reference
Local Type II singularity	Criterion. For every $\rho > 0$, $\mathfrak{T}_\rho(u; z_*) = \infty$. Status. Not excluded in this paper; this is the remaining alternative after local pointwise Type I is ruled out. Link. Theorem 1.25 .
Small Type I profile	Criterion. Inherited small-amplitude class; schematically $\ V\ _{L^\infty(\mathbb{R}^3 \times \mathbb{R})} \leq \varepsilon_{\text{sm}}$. Discharge. Small-data/regularity closure recorded in the setup package. Link. Theorem 1.4 .
Stationary L^3 Type I profile	Criterion. Inherited stationary class; representative condition $\partial_\tau V = 0$ and $V \in L^3(\mathbb{R}^3)$. Discharge. Stationary L^3 Liouville/regularity closure imported from the setup package. Link. Theorem 1.4 .
Tight Type I profile	Criterion. Inherited tight class; representative condition $\lim_{R \rightarrow \infty} \sup_\tau \int_{ y > R} V(y, \tau) ^3 dy = 0$. Discharge. Endpoint L^3 rigidity for uniformly tight ancient dynamics. Link. Theorem 1.4 .
Structured Type I profile	Criterion. Inherited closed structured/decay class, $V \in \mathcal{C}_{\text{struct/decay}}^{\text{setup}}$. Discharge. Closed structured/decay alternatives imported from the setup package. Link. Theorem 1.4 .
$\mathcal{C}_{\text{ax}}$ residual profile	Criterion. Axisymmetric residual profile with bounded centered circulation $\Gamma(\rho, Z, \tau) = \rho V_\theta$ and $\ \Gamma\ _{L^\infty} < \infty$. Discharge. Centered circulation equation plus axis-absorption Liouville theorem; the profile becomes no-swirl and falls into an earlier structured class. Link. Theorem 4.3 .
$\mathcal{C}_{\text{rot}}$ residual profile	Criterion. Rotational relative equilibrium $V(y, \tau) = Q(\tau)W(Q(\tau)^T y)$, $Q(\tau) = e^{\tau\Omega}$. Discharge. Localized Coriolis-flux routing into the terminal local concentration decomposition. Link. Theorem 5.8 .
$\mathcal{C}_{\text{stat}}^{\text{ret}}$ residual profile	Criterion. Degenerate stationary-hull retained profile: along time translations $T_{s_k} V \rightarrow W$, where W is stationary and $\mathcal{E}_1(W, \Pi; 0) \geq \varepsilon_*$. Discharge. Scale reselection reduces to a stationary retained concentration profile, then terminal concentration excludes it. Link. Theorem 7.6 .
$\mathcal{C}_{\text{aff}}$ lower stratum	Criterion. Affine-normalized oscillatory activity vanishes: $\Phi(S_c U) = 0$ for an admissible affine symmetry S_c . Discharge. Oscillatory entry normalization forces retained residual states to keep positive non-affine activity after affine normalization. Link. Theorem 9.53 .
$\mathcal{C}_{\text{log diff}}$ tail singularity	Criterion. Logarithmic escape component in the realized critical-tail compactification: $\Lambda \neq 0$. Discharge. Log-window descendant selection plus first-bad mesoscopic routing and hidden-scale compactness. Link. Theorem 9.26 .
$\mathcal{C}_{\text{Young}}$ tail	Criterion. Tight annular critical-tail state with non-Dirac Young measure: $\Lambda = 0$ and $\mathcal{Y}_{x,s} \neq \delta_{\tilde{W}(x,s)}$ on a set of positive measure. Discharge. No hidden-scale variance gives strong L_{loc}^3 compactness, hence a Dirac Young measure. Link. Theorem 9.54 .
$\mathcal{C}_{\text{homcrit}}$ coherent tail	Criterion. Coherent (-1) -homogeneous tail: $W(r\omega, s) = r^{-1}A(\omega, s)$. Discharge. Bounded-origin realization gives $W \in L_{\text{loc}}^\infty$ near $r = 0$, forcing $A = 0$. Link. Theorem 9.59 .

Terminal alternative	Criterion, discharge mechanism, and reference
$\mathcal{C}_{\log\text{per}}$ coherent tail	Criterion. Nonzero log-periodic coherent tail: $Z(\rho + L, \omega, s) = Z(\rho, \omega, s)$, $L \neq 0$, where $Z = e^\rho W(e^\rho \omega, s)$. Discharge. Bounded-origin realization gives $Z(\rho, \cdot, \cdot) \rightarrow 0$ as $\rho \rightarrow -\infty$; periodicity forces $Z \equiv 0$. Link. Theorem 9.59 .
$\mathcal{C}_{\text{ap}}$ coherent tail	Criterion. Nonzero aperiodic minimal logarithmic hull: $\mathcal{H}(Z)$ is minimal, nonperiodic, and nonzero. Discharge. Bounded-origin realization puts 0 in the compact log-translation hull; minimality forces the hull to be $\{0\}$. Link. Theorem 9.59 .
$\mathcal{C}_{\text{gen}}$ generic terminal residual	Criterion. Ordered complement $V \in \mathcal{R}(\mathcal{S}; I, J) \setminus \bigcup_{\mathcal{B} \in \mathfrak{B}_{\text{lower}}} \mathcal{B},$ where $\mathfrak{B}_{\text{lower}}$ is the list of all preceding Type I alternatives. Discharge. Paired active-locus decomposition, finite separated-family exclusion, terminal indecomposability, sequence-L^3, parasitic-free mildness, and the Albritton–Barker endpoint Liouville theorem. Link. Theorem 11.2.

3. SHARED LOCAL NOTATION

For $z_0 = (y_0, \tau_0)$ and $r > 0$, write

$$Q_r(z_0) := B_r(y_0) \times (\tau_0 - r^2, \tau_0).$$

For a centered profile (V, P) , the pressure-normalized local critical quantity is

$$(3.1) \quad \mathcal{E}_r(V, P; z_0) := \iint_{Q_r(z_0)} \left(|V|^3 + |P - c_{r,z_0}(\tau)|^{3/2} \right) dy d\tau,$$

where $c_{r,z_0}(\tau)$ is an allowed time-dependent pressure gauge, usually chosen as a spatial average over $B_r(y_0)$. All local pressure convergences below are understood after such gauges.

Definition 3.1 (Retained local concentration profile). Fix $\varepsilon_* > 0$. A terminal profile (U, Π) has retained local concentration if, after the terminal normalization under consideration,

$$\mathcal{E}_1(U, \Pi; 0) \geq \varepsilon_*.$$

Definition 3.2 (Threshold hierarchy). The constants used in the terminal package are chosen in the following order. The compact velocity lower bounds η_0 and η_v are supplied by the setup package in `proof_setup.tex` and the local concentration input therein, respectively. The local small-data threshold ε_{sd} is then fixed with

$$0 < \varepsilon_{\text{sd}} \leq \min\{\eta_0, \eta_v, \varepsilon_{\text{CKN}}\},$$

where ε_{CKN} denotes the universal CKN regularity threshold whenever that threshold is invoked. The retained descendant threshold ε_* is chosen after the pressure gauges and compactness topology are fixed, with

$$0 < \varepsilon_* \leq \frac{1}{2} \varepsilon_{\text{sd}}.$$

Whenever a compactness or lower-semicontinuity passage loses a harmless fixed factor, ε_* is replaced once and for all by a smaller positive value. No later argument chooses a new concentration threshold depending on a previously extracted profile tree.

Definition 3.3 (Refined residual candidate). A refined residual candidate is a bounded normalized Seregin ancient limit obtained from the admissible Seregin extraction theorem in `proof_setup.tex`, which carries the positive compact local velocity mass in (1.5), and which is not in any alternative already closed before the refined residual decomposition is entered.

The definition is local: the only persistent lower bound attached to the candidate by the setup package is the fixed compact velocity L^3 lower bound.

Remark 3.4 (No hidden global L^3 assertion). The terminal arguments below do not claim that a generic residual profile is globally L^3 -tight or globally bounded in $L_\tau^\infty L_y^3$. The only nonlocal critical-norm conclusion used at the endpoint is the weaker sequence- L^3 completeness of retained indecomposable concentration profiles, and that conclusion is derived from local terminal alternatives rather than from an a priori global estimate on the original residual class.

4. EXCLUSION OF THE AXISYMMETRIC BOUNDED-CIRCULATION RESIDUAL CLASS

In this section we prove the rigidity statement needed for the first refined residual class. The argument uses the scale-invariant angular circulation in centered variables. The essential point is that the centered Type I equation contains the Ornstein–Uhlenbeck drift. Thus the angular-circulation equation has an absorbing boundary at the symmetry axis together with a confining drift at spatial infinity. This gives a scalar Liouville theorem for bounded ancient solutions of the circulation equation.

Throughout the section we write cylindrical coordinates as (ρ, θ, Z) , and an axisymmetric vector field as

$$V = V_\rho e_\rho + V_\theta e_\theta + V_Z e_Z.$$

The centered Navier–Stokes equation is

$$(4.1) \quad \partial_\tau V - \Delta V + \frac{1}{2}y \cdot \nabla V + \frac{1}{2}V + (V \cdot \nabla)V + \nabla P = 0, \quad \nabla \cdot V = 0.$$

For an axisymmetric solution define the centered angular circulation

$$(4.2) \quad \Gamma(\rho, Z, \tau) := \rho V_\theta(\rho, Z, \tau).$$

This is the same scale-invariant quantity as the physical circulation rU_θ . If the physical representative is

$$U(x, t) = (-t)^{-1/2} V(x/\sqrt{-t}, -\log(-t)),$$

then

$$rU_\theta(r, z, t) = \rho V_\theta(\rho, Z, \tau), \quad r = \sqrt{-t} \rho, \quad z = \sqrt{-t} Z.$$

Lemma 4.1 (Centered circulation equation). *Let (V, P) be a smooth bounded axisymmetric solution of (4.1). Then $\Gamma = \rho V_\theta$ satisfies*

$$(4.3) \quad \partial_\tau \Gamma = \partial_\rho^2 \Gamma - \frac{1}{\rho} \partial_\rho \Gamma + \partial_Z^2 \Gamma - \left(V_\rho + \frac{\rho}{2}\right) \partial_\rho \Gamma - \left(V_Z + \frac{Z}{2}\right) \partial_Z \Gamma$$

on $(0, \infty) \times \mathbb{R} \times \mathbb{R}$. Moreover

$$(4.4) \quad \Gamma(0, Z, \tau) = 0 \quad \text{for all } (Z, \tau) \in \mathbb{R} \times \mathbb{R}.$$

Proof. The angular component of (4.1) is

$$\partial_\tau V_\theta - \left(\Delta - \frac{1}{\rho^2}\right) V_\theta + \frac{1}{2}(\rho \partial_\rho + Z \partial_Z) V_\theta + \frac{1}{2} V_\theta + V_\rho \partial_\rho V_\theta + V_Z \partial_Z V_\theta + \frac{V_\rho V_\theta}{\rho} = 0.$$

Multiplying by ρ , using

$$\begin{aligned} \rho \left(\Delta - \frac{1}{\rho^2}\right) V_\theta &= \partial_\rho^2 \Gamma - \frac{1}{\rho} \partial_\rho \Gamma + \partial_Z^2 \Gamma, \\ \rho \left[\frac{1}{2}(\rho \partial_\rho + Z \partial_Z) V_\theta + \frac{1}{2} V_\theta\right] &= \frac{1}{2}(\rho \partial_\rho + Z \partial_Z) \Gamma, \end{aligned}$$

and

$$\rho \left(V_\rho \partial_\rho V_\theta + V_Z \partial_Z V_\theta + \frac{V_\rho V_\theta}{\rho} \right) = V_\rho \partial_\rho \Gamma + V_Z \partial_Z \Gamma,$$

gives (4.3). Finally, smoothness of a bounded axisymmetric vector field at the axis implies $V_\theta(\rho, Z, \tau) = O(\rho)$ as $\rho \downarrow 0$, locally uniformly in (Z, τ) . Hence $\Gamma = \rho V_\theta = O(\rho^2)$, which gives (4.4). $\square$

We next record the scalar Liouville theorem used below. It is stated in a form that isolates precisely the PDE mechanism: bounded drift perturbations of the centered circulation operator cannot sustain a bounded ancient solution which vanishes on the symmetry axis.

Lemma 4.2 (Axis-absorption Liouville theorem). *Let $B = (B_\rho, B_Z)$ be a smooth vector field on $(0, \infty) \times \mathbb{R} \times \mathbb{R}$ satisfying*

$$\|B\|_{L^\infty} < \infty.$$

Let $G \in C^{2,1}((0, \infty) \times \mathbb{R} \times \mathbb{R}) \cap L^\infty$ solve

$$(4.5) \quad \partial_\tau G = \partial_\rho^2 G - \frac{1}{\rho} \partial_\rho G + \partial_Z^2 G - \left(B_\rho + \frac{\rho}{2} \right) \partial_\rho G - \left(B_Z + \frac{Z}{2} \right) \partial_Z G$$

for all $\tau \in \mathbb{R}$, and assume

$$(4.6) \quad \lim_{\rho \downarrow 0} G(\rho, Z, \tau) = 0 \quad \text{locally uniformly in } (Z, \tau).$$

Then $G \equiv 0$.

Proof. Set $M_B = \|B\|_{L^\infty}$. For $R > 1$ define

$$(4.7) \quad h_R(\rho) := \frac{\int_0^{\min\{\rho, R\}} s \exp\left(\frac{s^2}{4} - M_B s\right) ds}{\int_0^R s \exp\left(\frac{s^2}{4} - M_B s\right) ds}.$$

Then $0 \leq h_R \leq 1$, $h_R(0) = 0$, $h_R(R) = 1$, and, on $0 < \rho < R$,

$$(4.8) \quad h_R'' + \left(M_B - \frac{\rho}{2} - \frac{1}{\rho} \right) h_R' = 0, \quad h_R' > 0.$$

Let

$$\mathcal{L}_\tau f := \partial_\rho^2 f - \frac{1}{\rho} \partial_\rho f + \partial_Z^2 f - \left(B_\rho + \frac{\rho}{2} \right) \partial_\rho f - \left(B_Z + \frac{Z}{2} \right) \partial_Z f.$$

Since $B_\rho \geq -M_B$, (4.8) gives

$$(4.9) \quad \mathcal{L}_\tau h_R = h_R'' - \left(\frac{1}{\rho} + \frac{\rho}{2} + B_\rho \right) h_R' \leq h_R'' + \left(M_B - \frac{\rho}{2} - \frac{1}{\rho} \right) h_R' = 0.$$

Thus $(\partial_\tau - \mathcal{L}_\tau)h_R \geq 0$.

We now compare G to this barrier on the strip $\Omega_R = (0, R) \times \mathbb{R}$. Let $M_G = \|G\|_{L^\infty}$. We first record the killed-strip decay estimate used by the comparison argument. If $q_{s,R}$ solves

$$\partial_\tau q_{s,R} = \mathcal{L}_\tau q_{s,R}, \quad q_{s,R} = 0 \text{ on } \rho = 0, R, \quad q_{s,R}(\rho, Z, s) = 1,$$

then, for every compact $K \Subset \Omega_R$ and every fixed t ,

$$(4.10) \quad \lim_{s \rightarrow -\infty} \sup_{(\rho, Z) \in K} q_{s,R}(\rho, Z, t) = 0.$$

To prove (4.10), fix $K \Subset \Omega_R$. Choose $\delta > 0$, $Z_K > 0$, and $Z_1 > Z_K + 1$ so that

$$K \subset [\delta, R - \delta] \times [-Z_K, Z_K] \subset (0, R) \times (-Z_1, Z_1).$$

On the finite cylinder

$$\Omega_{R,Z_1} := (0, R) \times (-Z_1, Z_1)$$

with Dirichlet data on the full parabolic boundary, the operator $\mathcal{L}_\tau$ is uniformly parabolic on $[\delta/2, R - \delta/2] \times [-Z_1, Z_1]$, has bounded first-order coefficients there, and has absorbing boundary at $\rho = 0, R$. The boundary parabolic Harnack inequality and the strong maximum principle give a time $T_R > 0$ and a number $\vartheta_R \in (0, 1)$, depending only on R, M_B, δ, Z_K and not on Z_1 , such that every nonnegative solution with initial data bounded by 1 and zero lateral data satisfies

$$0 \leq q(\cdot, s + T_R) \leq 1 - \vartheta_R \quad \text{on } [\delta, R - \delta] \times [-Z_K, Z_K].$$

Applying the same estimate after each time translation and using the maximum principle between the compact set and the boundary gives

$$\sup_K q_{s,R}(\cdot, s + mT_R) \leq (1 - \vartheta_R)^m, \quad m = 1, 2, \dots$$

For a fixed $t > s$, choose m with $s + mT_R \leq t < s + (m + 1)T_R$. The maximum principle and interior Harnack estimate on the final interval $[s + mT_R, t]$ preserve the preceding bound up to a constant depending only on R, M_B, δ, Z_K, T_R . Letting $s \rightarrow -\infty$ forces $m \rightarrow \infty$, proving decay on K for the truncated problem. The solutions on Ω_{R,Z_1} increase monotonically to the killed-strip solution as $Z_1 \rightarrow \infty$; passing to this limit gives (4.10).

Fix $R > 1$, $s < t$, and set

$$W_R^+ := \frac{G}{M_G} - h_R, \quad W_R^- := -\frac{G}{M_G} - h_R,$$

with the convention that the conclusion is immediate if $M_G = 0$. By (4.5) and (4.9),

$$(\partial_\tau - \mathcal{L}_\tau)W_R^\pm \leq 0 \quad \text{in } \Omega_R \times (s, t).$$

On $\rho = 0$, (4.6) and $h_R(0) = 0$ give $W_R^\pm \leq 0$. On $\rho = R$, $h_R(R) = 1$ and $|G| \leq M_G$ give again $W_R^\pm \leq 0$. The parabolic comparison principle with the killed-strip kernel therefore yields, for $\tau = t$,

$$(W_R^\pm)_+(\rho, Z, t) \leq q_{s,R}(\rho, Z, t) \quad ((\rho, Z) \in \Omega_R).$$

Letting $s \rightarrow -\infty$ and using (4.10) gives $W_R^\pm \leq 0$ on compact subsets of Ω_R . Since the compact subset is arbitrary,

$$(4.11) \quad |G(\rho, Z, t)| \leq M_G h_R(\rho) \quad (0 < \rho < R, Z \in \mathbb{R}, t \in \mathbb{R}).$$

Finally, for each fixed $\rho < \infty$, the denominator in (4.7) tends to infinity super-exponentially as $R \rightarrow \infty$, while the numerator remains fixed. Hence $h_R(\rho) \rightarrow 0$. Letting $R \rightarrow \infty$ in (4.11) gives $G(\rho, Z, t) = 0$ for every $\rho > 0$, $Z \in \mathbb{R}$, and $t \in \mathbb{R}$. The boundary value at $\rho = 0$ is already zero by assumption. Therefore $G \equiv 0$. $\square$

Theorem 4.3 (Exclusion of the axisymmetric bounded-circulation residual class). *Let $\mathcal{C}_{\text{ax}}(\mathcal{S}; I, J)$ be the axisymmetric bounded-circulation residual class in the refined residual decomposition. Then*

$$(4.12) \quad \mathcal{C}_{\text{ax}}(\mathcal{S}; I, J) = \emptyset.$$

Proof. Assume, for contradiction, that $V \in \mathcal{C}_{\text{ax}}(\mathcal{S}; I, J)$. By definition of $\mathcal{C}_{\text{ax}}$, the associated physical ancient representative is axisymmetric and has bounded angular circulation. Equivalently, in centered variables,

$$\Gamma(\rho, Z, \tau) = \rho V_\theta(\rho, Z, \tau)$$

is bounded on $(0, \infty) \times \mathbb{R} \times \mathbb{R}$. Since V is a Seregin ancient limit, it is smooth and bounded in centered variables. Hence $B = (V_\rho, V_Z)$ is bounded.

By [Theorem 4.1](#), Γ satisfies (4.3) and vanishes at the axis. Applying [Theorem 4.2](#) with $G = \Gamma$ and $B = (V_\rho, V_Z)$ gives

$$\Gamma \equiv 0.$$

Therefore $V_\theta \equiv 0$ for $\rho > 0$, and by smoothness also on the axis. Thus the associated ancient solution is axisymmetric without swirl. But the axisymmetric no-swirl class is one of the structured alternatives removed before the residual class is defined. This contradicts $V \in \mathcal{R}(\mathcal{S}; I, J)$, and therefore no such V exists. $\square$

Remark 4.4 (Why this does not assert the full bounded-swirl Liouville theorem). The proof uses the centered Type I equation, specifically the Ornstein–Uhlenbeck drift terms $-\rho\partial_\rho/2$ and $-Z\partial_Z/2$ in the scalar equation for Γ . It is therefore a Type I Seregini limit argument. It does not prove that every bounded ancient axisymmetric solution of the unrenormalized Navier–Stokes equations with bounded physical circulation is swirl-free. The latter lacks the confining centered drift and remains a different shell-transport problem.

5. THE ROTATIONAL RELATIVE-EQUILIBRIUM RESIDUAL CLASS

We next isolate the second refined residual class. This class consists of centered Type I ancient limits whose time dependence is a rigid rotation of a fixed spatial profile. Passing to the co-rotating frame converts the centered Navier–Stokes equation into a stationary Leray profile equation with a Coriolis transport term. A whole-space Coriolis–Leray Liouville theorem for all bounded profiles is not used here. Instead, the Coriolis term is localized as an annular flux and is then treated by the local concentration analysis proved in [Sections 9](#) and [10](#).

Throughout this section the centered equation is

$$(5.1) \quad \partial_\tau V - \Delta V + \frac{1}{2}y \cdot \nabla V + \frac{1}{2}V + (V \cdot \nabla)V + \nabla P = 0, \quad \nabla \cdot V = 0.$$

For $\Omega \in \mathfrak{so}(3)$, set

$$(5.2) \quad Q(\tau) := \exp(\tau\Omega) \in SO(3).$$

Definition 5.1 (Rotational relative-equilibrium profile). A bounded smooth ancient solution (V, P) of (5.1) is a rotational relative equilibrium if there exist $\Omega \in \mathfrak{so}(3)$, a bounded smooth divergence-free vector field W , and a pressure profile Π , such that

$$(5.3) \quad V(y, \tau) = Q(\tau)W(Q(\tau)^T y), \quad P(y, \tau) = \Pi(Q(\tau)^T y)$$

up to the usual time-dependent pressure gauge. The class $\mathcal{C}_{\text{rot}}$ consists of such residual Seregini profiles, after the axisymmetric bounded-circulation class and the previously closed alternatives in the setup package have been removed.

Lemma 5.2 (Co-rotating stationary equation). *Let (V, P) be a rotational relative equilibrium. Then the co-rotating profile (W, Π) satisfies*

$$(5.4) \quad -\Delta W + \frac{1}{2}y \cdot \nabla W + \frac{1}{2}W + (W \cdot \nabla)W + \Omega W - (\Omega y) \cdot \nabla W + \nabla \Pi = 0, \quad \nabla \cdot W = 0.$$

Equivalently,

$$(5.5) \quad \mathbb{P} \left[-\Delta W + \frac{1}{2}y \cdot \nabla W + \frac{1}{2}W + (W \cdot \nabla)W + \Omega W - (\Omega y) \cdot \nabla W \right] = 0,$$

where $\mathbb{P}$ is the whole-space Leray projector.

Proof. Let $z = Q(\tau)^T y$. Since $Q'(\tau) = \Omega Q(\tau)$ and $Q(\tau)^T \Omega Q(\tau) = \Omega$, one has

$$\partial_\tau z = -\Omega z.$$

Therefore

$$\partial_\tau V(y, \tau) = Q(\tau) \left(\Omega W(z) - (\Omega z) \cdot \nabla W(z) \right).$$

The Laplacian, divergence, convection term, and pressure gradient are invariant under the orthogonal change of variables. Substitution into (5.1), followed by multiplication by $Q(\tau)^T$, gives (5.4). Applying the Leray projector gives (5.5). $\square$

5.1. The Bernoulli identity and the local Coriolis flux. Write

$$(5.6) \quad F(y) := \frac{1}{2}y - \Omega y, \quad A(y) := W(y) + F(y), \quad \omega := \nabla \times W.$$

Since Ω is skew-symmetric, there is a unique vector $\varpi \in \mathbb{R}^3$ such that

$$(5.7) \quad \Omega y = \varpi \times y.$$

Then

$$(5.8) \quad \nabla \times F = -2\varpi.$$

Lemma 5.3 (Coriolis Bernoulli identity). *Let (W, Π) be a smooth solution of (5.4). Define*

$$(5.9) \quad B := \frac{1}{2}|W|^2 + \Pi + F \cdot W.$$

Then

$$(5.10) \quad \nabla B + \nabla \times \omega - A \times \omega = 0,$$

and consequently

$$(5.11) \quad -\Delta B + A \cdot \nabla B = -|\omega|^2 - \omega \cdot (\nabla \times F) = -|\omega|^2 + 2\varpi \cdot \omega.$$

Proof. The vector identity

$$(W \cdot \nabla)W = \nabla \left(\frac{1}{2}|W|^2 \right) - W \times \omega$$

and the identity

$$(F \cdot \nabla)W + (\nabla F)^T W = \nabla(F \cdot W) - F \times \omega$$

apply because $(\nabla F)^T W = \frac{1}{2}W + \Omega W$. Thus (5.4) can be rewritten as

$$-\Delta W + \nabla \left(\frac{1}{2}|W|^2 + \Pi + F \cdot W \right) - (W + F) \times \omega = 0.$$

Since $\nabla \cdot W = 0$, one has $-\Delta W = \nabla \times \omega$, which gives (5.10).

Taking divergence in (5.10) gives

$$\Delta B = \nabla \cdot (A \times \omega),$$

because the divergence of a curl vanishes. Using

$$\nabla \cdot (A \times \omega) = \omega \cdot (\nabla \times A) - A \cdot (\nabla \times \omega), \quad \nabla \times A = \omega + \nabla \times F,$$

we obtain

$$\Delta B = |\omega|^2 + \omega \cdot (\nabla \times F) - A \cdot (\nabla \times \omega).$$

On the other hand, by (5.10),

$$A \cdot \nabla B = A \cdot (A \times \omega) - A \cdot (\nabla \times \omega) = -A \cdot (\nabla \times \omega).$$

Combining the last two identities gives (5.11). Finally, (5.8) follows from (5.7). $\square$

Lemma 5.4 (Localized Coriolis flux identity). *Let (W, Π) be as in [Theorem 5.3](#), and let $\chi \in C_c^\infty(\mathbb{R}^3)$. Then*

$$(5.12) \quad \int_{\mathbb{R}^3} \chi (-\Delta B + A \cdot \nabla B + |\omega|^2) dy = -2 \int_{\mathbb{R}^3} (W \times \varpi) \cdot \nabla \chi dy.$$

In particular the Coriolis contribution is supported only where the cutoff varies.

Proof. By (5.11),

$$-\Delta B + A \cdot \nabla B + |\omega|^2 = 2\varpi \cdot \omega.$$

Because ϖ is constant and $\omega = \nabla \times W$,

$$\varpi \cdot \omega = \nabla \cdot (W \times \varpi).$$

Multiplying by χ , integrating over $\mathbb{R}^3$, and integrating by parts gives

$$2 \int \chi \varpi \cdot \omega = 2 \int \chi \nabla \cdot (W \times \varpi) = -2 \int (W \times \varpi) \cdot \nabla \chi,$$

which proves (5.12). $\square$

Lemma 5.5 (Non-small Coriolis flux creates a retained local concentration). *Let $\chi \in C_c^\infty(\mathbb{R}^3)$, let $A_\chi := \text{supp } \nabla \chi$, and put*

$$C_\chi := \int_{A_\chi} |\nabla \chi|^{3/2} dy.$$

Fix $T \in (0, 1]$, and let

$$\mathcal{T}_{\chi, T} := \{(y, \tau) : -T < \tau < 0, Q(\tau)^T y \in A_\chi\}$$

be the swept annular tube in the original centered variables. Cover $\mathcal{T}_{\chi, T}$ by finitely many unit terminal cylinders

$$\mathcal{T}_{\chi, T} \subset \bigcup_{m=1}^{N_{\chi, T}} Q_1(z_m).$$

Let $\varepsilon_{\text{sd}} > 0$ be the local concentration threshold used in [Theorem 9.3](#). If $\varpi \neq 0$ and

$$(5.13) \quad \left| 2 \int_{\mathbb{R}^3} (W \times \varpi) \cdot \nabla \chi dy \right| \geq 2|\varpi| C_\chi^{2/3} \left(\frac{N_{\chi, T} \varepsilon_{\text{sd}}}{T} \right)^{1/3},$$

then at least one cylinder $Q_1(z_m)$ is ε_{sd} -concentrating for the original centered solution V .

Proof. Write

$$\delta_\chi := \left| 2 \int_{\mathbb{R}^3} (W \times \varpi) \cdot \nabla \chi dy \right|.$$

Hölder's inequality gives

$$\begin{aligned} \delta_\chi &\leq 2|\varpi| \int_{A_\chi} |W| |\nabla \chi| dy \\ &\leq 2|\varpi| \left(\int_{A_\chi} |W|^3 dy \right)^{1/3} C_\chi^{2/3}. \end{aligned}$$

Thus (5.13) implies

$$\int_{A_\chi} |W|^3 dy \geq \frac{N_{\chi, T} \varepsilon_{\text{sd}}}{T}.$$

Since $V(y, \tau) = Q(\tau)W(Q(\tau)^T y)$ and $Q(\tau)$ is an isometry,

$$\begin{aligned} \iint_{\mathcal{T}_{\chi, T}} |V(y, \tau)|^3 dy d\tau &= \int_{-T}^0 \int_{Q(\tau)A_\chi} |W(Q(\tau)^T y)|^3 dy d\tau \\ &= T \int_{A_\chi} |W(z)|^3 dz \geq N_{\chi, T} \varepsilon_{\text{sd}}. \end{aligned}$$

Because the tube is covered by $N_{\chi, T}$ unit terminal cylinders, one of them satisfies

$$\iint_{Q_1(z_m)} |V|^3 dy d\tau \geq \varepsilon_{\text{sd}}.$$

The pressure contribution to the local CKN measure is nonnegative after the allowed pressure gauge. Hence

$$\mu(Q_1(z_m)) \geq \varepsilon_{\text{sd}},$$

so the corresponding recentering sequence is ε_{sd} -concentrating in the sense of [Theorem 9.3](#). $\square$

Lemma 5.6 (Local Coriolis-flux dichotomy). *Fix $\chi \in C_c^\infty(\mathbb{R}^3)$ and $T \in (0, 1]$. Along every terminal extraction of a rotational relative equilibrium, exactly one of the following local alternatives occurs:*

- (i) $\varpi \neq 0$, the flux across $A_\chi = \text{supp } \nabla \chi$ satisfies the threshold (5.13), and the swept annular tube contains a unit-scale concentration;
- (ii) either $\varpi = 0$, or the threshold (5.13) fails; then the flux is a bounded compactly supported functional of W on A_χ , passes to terminal limits by local L^3 convergence, and creates no terminal alternative other than the single-profile, finite separated-profile, exterior-escape, diffuse-exterior-concentration, local-vanishing, or local-compactness-failure alternatives of [Theorem 9.9](#).

Proof. Alternative (i) is precisely [Theorem 5.5](#). If $\varpi = 0$, then the flux vanishes identically. If $\varpi \neq 0$ and (5.13) fails, the flux is bounded by the right hand side of that threshold. In both cases the functional

$$W \mapsto -2 \int_{\mathbb{R}^3} (W \times \varpi) \cdot \nabla \chi dy$$

depends only on W on the compact annular set A_χ . Under the local Seregin compactness supplied by the admissible setup package, velocities converge strongly in L_{loc}^3 ; since $\nabla \chi \in L^{3/2}$ and ϖ is fixed, the functional is continuous along every terminal subsequence:

$$\int (W_n \times \varpi) \cdot \nabla \chi \longrightarrow \int (W_\infty \times \varpi) \cdot \nabla \chi.$$

Thus a subthreshold flux is retained only as a local term in the limiting localized Bernoulli identity (5.12). It is not a new location of local velocity mass. Every actual location of compact velocity L^3 -mass is still one of the terminal local concentration alternatives listed in [Theorem 9.9](#). $\square$

Remark 5.7 (Why this is not a global Liouville argument). For $\Omega = 0$, the Coriolis flux in [Theorem 5.4](#) vanishes. For $\Omega \neq 0$, the same lemma does not give a sign-definite maximum principle for B . It gives only the local dichotomy in [Theorem 5.6](#): either a chosen annulus already contains a recentered local concentration, or the annular flux is a subthreshold compact functional which passes to the local terminal limit. No tail decay, whole-space L^p bound, Gaussian energy estimate, or global pressure normalization is used.

5.2. Local closure of the rotational relative-equilibrium class.

Theorem 5.8 (Stratified exclusion of rotational relative equilibria). *The ordered rotational relative-equilibrium residual class is empty:*

$$(5.14) \quad \mathcal{C}_{\text{rot}}(\mathcal{S}; I, J) = \emptyset.$$

Proof. Suppose, for contradiction, that $V \in \mathcal{C}_{\text{rot}}(\mathcal{S}; I, J)$. By [Theorem 5.1](#) and [Theorem 5.2](#), there are $\Omega \in \mathfrak{so}(3)$ and a bounded smooth divergence-free profile (W, Π) satisfying the Coriolis–Leray equation (5.4). The centered representative V is a normalized Seregin ancient limit, so by [Theorem 1.3](#) it carries a fixed positive compact velocity L^3 -mass.

Choose a countable family of smooth cutoffs which is cofinal among compact annuli in the terminal variables. For each cutoff and each rational $T \in (0, 1]$, apply the local dichotomy [Theorem 5.6](#). A superthreshold Coriolis flux produces a unit-scale concentration and is therefore already part of the local concentration decomposition. A subthreshold Coriolis flux is a compactly supported functional of W , continuous under the local convergence supplied by the admissible setup package, and hence remains inside the localized limiting equation; it carries no independent CKN mass and creates no additional terminal alternative.

We now apply the terminal local concentration decomposition to the inherited compact density. Loss of local compactness, local vanishing, exterior escape, and diffuse exterior concentration cannot be the sole locations of this density by [Theorem 9.12](#). By the preceding paragraph, the Coriolis flux has already led either to a local concentration or to a compact local term in the selected terminal equation. Thus, after local concentration extraction, the retained density is located either in a finite separated family of retained concentration profiles or in a single retained concentration terminal profile.

The finite separated-family alternative is impossible by [Theorem 10.23](#). In the single-profile case, [Theorem 10.4](#) makes the terminal profile terminally indecomposable, and [Theorem 10.9](#) excludes retained indecomposable concentration profiles. Both possible locations of the inherited compact density are impossible. This contradicts the normalization of V , and therefore $\mathcal{C}_{\text{rot}}(\mathcal{S}; I, J) = \emptyset$. $\square$

Proposition 5.9 (Already-closed integrable subcase). *Let V be a rotational relative equilibrium with co-rotating profile W . If*

$$(5.15) \quad W \in L^3(\mathbb{R}^3),$$

then $V \equiv 0$. More generally, the same conclusion holds under any condition which places the physical representative in $L_t^\infty L_x^3$ on its ancient time interval.

Proof. Rotations and the Navier–Stokes critical scaling preserve the L^3 norm. Thus $W \in L^3$ implies

$$\sup_{\tau \in \mathbb{R}} \|V(\cdot, \tau)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Equivalently, the physical representative lies in the critical class $L_t^\infty L_x^3$. The endpoint L^3 class is one of the already closed alternatives used in the setup package, by the Escauriaza–Seregin–Šverák endpoint regularity theorem [3] and the stationary Nečas–Růžička–Šverák Liouville theorem [7]. Therefore $V \equiv 0$. A zero profile has zero local velocity mass, so no such profile can belong to the residual nonvanishing class. $\square$

6. THE DEGENERATE STATIONARY-HULL RESIDUAL CLASS

We now isolate the third refined residual class. This class is designed to capture Type I ancient limits whose time-translation hull contains a stationary profile with retained local concentration lying on a non-isolated family of stationary centered profiles. The role of the stationary-family

condition is to separate genuinely degenerate stationary profile families from isolated stationary profiles and from the symmetry-generated classes already removed in $\mathcal{C}_{\text{ax}}$ and $\mathcal{C}_{\text{rot}}$.

A small point of formulation is important. It is not enough to assume that the hull of a residual ancient solution contains some stationary-family element. A stationary element may occur as a limit without retained local concentration and therefore need not retain the local CKN concentration inherited from the original singular point. Such a hull element should not be placed in a class that is to be closed by a stationary-profile exclusion. Therefore the class below is defined using *retained-concentration occurrence*: the stationary-family profile must itself occur as a Seregim limit carrying a positive local CKN concentration.

Throughout this section the centered equation is

$$(6.1) \quad \partial_\tau V - \Delta V + \frac{1}{2}y \cdot \nabla V + \frac{1}{2}V + (V \cdot \nabla)V + \nabla P = 0, \quad \nabla \cdot V = 0.$$

Its projected form is

$$(6.2) \quad \partial_\tau V = \mathcal{F}(V), \quad \mathcal{F}(W) := \mathbb{P} \left[\Delta W - \frac{1}{2}(W + y \cdot \nabla W) - (W \cdot \nabla)W \right],$$

where $\mathbb{P}$ is the whole-space Leray projector.

6.1. Stationary families and retained-concentration hull limits.

Definition 6.1 (Admissible stationary phase space). An admissible stationary phase space is a Banach space $\mathfrak{B}$ of solenoidal vector fields on $\mathbb{R}^3$ such that:

- (i) $\mathfrak{B} \hookrightarrow C_{\text{loc}}^{2,\alpha}(\mathbb{R}^3) \cap L^\infty(\mathbb{R}^3)$ continuously for some $\alpha \in (0, 1)$;
- (ii) the map $\mathcal{F}$ in (6.2) is well-defined on an open subset $\mathcal{U} \subset \mathfrak{B}$ and maps $\mathcal{U}$ into a Banach space $\mathfrak{V}$;
- (iii) $\mathcal{F} : \mathcal{U} \rightarrow \mathfrak{V}$ is C^1 .

Definition 6.2 (Nontrivial stationary family). Let $W \in \mathcal{U} \subset \mathfrak{B}$ satisfy $\mathcal{F}(W) = 0$. We say that W lies on an admissible nontrivial stationary family if there exist $\varepsilon > 0$ and a C^1 curve

$$a \mapsto W_a \in \mathcal{U}, \quad |a| < \varepsilon,$$

such that

$$(6.3) \quad W_0 = W, \quad \mathcal{F}(W_a) = 0 \quad (|a| < \varepsilon),$$

and such that the family is not locally generated only by the action of the rigid rotation group. Equivalently, after possibly reparametrizing the curve, one may assume that

$$(6.4) \quad \dot{W}_0 \notin \{\Omega W - (\Omega y) \cdot \nabla W : \Omega \in \mathfrak{so}(3)\}$$

whenever the tangent $\dot{W}_0$ exists and is nonzero.

Definition 6.3 (Retained local CKN concentration). Let (W, Π) be a stationary centered profile. We say that (W, Π) is ε_* -concentrating if there exist a compact parabolic cylinder $Q_r(z_0)$ and an admissible pressure gauge c_{r,z_0} such that

$$(6.5) \quad \iint_{Q_r(z_0)} \left(|W|^3 + |\Pi - c_{r,z_0}|^{3/2} \right) dy d\tau \geq \varepsilon_*.$$

For a stationary profile the integrand is independent of τ , but the parabolic-cylinder notation is retained to match the CKN normalization used for Seregim limits.

Definition 6.4 (Degenerate stationary-hull residual class with retained concentration). Fix an admissible stationary phase space $\mathfrak{B}$. The degenerate stationary-hull residual class with retained local concentration

$$\mathcal{C}_{\text{stat}}^{\text{ret}}(\mathcal{S}; I, J; \mathfrak{B})$$

consists of those $V \in \mathcal{R}(\mathcal{S}; I, J)$, after the ordered removal of $\mathcal{C}_{\text{ax}}$ and $\mathcal{C}_{\text{rot}}$, for which there exists a sequence $s_k \in \mathbb{R}$ and a stationary profile (W, Π) such that

$$(6.6) \quad (V(\cdot, \cdot + s_k), P(\cdot, \cdot + s_k) - c_k(\cdot)) \longrightarrow (W, \Pi)$$

locally smoothly for the velocity and locally in $L^{3/2}$ for the pressure modulo admissible gauges, and such that:

- (i) $W \in \mathfrak{B}$ and $\mathcal{F}(W) = 0$;
- (ii) W lies on an admissible nontrivial stationary family in the sense of [Theorem 6.2](#);
- (iii) (W, Π) is ε_* -concentrating in the sense of [Theorem 6.3](#);
- (iv) (W, Π) is not in any previously closed stationary, structured, axisymmetric bounded-circulation, or rotational relative-equilibrium class.

In the ordered residual decomposition one keeps this class of stationary-hull limits with retained concentration as the degenerate stationary-hull alternative. The hull cases without retained local concentration should be assigned to the generic terminal residual class, where they are handled by the terminal local concentration decomposition rather than by a stationary-profile exclusion.

Remark 6.5 (Why retained concentration is necessary). The condition (6.5) prevents the following logical error. A nonstationary ancient solution may have a stationary profile in its hull with zero local CKN mass on the relevant compact cylinders. Excluding that stationary profile would not exclude the original ancient solution. The stationary-profile exclusion used below applies only to stationary profiles that themselves occur as nontrivial Seregin limits with positive local concentration.

6.2. Hull limits are again Seregin limits. The next lemma is the basic compatibility between time-translation hulls in centered variables and rescaling sequences in physical variables. It is the reason that a stationary profile in the hull which retains local concentration may be treated as a blow-up profile of the original solution, not merely as a limit inside the already extracted ancient solution.

Lemma 6.6 (Scale reselection for centered hull limits). *Let (V, P) be a centered Type I Seregin limit obtained from a finite-energy suitable weak solution (u, p) at a singular point $z_* = (x_*, t_*)$. Suppose that, for some sequence $s_k \in \mathbb{R}$,*

$$(V(\cdot, \cdot + s_k), P(\cdot, \cdot + s_k) - c_k(\cdot)) \rightarrow (W, \Pi)$$

locally in the Seregin topology. Then, after passing to a diagonal subsequence and reselecting the physical blow-up scales, (W, Π) is itself a centered Type I Seregin limit of (u, p) at z_ . If the convergence to (W, Π) retains a positive local CKN lower bound, then (W, Π) is a nontrivial Seregin limit with the same type of inherited nonvanishing condition.*

Proof. Let $\lambda_n \downarrow 0$ be a sequence of scales producing (V, P) . In centered variables, a time translation by s is exactly a change of the physical blow-up scale by a factor depending only on s . With the convention used in (6.1), write this adjusted scale as

$$(6.7) \quad \lambda_{n,s} := e^{-s/2} \lambda_n.$$

The rescaled fields built from (u, p) at scale $\lambda_{n,s}$ are the same as the fields at scale λ_n , translated by s in the centered time variable, up to the corresponding pressure gauge.

Fix an exhaustion of compact cylinders $K_m \subset \mathbb{R}^3 \times \mathbb{R}$. For each k , choose $n(k)$ so large that $\lambda_{n(k), s_k} \rightarrow 0$ and the rescaled fields at scale $\lambda_{n(k)}$, shifted by s_k , are within $1/k$ of $(V(\cdot, \cdot + s_k), P(\cdot, \cdot + s_k) - c_k)$ on K_k in the local Seregin topology. This is possible because $\lambda_n \downarrow 0$ and because the original sequence converges locally to (V, P) .

By the assumed hull convergence, the shifted limit $(V(\cdot, \cdot + s_k), P(\cdot, \cdot + s_k) - c_k)$ converges to (W, Π) on each fixed compact cylinder. The diagonal sequence of physical rescalings with scales $\lambda_{n(k), s_k}$ therefore converges locally to (W, Π) . Hence (W, Π) is again a centered Type I Seregin limit at z_* . The assertion about nonvanishing follows from lower semicontinuity and local convergence of the CKN quantities after the prescribed pressure gauges. $\square$

6.3. Elementary consequences of stationary-family degeneracy. The stationary-family condition has two immediate PDE consequences. First, a differentiable nontrivial family produces a nontrivial element of the kernel of the linearized stationary Leray operator. Second, if an ancient trajectory is exactly constrained to a stationary family, then it is stationary.

Lemma 6.7 (Stationary family implies linearized degeneracy). *Let $a \mapsto W_a \in \mathfrak{B}$ be a C^1 stationary family satisfying (6.3). Then*

$$(6.8) \quad D\mathcal{F}(W_0)\dot{W}_0 = 0.$$

In particular, if $\dot{W}_0 \neq 0$ modulo infinitesimal rotations, the linearized stationary Leray operator has a genuine non-symmetry kernel element.

Proof. Differentiate $\mathcal{F}(W_a) = 0$ at $a = 0$. Since $\mathcal{F}$ is C^1 on $\mathfrak{B}$, this gives (6.8). The final statement is exactly the nontriviality condition in Theorem 6.2. $\square$

Lemma 6.8 (Trajectories constrained to a stationary family are stationary). *Let $a \mapsto W_a \in \mathfrak{B}$ be a C^1 stationary family with $\dot{W}_a \neq 0$ on an interval. Suppose that a smooth solution of (6.2) has the form*

$$(6.9) \quad V(\tau) = W_{a(\tau)}$$

for a C^1 function $a(\tau)$. Then a is constant, and hence V is stationary.

Proof. Since $\mathcal{F}(W_a) = 0$ for every a , equation (6.2) gives

$$a'(\tau)\dot{W}_{a(\tau)} = \partial_\tau V(\tau) = \mathcal{F}(W_{a(\tau)}) = 0.$$

Because $\dot{W}_{a(\tau)} \neq 0$, we obtain $a'(\tau) = 0$. Thus a is constant on the interval. $\square$

Corollary 6.9 (Exact stationary-family subcase). *If an element of the residual class is itself an exact trajectory on an admissible stationary family, then it is a stationary centered profile. Hence it is excluded whenever it satisfies the stationary L^3 Liouville criterion, or more generally whenever the stationary profile belongs to one of the previously closed stationary or structured classes.*

Proof. Let $V(\tau) = W_{a(\tau)}$ be such an exact stationary-family trajectory. By Theorem 6.8, $a'(\tau) = 0$ on every interval on which the stationary-family parametrization is nondegenerate. Hence a is constant and V is a single stationary centered profile. If that stationary profile satisfies the stationary L^3 Liouville criterion or lies in one of the previously closed structured or ordered lower residual classes, the corresponding prior exclusion removes it from the residual class. $\square$

6.4. Stationary-hull profiles which occur as Seregin limits. The remaining issue is not the internal dynamics along a stationary family; exact stationary-family motion is stationary by [Theorem 6.8](#). The real issue is occurrence as a blow-up limit: can a degenerate stationary-family profile occur as a Seregin limit of a finite-energy suitable weak solution while retaining local concentration? The following statement is the precise Seregin limit occurrence statement required to close the degenerate stationary-hull class with retained local concentration. It is proved in the following section by the terminal local concentration theorem.

Theorem 6.10 (No degenerate stationary-family concentration profile occurs as a Seregin limit). *Fix an admissible stationary phase space $\mathfrak{B}$. Let (W, Π) be a stationary centered profile satisfying:*

- (i) $W \in \mathfrak{B}$, $\mathcal{F}(W) = 0$, and W lies on an admissible nontrivial stationary family;
- (ii) (W, Π) is a centered Type I Seregin limit of a finite-energy suitable weak solution at a singular point;
- (iii) (W, Π) is ε_* -concentrating;
- (iv) (W, Π) is not in the small, stationary L^3 , tight, closed structured/decay, axisymmetric bounded-circulation, or rotational relative-equilibrium classes already removed earlier in the stratification.

Then no such (W, Π) exists.

Proof. This is exactly [Theorem 7.5](#), proved below from the local terminal stratification theorem. $\square$

Remark 6.11 (Interpretation of the Seregin limit occurrence theorem). [Theorem 6.10](#) excludes occurrence as a Type I Seregin limit. It does not assert that degenerate stationary profiles cannot exist as elliptic solutions of the stationary centered equation. It asserts only that such profiles cannot occur as Type I blow-up limits with retained local concentration after all previously closed classes have been removed. This is the correct formulation for the residual problem: the elliptic family may exist, but it cannot be reached by the local blow-up procedure coming from a finite-energy suitable weak solution.

6.5. Closure of the degenerate stationary-hull concentration class.

Theorem 6.12 (Exclusion of the degenerate stationary-hull concentration class).

$$(6.10) \quad \mathcal{C}_{\text{stat}}^{\text{ret}}(\mathcal{S}; I, J; \mathfrak{B}) = \emptyset.$$

Consequently the ordered third residual class is empty, provided it is defined with the retained-concentration Seregin limit condition in [Theorem 6.4](#).

Proof. Suppose, for contradiction, that $V \in \mathcal{C}_{\text{stat}}^{\text{ret}}(\mathcal{S}; I, J; \mathfrak{B})$. By [Theorem 6.4](#), there is a stationary profile (W, Π) in the centered time-translation hull of (V, P) , and (W, Π) is ε_* -concentrating. The profile W belongs to $\mathfrak{B}$, satisfies $\mathcal{F}(W) = 0$, and lies on an admissible nontrivial stationary family. Moreover, by the ordered definition of $\mathcal{C}_{\text{stat}}^{\text{ret}}$, it is not in any of the previously closed lower classes.

By [Theorem 6.6](#), the hull profile (W, Π) is itself a centered Type I Seregin limit of the original finite-energy suitable weak solution at the same singular point, after reselecting scales. Since local concentration is retained, (W, Π) satisfies every condition in [Theorem 6.10](#). This contradicts the Seregin limit occurrence theorem. Therefore no such V exists, proving (6.10). $\square$

Proposition 6.13 (Already-closed integrable stationary-family subcase). *Let (W, Π) be a stationary centered profile lying on an admissible stationary family. If*

$$(6.11) \quad W \in L^3(\mathbb{R}^3),$$

then $W \equiv 0$. Hence no retained-concentration hull profile satisfying (6.11) can occur in $\mathcal{C}_{\text{stat}}^{\text{ret}}$.

Proof. The stationary physical representative associated with W is a backward self-similar Navier–Stokes solution. The assumption $W \in L^3(\mathbb{R}^3)$ places it in the stationary L^3 class already excluded by the classical Nečas–Růžička–Šverák Liouville theorem [7], equivalently by the endpoint ancient Liouville theorem already recorded in the setup package and cited below as [Theorem 10.8](#). Hence $W \equiv 0$. A zero profile has zero pressure-normalized local CKN mass after choosing the admissible pressure gauge, so it cannot be ε_* -concentrating. $\square$

Remark 6.14 (Closure after terminal local concentration analysis). The preceding argument reduces the stationary-hull class with retained local concentration from a time-translation hull question to the question whether a stationary centered profile in a nontrivial stationary family can occur as a Seregin limit. [Theorem 6.10](#) is supplied by the terminal local concentration analysis in the next section.

Remark 6.15 (Recommended statement for the residual decomposition). For the residual decomposition, the third exclusion line should be stated as follows:

$$\begin{array}{c} \text{stationary-hull residual with retained concentration} \\ \xrightarrow{\text{stationary-family Seregin limit exclusion}} \emptyset. \end{array}$$

The precise class is the *degenerate stationary-hull residual class with retained local concentration*.

7. EXCLUSION OF DEGENERATE STATIONARY-HULL PROFILES WITH RETAINED CONCENTRATION

We close the remaining analytic statement in the degenerate stationary-hull class by the terminal local concentration theorem proved in the main paper. The argument is not specific to the existence of a non-isolated stationary family. Once a stationary-family element occurs as a Type I Seregin limit carrying retained local CKN concentration, it is a stationary terminal concentration profile. Such profiles are excluded by the same local terminal concentration analysis and separated-profile exclusion used for the final residual class. No global coercive estimate for the original solution is used.

Throughout this section, a centered Type I profile satisfies

$$(7.1) \quad \partial_\tau U - \Delta U + \frac{1}{2}y \cdot \nabla U + \frac{1}{2}U + (U \cdot \nabla)U + \nabla \Pi = 0, \quad \nabla \cdot U = 0$$

on $\mathbb{R}^3 \times \mathbb{R}$, with the usual local suitability and pressure-gauge conventions.

7.1. Terminal local concentration theorem. We use the following specialization of the terminal local concentration theorem proved below. It is stated here exactly in the form needed for the stationary-hull closure with retained local concentration.

Theorem 7.1 (Terminal local concentration exclusion theorem). *Let (U, Π) be a bounded centered Type I Seregin profile satisfying local suitability and carrying a retained local CKN concentration, i.e. for some compact parabolic cylinder Q and some $\varepsilon_* > 0$,*

$$(7.2) \quad \iint_Q (|U|^3 + |\Pi - c_Q(\tau)|^{3/2}) \, dy \, d\tau \geq \varepsilon_*.$$

Assume that (U, Π) is outside the previously closed small-amplitude, stationary L^3 , tight, closed structured/decay, axisymmetric bounded-circulation, and rotational relative-equilibrium classes. Then the following hold.

- (i) *Every terminal extraction of (U, Π) has the paired decomposition of [Theorem 9.9](#): an active retained profile locus together with a residual remainder which is locally vanishing, diffuse, or noncompact after neighborhoods of the active locus are removed.*

- (ii) *Local vanishing, exterior escape, diffuse exterior concentration, and loss of local compactness cannot be the sole locations of the retained compact CKN concentration (7.2).*
 - (iii) *Mixed compact–diffuse and compact–noncompact residual states, and infinite active descendant chains, are impossible in the refined residual class.*
 - (iv) *Finite separated families of retained concentration profiles are impossible.*
 - (v) *Any remaining single retained concentration profile is terminally indecomposable: it has no retained concentrating tail limit, no diffuse exterior concentration, and no loss of local compactness along escaping recenterings.*
 - (vi) *Every retained terminally indecomposable concentration profile has a backward sequence of finite critical norm: there exist $\tau_k \rightarrow -\infty$ such that*
- $$(7.3) \quad \sup_k \|U(\cdot, \tau_k)\|_{L^3(\mathbb{R}^3)} < \infty.$$
- (vii) *Retained terminal concentration profiles are either affine/parasitic lower-stratum profiles or have parasitic-free finite-shift mild physical pullbacks. Therefore the Albritton–Barker endpoint Liouville theorem applies to any normalized residual profile satisfying (7.3).*

Proof. This is Theorem 10.24 applied to (U, Π) . The local alternatives in (i) are those of Theorem 9.9. Item (ii) is the sole-carrier exclusion in Theorem 9.12. Item (iii) is Theorems 9.61 and 10.20. Item (iv) is Theorem 10.23. Item (v) is Theorem 10.4. Item (vi) is Theorem 10.5, and item (vii) is Theorems 10.7 and 10.8. The active-locus, active path-space recurrence, and diffuse-defect compactness statements are proved below from local compactness. The critical-tail boundary is handled by Theorems 9.52 and 9.60. No global bound is assumed. $\square$

Remark 7.2 (Local nature of the theorem). The theorem above is a local concentration statement. It uses the local suitable-weak-solution compactness and pressure normalization developed from the Caffarelli–Kohn–Nirenberg theory [2] together with the unified setup package in `proof_setup.tex`: local positive concentration and the Type I Seregin compactness package. It then uses paired terminal profile decomposition, sole-carrier exclusions, critical-tail compactification with realized critical-tail rigidity, active path-space recurrence, and the finite separated-profile theorem. It does not assert or require a global a priori bound for the original physical solution.

For completeness, and to make the dependence on local estimates explicit, we record the elementary part of the argument which converts terminal indecomposability into sequence- L^3 control.

Lemma 7.3 (Indecomposable local profiles have backward sequence- L^3 control). *Let (U, Π) be a bounded smooth ancient solution of (7.1). Suppose that (U, Π) has retained local concentration and is terminally indecomposable: it has no retained concentrating tail limit, no diffuse exterior concentration in the parabolic CKN sense, and no escaping recentering sequence along which local compactness, suitability, or pressure control fails. Then there exists a sequence $\tau_k \rightarrow -\infty$ such that*

$$(7.4) \quad \sup_k \|U(\cdot, \tau_k)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Proof. We argue by contradiction. Assume that no sequence satisfying (7.4) exists. Then

$$(7.5) \quad \liminf_{\tau \rightarrow -\infty} \|U(\cdot, \tau)\|_{L^3(\mathbb{R}^3)} = \infty.$$

Choose $\tau_n \rightarrow -\infty$ such that $\int_{\mathbb{R}^3} |U(y, \tau_n)|^3 dy \rightarrow \infty$. Since U is bounded, we may choose radii $R_n \rightarrow \infty$ so that

$$(7.6) \quad \int_{|y| > R_n} |U(y, \tau_n)|^3 dy \geq 1.$$

Define the parabolic exterior activity level

$$m_n^{\text{par}} := \sup_{\substack{|x| > R_n - 2 \\ |s - \tau_n| \leq 1}} \mathcal{E}_1(\mathcal{T}_{(x,s)}(U, \Pi); 0),$$

where the local CKN quantity is computed after the admissible pressure gauge in the recentered variables. This is the terminal quantity relevant for indecomposability: it tests unit parabolic cylinders, not only spatial balls at the distinguished time slice.

If $\limsup_n m_n^{\text{par}} \geq \varepsilon_*$, then for a subsequence there are (x_n, s_n) , with $|x_n| > R_n - 2$ and $|s_n - \tau_n| \leq 1$, such that

$$\mathcal{E}_1(\mathcal{T}_{(x_n, s_n)}(U, \Pi); 0) \geq \frac{1}{2} \varepsilon_*.$$

Since $R_n \rightarrow \infty$ and $\tau_n \rightarrow -\infty$, these are escaping covariant tail recenterings. Either local compactness and pressure control fail, or a subsequence converges in the terminal topology to a profile (W, Π_W) . In the compact case lower semicontinuity of $\mathcal{E}_1$ gives, after lowering the fixed retained threshold by a harmless factor,

$$\mathcal{E}_1(W, \Pi_W; 0) \geq c\varepsilon_*.$$

Thus the compact case gives a retained concentrating tail limit, while the noncompact case is an escaping noncompact recentering. Both contradict terminal indecomposability.

It remains to consider the case $m_n^{\text{par}} < \varepsilon_*$ for all sufficiently large n , after lowering the threshold if necessary. If some unit exterior covariant recentering sequence in

$$\{|y| > R_n - 2\} \times (\tau_n - 1, \tau_n + 1)$$

has loss of local compactness, terminal indecomposability is contradicted. If no such recentering loses local compactness, then (7.6) together with the parabolic smallness $m_n^{\text{par}} < \varepsilon_*$ is exactly the parabolic diffuse exterior concentration alternative. This again contradicts terminal indecomposability. Thus (7.5) is impossible, and (7.4) follows. $\square$

7.2. Stationary concentration profiles are impossible.

Theorem 7.4 (No stationary residual concentration profile occurs as a Seregin limit). *Let (W, Π) be a stationary solution of the centered equation,*

$$(7.7) \quad -\Delta W + \frac{1}{2} y \cdot \nabla W + \frac{1}{2} W + (W \cdot \nabla) W + \nabla \Pi = 0, \quad \nabla \cdot W = 0.$$

Assume that:

- (i) (W, Π) is obtained as a centered Type I Seregin limit of a finite-energy suitable weak solution at a singular point;
- (ii) (W, Π) has retained local concentration in the sense of (7.2);
- (iii) (W, Π) is outside all previously closed lower classes listed in Theorem 7.1;

Then no such (W, Π) exists.

Proof. Suppose that such a stationary profile (W, Π) exists. Since it is a centered Type I Seregin limit and has retained local concentration, it is an admissible input for the terminal local concentration theorem. Apply Theorem 7.1 to (W, Π) .

Finite separated terminal concentration profiles, infinite active descendant chains, and mixed compact–diffuse or compact–noncompact remainders are impossible by Theorem 7.1. Local vanishing, exterior escape, diffuse exterior concentration, and local compactness failure cannot be the only locations of the retained compact CKN concentration. Hence the retained concentration is represented by a single terminal concentration profile with no surviving tail remainder. By the indecomposability conclusion in Theorem 7.1, this single profile is terminally indecomposable.

Because (W, Π) is stationary in τ , the single terminal concentration profile may be represented by (W, Π) itself after the terminal normalization. Therefore [Theorem 7.3](#) gives a sequence $\tau_k \rightarrow -\infty$ such that

$$(7.8) \quad \sup_k \|W\|_{L^3(\mathbb{R}^3)} = \sup_k \|W(\cdot, \tau_k)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Thus $W \in L^3(\mathbb{R}^3)$.

Let

$$(7.9) \quad w(x, t) = (-t)^{-1/2} W\left(\frac{x}{\sqrt{-t}}\right), \quad q(x, t) = (-t)^{-1} \Pi\left(\frac{x}{\sqrt{-t}}\right), \quad t < 0.$$

By [Theorem 10.7](#), either (W, Π) is in the affine/parasitic lower stratum, already excluded from the residual class, or every finite terminal shift $w^T(x, s) := w(x, T + s)$, $s < 0$, is a bounded mild ancient Navier–Stokes solution. The critical scaling gives

$$(7.10) \quad \|w(\cdot, t)\|_{L^3(\mathbb{R}^3)} = \|W\|_{L^3(\mathbb{R}^3)} < \infty \quad (t < 0).$$

Fix $T < 0$, choose any sequence $t_k \downarrow -\infty$ with $t_k < T$, and put $s_k = t_k - T$. Applying the Albritton–Barker endpoint Liouville theorem to w^T along $s_k \downarrow -\infty$, [Theorem 10.8](#), we obtain $w^T \equiv 0$. Since $T < 0$ is arbitrary, $w \equiv 0$, and hence $W \equiv 0$. This contradicts retained local concentration, since the pressure-normalized local CKN quantity of the zero profile is zero. Therefore no stationary residual concentration profile can occur as such a Seregin limit. $\square$

7.3. Closure of the degenerate stationary-hull residual class.

Corollary 7.5 (No degenerate stationary-family concentration profile occurs as a Seregin limit). *Fix an admissible stationary phase space $\mathfrak{B}$. There is no stationary centered profile (W, Π) satisfying all of the following conditions:*

- (i) $W \in \mathfrak{B}$, W solves the stationary centered equation (7.7), and W lies on an admissible nontrivial stationary family;
- (ii) (W, Π) is a centered Type I Seregin limit of a finite-energy suitable weak solution at a singular point;
- (iii) (W, Π) has retained local concentration;
- (iv) (W, Π) is outside the previously closed lower classes;

Proof. The stationary-family condition in (i) is additional structure, but the exclusion in [Theorem 7.4](#) applies to every stationary residual profile carrying retained local concentration. Hence it applies in particular to stationary profiles lying in admissible nontrivial stationary families. $\square$

Theorem 7.6 (Closure of the degenerate stationary-hull concentration class). *Let $\mathcal{C}_{\text{stat}}^{\text{ret}}(\mathcal{S}; I, J; \mathfrak{B})$ be the degenerate stationary-hull residual class with retained local concentration. Then*

$$(7.11) \quad \mathcal{C}_{\text{stat}}^{\text{ret}}(\mathcal{S}; I, J; \mathfrak{B}) = \emptyset.$$

Proof. Suppose, for contradiction, that $V \in \mathcal{C}_{\text{stat}}^{\text{ret}}(\mathcal{S}; I, J; \mathfrak{B})$. By the definition of the degenerate stationary-hull residual class with retained local concentration, the centered translation hull of (V, P) contains a stationary profile (W, Π) with retained local concentration which lies in an admissible nontrivial stationary family and is outside the previously closed lower classes. The scale-reselection lemma for centered hull limits shows that (W, Π) is itself a centered Type I Seregin limit of the original finite-energy suitable weak solution at the same singular point.

Thus (W, Π) satisfies the hypotheses of [Theorem 7.5](#). This is impossible. Hence no such V exists, and (7.11) follows. $\square$

Remark 7.7 (How this enters the residual decomposition). The third residual line should be recorded as a theorem closed by the terminal local concentration analysis:

$$\begin{aligned} & \text{degenerate stationary-hull residual with retained concentration} \\ & \implies \text{stationary profile with retained concentration} \implies \emptyset. \end{aligned}$$

The second implication is proved by terminal local concentration analysis: decomposable profiles produce either an excluded separated concentration family, diffuse exterior concentration, or loss of local compactness; indecomposable profiles have backward sequence- L^3 control and are ruled out by the endpoint ancient Liouville theorem.

8. TAIL GEOMETRY OF THE GENERIC TERMINAL RESIDUAL

We now pass to the generic terminal residual class. The preliminary point is that the centered equation is not invariant under raw fixed spatial translations. The correct far-field recenterings are self-similar covariant translations.

Definition 8.1 (Covariant translations). For $a \in \mathbb{R}^3$, define

$$(T_a V)(y, \tau) := V(y + e^{\tau/2} a, \tau), \quad (T_a P)(y, \tau) := P(y + e^{\tau/2} a, \tau).$$

Lemma 8.2 (Covariant translation invariance). *If (V, P) solves (1.2), then $(T_a V, T_a P)$ also solves (1.2).*

Proof. Set $z = y + e^{\tau/2} a$. Then

$$\partial_\tau (T_a V)(y, \tau) = (\partial_\tau V)(z, \tau) + \frac{1}{2} e^{\tau/2} a \cdot \nabla V(z, \tau).$$

Moreover,

$$\frac{1}{2} y \cdot \nabla (T_a V)(y, \tau) = \frac{1}{2} z \cdot \nabla V(z, \tau) - \frac{1}{2} e^{\tau/2} a \cdot \nabla V(z, \tau).$$

The two additional transport terms cancel in $\partial_\tau T_a V + \frac{1}{2} y \cdot \nabla T_a V$. The Laplacian, the zeroth-order term, the convection term, the pressure gradient, and the divergence constraint are translation covariant. Substitution into (1.2) proves the claim. $\square$

Remark 8.3 (Raw translates create a spurious drift). If one sets $V_n(y, \tau) = V(y + x_n, \tau)$ with fixed x_n , then the centered equation acquires the additional transport term

$$-\frac{1}{2} x_n \cdot \nabla V_n.$$

Thus fixed translates do not define a compact invariant far-field hull for the renormalized dynamics.

Definition 8.4 (Spacetime covariant action). For $s \in \mathbb{R}$, define the centered time translation

$$(\Theta_s V)(y, \tau) := V(y, \tau + s), \quad (\Theta_s P)(y, \tau) := P(y, \tau + s).$$

For a tail center $x \in \mathbb{R}^3$ observed at centered time σ , define the covariant tail recentering

$$\mathcal{T}_{(x, \sigma)} V(y, s) := V(y + e^{s/2} x, \sigma + s), \quad \mathcal{T}_{(x, \sigma)} P(y, s) := P(y + e^{s/2} x, \sigma + s).$$

Equivalently,

$$\mathcal{T}_{(x, \sigma)} = \Theta_\sigma T_{e^{-\sigma/2} x}.$$

Thus $\mathcal{T}_{(x, \sigma)}$ is a composition of exact symmetries of the centered equation. Raw parabolic recenterings are used below only for local terminal extraction before one has passed to a centered ancient profile; all escaping tail limits of centered profiles use $\mathcal{T}_{(x, \sigma)}$.

Lemma 8.5 (Semidirect covariance of the centered action). *The centered time translations and spatial covariant translations satisfy*

$$\Theta_s T_a \Theta_{-s} = T_{e^{s/2}a}, \quad s \in \mathbb{R}, \quad a \in \mathbb{R}^3.$$

In particular the subgroup $\{T_a : a \in \mathbb{R}^3\}$ is normal in the full spacetime covariant action, and the subspace of observables invariant under all spatial covariant translations is invariant under every Θ_s .

Proof. For a profile V ,

$$(\Theta_s T_a \Theta_{-s} V)(y, \tau) = V(y + e^{(\tau+s)/2}a, \tau) = (T_{e^{s/2}a} V)(y, \tau).$$

The pressure transforms identically. The asserted invariance of the spatially invariant observable subspace is the same identity written on the induced L^2 -representation. $\square$

Theorem 8.6 (Covariant observer calculus). *Let $Z := \mathbb{R}^3 \times \mathbb{R}$ denote the centered observer space, with points $z = (\xi, \sigma)$. For $a \in \mathbb{R}^3$ and $\sigma \in \mathbb{R}$,*

$$(T_a U)(y, \tau) = U(y + e^{\tau/2}a, \tau), \quad (\Theta_\sigma U)(y, \tau) = U(y, \tau + \sigma).$$

Then

$$\Theta_\sigma T_a = T_{e^{\sigma/2}a} \Theta_\sigma, \quad \Theta_\sigma T_a \Theta_{-\sigma} = T_{e^{\sigma/2}a}.$$

For $z = (\xi, \sigma)$, define the covariant observer

$$\mathcal{R}_z U := \Theta_\sigma T_{e^{-\sigma/2}\xi} U.$$

Equivalently,

$$(\mathcal{R}_{(\xi, \sigma)} U)(y, s) = U(y + e^{s/2}\xi, \sigma + s).$$

The covariant tube pulled back from $Q_r(0, 0)$ is

$$\mathcal{Q}_r^{\text{cov}}(\xi, \sigma) := \left\{ (Y, \tau) : |\tau - \sigma| < r^2, \quad |Y - e^{(\tau-\sigma)/2}\xi| < r \right\}.$$

The map

$$(y, s) \mapsto (Y, \tau) = (y + e^{s/2}\xi, \sigma + s)$$

is a bijection from $Q_r(0, 0)$ onto $\mathcal{Q}_r^{\text{cov}}(\xi, \sigma)$ with spacetime Jacobian one. The same formulas apply to the pressure, modulo the admissible time-dependent gauges.

Proof. The semidirect identities are exactly [Theorem 8.5](#); written without conjugation,

$$(\Theta_\sigma T_a U)(y, \tau) = U(y + e^{(\tau+\sigma)/2}a, \tau + \sigma) = (T_{e^{\sigma/2}a} \Theta_\sigma U)(y, \tau).$$

Taking $a = e^{-\sigma/2}\xi$ gives

$$(\Theta_\sigma T_{e^{-\sigma/2}\xi} U)(y, s) = U(y + e^{(s+\sigma)/2}e^{-\sigma/2}\xi, \sigma + s) = U(y + e^{s/2}\xi, \sigma + s).$$

If $Y = y + e^{s/2}\xi$ and $\tau = \sigma + s$, then

$$y = Y - e^{(\tau-\sigma)/2}\xi, \quad s = \tau - \sigma,$$

so $|s| < r^2$ and $|y| < r$ are equivalent to the defining inequalities for $\mathcal{Q}_r^{\text{cov}}(\xi, \sigma)$. The transformation is a time-dependent spatial translation, hence has Jacobian one. $\square$

Definition 8.7 (Terminal state space and local topology). For $M < \infty$, let $\mathfrak{X}_M$ be the class of pairs (U, Π) , modulo addition of arbitrary functions of time to Π , such that:

- (i) U is a smooth bounded ancient solution of (1.2) on $\mathbb{R}^3 \times \mathbb{R}$, with $\|U\|_{L^\infty} \leq M$;
- (ii) (U, Π) is locally suitable on every compact terminal cylinder;
- (iii) on every compact cylinder the pressure has an admissible $L^{3/2}$ -representative after subtracting a time-dependent gauge.

The topology on $\mathfrak{X}_M$ is

$$U_n \rightarrow U \quad \text{in } C_{\text{loc}}^\infty, \quad \Pi_n - c_n(\tau) \rightarrow \Pi \quad \text{in } L_{\text{loc}}^{3/2}$$

for suitable time-dependent gauges c_n . All profile limits, tail descendants, active descendant hulls, and recurrent-core limits below are taken in this topology.

Definition 8.8 (Affine-normalized observer oscillation). Let Q be a bounded covariant terminal cylinder. For a velocity field U set

$$\text{osc}_3(U; Q) := \inf_{c \in \mathbb{R}^3} \iint_Q |U - c|^3 dy d\tau.$$

A covariant observer $z \in Z$ is η -oscillation-active for U if

$$\text{osc}_3(U; \mathcal{Q}_1^{\text{cov}}(z)) \geq \eta.$$

On each compact observer window we fix a uniformly locally finite family of observer centers $\{z_k\} \subset Z$, subordinate to a bounded-overlap covariant unit-cylinder cover. The corresponding local oscillation measure on observer-center sets $E \subset Z$ is

$$\mu_U^{\text{osc}}(E) := \sum_{z_k \in E} \text{osc}_3(U; \mathcal{Q}_1^{\text{cov}}(z_k)).$$

For a sequence U_n we write μ_n^{osc} for this measure. All uses of μ_n^{osc} are on compact observer-center windows or on specified escaping windows after restriction, so the bounded overlap gives uniform local finiteness. Replacing the cover by another bounded-overlap covariant unit cover changes only harmless constants and never changes the alternatives “positive threshold,” “vanishing threshold,” or “diffuse defect.”

Lemma 8.9 (Local compactness and continuity of the state actions). *Let $(U_n, \Pi_n) \subset \mathfrak{X}_M$ be a sequence with the local suitability and pressure-gauge bounds supplied by the setup package on every fixed compact cylinder. Then every subsequence has a further subsequence converging in the topology of $\mathfrak{X}_M$. Moreover the actions T_a , Θ_s , the covariant tail recenterings $\mathcal{T}_{(x,\sigma)}$ are continuous on $\mathfrak{X}_M$. The affine maps satisfy the bound-tracking continuity*

$$S_c : \mathfrak{X}_M \longrightarrow \mathfrak{X}_{M+|c|},$$

again with the pressure interpreted modulo the corresponding transformed time gauges.

Proof. The compactness assertion is the standard local Seregin compactness theorem applied on an exhausting sequence of compact cylinders, followed by a diagonal argument. The velocity convergence is upgraded to C_{loc}^∞ for bounded smooth centered profiles by interior parabolic estimates for (1.2); the pressure representatives converge weakly in $L_{\text{loc}}^{3/2}$ after the local gauges. The formulas defining T_a , Θ_s , $\mathcal{T}_{(x,\sigma)}$, and S_c are smooth changes of variables and affine pressure changes on each fixed compact cylinder. The first three preserve the L^∞ bound M . The affine map S_c subtracts the constant vector c , so its velocity bound is at most $M + |c|$. All four maps preserve the local topology and transform admissible gauges into admissible gauges. No global spatial norm or global pressure representative is used. $\square$

Definition 8.10 (Compact normalized descendant state space). Let $\mathfrak{X}_M^\# \subset \mathfrak{X}_M$ denote the sequential closure, in the local topology of $\mathfrak{X}_M$, of all normalized Seregin profiles from the setup package and all profiles obtained from them by the state-space operations used in this paper: centered time translations, covariant translations, covariant tail recenterings, terminal profile limits, active descendant limits, recurrent-core limits, and the canonical affine normalization whenever the affine parameter is bounded. The closure is always taken with the local pressure gauges supplied by the normalized Seregin compactness theorem.

Lemma 8.11 (Compactness and continuity on the normalized state space). *The space $\mathfrak{X}_M^\#$ is compact in the local topology inherited from $\mathfrak{X}_M$. The maps T_a , Θ_s , and the covariant recenterings used to form descendants act continuously on $\mathfrak{X}_M^\#$. The affine map satisfies the bound-tracking continuity*

$$S_c : \mathfrak{X}_M^\# \longrightarrow \mathfrak{X}_{M+|c|}^\#.$$

Proof. By definition, every sequence in $\mathfrak{X}_M^\#$ is a diagonal limit of normalized Seregin descendants with the same local L^∞ , local suitability, pressure-gauge, and interior smooth bounds on each fixed compact cylinder. Applying [Theorem 8.9](#) on an exhaustion of compact cylinders gives a further subsequential limit in $\mathfrak{X}_M$. The closure defining $\mathfrak{X}_M^\#$ contains that limit, so $\mathfrak{X}_M^\#$ is sequentially compact; the local topology is metrizable on bounded sets by the standard countable exhaustion, hence compact.

Continuity of the state-space operations is the continuity already proved in [Theorem 8.9](#), restricted to the closed normalized state space. The affine map changes only the velocity bound and the explicit affine pressure gauge, so it maps $\mathfrak{X}_M^\#$ continuously into $\mathfrak{X}_{M+|c|}^\#$. $\square$

Definition 8.12 (Affine homogeneous symmetry). For $c \in \mathbb{R}^3$, define

$$(S_c V)(y, \tau) := V(y - 2c, \tau) - c, \quad (S_c P)(y, \tau) := P(y - 2c, \tau) + \frac{1}{2}c \cdot y.$$

Lemma 8.13 (Affine renormalized Galilean symmetry). *If (V, P) solves (1.2), then $(S_c V, S_c P)$ also solves (1.2).*

Proof. Set $U(y, \tau) = V(y - 2c, \tau) - c$, $\Pi(y, \tau) = P(y - 2c, \tau) + \frac{1}{2}c \cdot y$, and $z = y - 2c$. Then

$$\partial_\tau U = (\partial_\tau V)(z, \tau), \quad \Delta U = (\Delta V)(z, \tau), \quad \nabla U = (\nabla V)(z, \tau).$$

The nonlinear term is

$$(U \cdot \nabla)U = (V - c) \cdot \nabla V = (V \cdot \nabla)V - c \cdot \nabla V,$$

where all quantities on the right are evaluated at (z, τ) . The drift term satisfies

$$\frac{1}{2}y \cdot \nabla U = \frac{1}{2}(z + 2c) \cdot \nabla V = \frac{1}{2}z \cdot \nabla V + c \cdot \nabla V.$$

The two $c \cdot \nabla V$ terms cancel. Finally,

$$\frac{1}{2}U + \nabla \Pi = \frac{1}{2}(V - c) + \nabla P + \frac{1}{2}c = \frac{1}{2}V + \nabla P.$$

Substitution into (1.2) at the point (z, τ) gives the equation for (U, Π) . The divergence condition is unchanged by translation and subtracting a constant vector. $\square$

Definition 8.14 (Affine/parasitic lower stratum and residual normalization). The affine/parasitic lower stratum $\mathcal{C}_{\text{aff}}$ consists of the spatially homogeneous modes which are invisible to affine-normalized oscillation. It has two equivalent realizations.

- (i) A genuine centered profile belongs to $\mathcal{C}_{\text{aff}}$ if, after one affine homogeneous symmetry S_c , its velocity is spatially constant. In the parasitic-free representative this is the zero profile.
- (ii) A quotient limit belongs to $\mathcal{C}_{\text{aff}}$ if it is represented by a locally bounded spatially homogeneous velocity $b(\tau)$. In the weak centered equation this mode is paired with the affine pressure functional

$$(-\partial_\tau b(\tau) - \frac{1}{2}b(\tau)) \cdot y$$

in $\mathcal{D}'_\tau \mathcal{D}'_y$. If the mode is realized as a smooth suitable profile, this affine pressure is an ordinary local pressure; if it arises only as a measurable homogeneous gauge $c_n(\tau)$, it is used only in the quotient weak formulation and never as an independent $L^3_{\text{loc}}/2$ pressure.

In the ordered residual decomposition this stratum is part of the previously closed lower classes, not part of $\mathcal{C}_{\text{gen}}$. Equivalently, every generic residual profile is considered after the canonical affine normalization which removes its spatially homogeneous mild mode. Retained activity in the residual class is therefore always measured after this normalization.

Remark 8.15 (How affine constants are used). When a compact recurrent tail core is shown below to consist only of spatially homogeneous parasitic modes, the contradiction is not that an unnormalized nonzero constant has zero local L^3 -mass. The contradiction is that these modes have already been assigned to the closed lower stratum in [Theorem 8.14](#); after the canonical affine normalization they are the zero profile and cannot represent a generic residual state carrying retained activity.

Lemma 8.16 (Homogeneous gauges generate only the affine/parasitic stratum). *Let $c_n : I \rightarrow \mathbb{R}^3$ be bounded in $L^\infty(I)$, and let*

$$C_n(y, \tau) := c_n(\tau).$$

Every local weak limit, time translate, or normalized noncompactness limit of (C_n) has zero affine-normalized spatial oscillation on every compact cylinder and belongs to $\mathcal{C}_{\text{aff}}$. Conversely, after the canonical parasitic-free normalization, no nonzero spatially homogeneous mode remains in the residual state space.

Proof. For every compact cylinder Q ,

$$\inf_{c(\tau)} \iint_Q |C_n(y, \tau) - c(\tau)|^3 dy d\tau = 0,$$

with the choice $c(\tau) = c_n(\tau)$. The same identity is stable under local weak limits and time recenterings. Hence every such limit has zero affine-normalized spatial oscillation on every compact cylinder and is, by [Theorem 8.14](#), an affine/parasitic quotient mode. If a homogeneous mode is realized as a smooth centered solution, substituting $U = b(\tau)$ into [\(1.2\)](#) gives

$$\partial_\tau b + \frac{1}{2}b + \nabla P = 0,$$

so the pressure is affine in y up to a time gauge, exactly as in the definition of $\mathcal{C}_{\text{aff}}$. If b is only a bounded measurable gauge, the same identity is understood distributionally and only in the quotient formulation. The residual representative is defined after removal of this quotient direction; therefore a nonzero retained effect of b is assigned to the closed lower stratum, while the parasitic-free residual state has no such mode. $\square$

Theorem 8.17 (Closed non-affine activity functional). *Let $Q \Subset \mathbb{R}^3 \times \mathbb{R}$ be a bounded connected parabolic cylinder. For $U \in \mathfrak{X}_M$, define*

$$d_Q(U) := \inf_{c \in \mathbb{R}^3} \left(\iint_Q |U - c|^3 dy d\tau \right)^{1/3}.$$

Then

$$|d_Q(U) - d_Q(V)| \leq \|U - V\|_{L^3(Q)}.$$

In particular d_Q is continuous under the local topology of $\mathfrak{X}_M$. Moreover,

$$d_Q(U) = 0 \iff U \text{ is spatially and temporally constant on } Q.$$

Fix once and for all a nested exhaustion

$$Q_1 \Subset Q_2 \Subset \cdots, \quad \bigcup_{j=1}^{\infty} Q_j = \mathbb{R}^3 \times \mathbb{R},$$

by bounded connected parabolic cylinders. Define

$$\Phi(U) := \sum_{j=1}^{\infty} 2^{-j} \min\{1, d_{Q_j}(U)\}.$$

Then Φ is continuous on $\mathfrak{X}_M$, and

$$\Phi(U) = 0 \iff U \text{ is globally constant.}$$

Consequently, for every $\eta > 0$, the non-affine activity set

$$\mathfrak{A}_\eta := \{(U, \Pi) \in \mathfrak{X}_M : \Phi(U) \geq \eta\}$$

is closed in the local state-space topology.

Proof. For every $c \in \mathbb{R}^3$, the triangle inequality in $L^3(Q)$ gives

$$\|U - c\|_{L^3(Q)} \leq \|U - V\|_{L^3(Q)} + \|V - c\|_{L^3(Q)}.$$

Taking the infimum over c gives

$$d_Q(U) \leq \|U - V\|_{L^3(Q)} + d_Q(V).$$

Interchanging U and V proves the Lipschitz estimate. Local smooth convergence implies local L^3 -convergence, so d_Q is continuous in the topology of $\mathfrak{X}_M$.

If $d_Q(U) = 0$, choose $c_k \in \mathbb{R}^3$ such that $\|U - c_k\|_{L^3(Q)} \rightarrow 0$. The sequence c_k is bounded; otherwise

$$\|U - c_k\|_{L^3(Q)} \geq |c_k| |Q|^{1/3} - \|U\|_{L^3(Q)} \rightarrow \infty,$$

which is impossible. Passing to a subsequence, $c_k \rightarrow c$, and hence $U = c$ almost everywhere on Q . Since U is smooth, $U \equiv c$ pointwise on Q . The converse is immediate.

Each summand in the definition of Φ is continuous and bounded by 2^{-j} . The Weierstrass test gives uniform convergence of the defining series, hence continuity of Φ . If $\Phi(U) = 0$, then $d_{Q_j}(U) = 0$ for every j , so U is constant on every Q_j . The cylinders are nested and overlap, so the constants agree and U is globally constant. The converse is again immediate. Closedness of $\mathfrak{A}_\eta$ follows from continuity of Φ . $\square$

Corollary 8.18 (Compact normalized activity sets). *For every $\eta > 0$,*

$$\mathfrak{A}_\eta^\# := \{(U, \Pi) \in \mathfrak{X}_M^\# : \Phi(U) \geq \eta\}$$

is compact in the normalized state-space topology.

Proof. By [Theorem 8.11](#), $\mathfrak{X}_M^\#$ is compact. By [Theorem 8.17](#), Φ is continuous in the local topology. Therefore $\mathfrak{A}_\eta^\#$ is a closed subset of a compact space. $\square$

Theorem 8.19 (Affine normalization dichotomy). *Let S_c be the affine homogeneous symmetry of [Theorem 8.12](#), and suppose the ordered lower strata include all constant centered profiles as in [Theorem 8.14](#). For every $(U, \Pi) \in \mathfrak{X}_M$ and every $c \in \mathbb{R}^3$, exactly one of the following alternatives holds:*

- (i) $\Phi(S_c U) > 0$, so the affine-normalized representative has positive non-affine retained activity;
- (ii) $\Phi(S_c U) = 0$, so $S_c U$ is globally constant and (U, Π) belongs to the affine/parasitic lower stratum.

Moreover, if $K \subset \mathfrak{X}_M$ is compact and

$$K \cap \{(U, \Pi) : \Phi(U) = 0\} = \emptyset,$$

then there exists $\eta_K > 0$ such that

$$\Phi(U) \geq \eta_K \quad \text{for every } (U, \Pi) \in K.$$

Proof. By [Theorem 8.17](#), $\Phi(S_c U) = 0$ if and only if $S_c U$ is globally constant. That is exactly the affine/parasitic lower stratum recorded in [Theorem 8.14](#). If this does not happen, $\Phi(S_c U) > 0$. The two alternatives are mutually exclusive and exhaust all cases. For the compact statement, Φ is continuous and strictly positive on K , so it attains a positive minimum

$$\eta_K := \min_{(U, \Pi) \in K} \Phi(U) > 0.$$

□

Definition 8.20 (Asymptotic hull at infinity). Let V be a bounded smooth ancient solution of (1.2). Define

$$\mathcal{A}_\infty(V) := \{W : \exists a_n \in \mathbb{R}^3, |a_n| \rightarrow \infty, T_{a_n} V \rightarrow W \text{ in } C_{\text{loc}}^\infty(\mathbb{R}^3 \times \mathbb{R})\}.$$

Theorem 8.21 (Compactness of the asymptotic hull). *Let V be a bounded smooth ancient solution of (1.2). Then $\mathcal{A}_\infty(V)$ is nonempty and compact in $C_{\text{loc}}^\infty(\mathbb{R}^3 \times \mathbb{R})$. Every $W \in \mathcal{A}_\infty(V)$ is a bounded smooth ancient solution of (1.2) with $\|W\|_{L^\infty} \leq \|V\|_{L^\infty}$.*

Proof. For any sequence a_n with $|a_n| \rightarrow \infty$, the translated solutions $T_{a_n} V$ solve the same equation and have the same L^∞ bound by [Theorem 8.2](#). On every compact cylinder, the coefficients of (1.2) are fixed and smooth. Interior parabolic estimates for the Stokes system with bounded smooth nonlinear data, combined with the pressure equation $-\Delta P = \partial_i \partial_j (V_i V_j)$ after local gauges, give uniform C^k -bounds on smaller compact cylinders for every k . A diagonal Arzela–Ascoli argument gives a subsequential limit in C_{loc}^∞ . Passing to the limit in the equation gives another bounded ancient solution. The same compactness argument applied to an arbitrary sequence of elements of $\mathcal{A}_\infty(V)$ proves compactness. □

Definition 8.22 (Tail alternatives). Let $\mathcal{C}$ denote the set of constant vector fields. Define

$$\mathcal{C}_{\text{gentail-const}} := \{V \in \mathcal{C}_{\text{gen}} : \mathcal{A}_\infty(V) \subset \mathcal{C}\},$$

and

$$\mathcal{C}_{\text{gentail-prof}} := \{V \in \mathcal{C}_{\text{gen}} : \mathcal{A}_\infty(V) \text{ contains a nonconstant profile}\}.$$

Proposition 8.23 (Tail dichotomy and tail-zero normalization). *Every $V \in \mathcal{C}_{\text{gen}}$ belongs to $\mathcal{C}_{\text{gentail-const}} \cup \mathcal{C}_{\text{gentail-prof}}$. If $\mathcal{A}_\infty(V) = \{C_\infty\}$ is a singleton constant hull, then*

$$U := S_{C_\infty} V$$

solves (1.2) and satisfies

$$\mathcal{A}_\infty(U) = \{0\}.$$

Proof. The dichotomy is the definition. Since T_a and S_c commute,

$$T_a(S_{C_\infty} V) = S_{C_\infty}(T_a V).$$

Thus every far-field limit of $S_{C_\infty} V$ is

$$S_{C_\infty}(C_\infty) = 0.$$

The equation is preserved by [Theorem 8.13](#). □

Definition 8.24 (Tail-minimal recurrent core). A nonempty compact set $\mathcal{M} \subset \mathcal{A}_\infty(V)$ is a tail-minimal recurrent core if it is invariant under all covariant translations T_a , invariant under all time translations

$$(\Theta_s W)(y, \tau) := W(y, \tau + s),$$

and contains no proper nonempty compact subset with the same two invariance properties.

Lemma 8.25 (Pressure gradients in invariant local averages). *Let $\mathcal{M}$ be a compact set of smooth bounded solutions of (1.2), compact in the locally smooth topology and invariant under the covariant translations T_a and the time translations Θ_s . Let μ be a Borel probability measure on $\mathcal{M}$ invariant under both actions. For each $W \in \mathcal{M}$, let P_W be any smooth local pressure near $(0, 0)$ satisfying (1.2). Put*

$$u_i(W) := W_i(0, 0), \quad p_i(W) := \partial_i P_W(0, 0), \quad b_{ij}(W) := u_i(W)u_j(W).$$

Let Π_0 denote the orthogonal projection in $L^2(\mathcal{M}, \mu)$ onto the closed subspace of functions invariant under all spatial covariant translations T_a , and set

$$u_i^0 := \Pi_0 u_i, \quad p_i^0 := \Pi_0 p_i.$$

Then the pressure contribution in the recurrent-core energy average is gauge-independent and satisfies

$$(8.1) \quad \int_{\mathcal{M}} u_i(W) p_i(W) d\mu(W) = \int_{\mathcal{M}} u_i^0(W) p_i^0(W) d\mu(W).$$

Equivalently, the nonzero spatial-frequency part of the pressure gradient does no work in invariant average:

$$(8.2) \quad \int_{\mathcal{M}} u_i(W) (p_i(W) - p_i^0(W)) d\mu(W) = 0.$$

Proof. The additive pressure gauge is irrelevant because only ∇P_W enters. The pressure gradient is fixed by the velocity through the equation

$$\nabla P_W = \Delta W - \frac{1}{2}(W + y \cdot \nabla W) - \partial_\tau W - (W \cdot \nabla)W.$$

Thus $W \mapsto p_i(W)$ is a continuous bounded local observable on $\mathcal{M}$. The local pressure reconstruction step in `proof_setup.tex` supplies the local pressure representatives; the present argument uses only their gradients, so the additive gauges never enter.

Let U_a be the unitary representation of spatial covariant translations on $L^2(\mathcal{M}, \mu)$,

$$(U_a F)(W) := F(T_a W).$$

Let D_1, D_2, D_3 be its skew-adjoint infinitesimal generators. For a smooth local observable F , $D_k F$ is obtained by differentiating $F(T_{he_k} W)$ at $h = 0$. In particular, at $\tau = 0$,

$$D_k u_i(W) = \partial_k W_i(0, 0).$$

Hence

$$(8.3) \quad D_i u_i = 0 \quad \text{in } L^2(\mathcal{M}, \mu),$$

because each W is divergence-free.

Taking the spatial divergence of (1.2) gives the local pressure Poisson equation

$$(8.4) \quad -\Delta_y P_W = \partial_i \partial_j (W_i W_j).$$

Indeed, the divergence of $\partial_\tau W$, ΔW , and W is zero; the divergence of $y \cdot \nabla W$ is $\partial_i(y_j \partial_j W_i) = \partial_j W_j + y_j \partial_j \partial_i W_i = 0$; and $(W \cdot \nabla)W_i = \partial_j(W_j W_i)$. Applying the infinitesimal translation generators at $(0, 0)$ gives

$$(8.5) \quad D_i p_i = -D_i D_j b_{ij}.$$

The pressure gradient is curl-free, hence

$$(8.6) \quad D_k p_i = D_i p_k.$$

Let

$$A := -\sum_{k=1}^3 D_k^2.$$

By the spectral theorem for the commuting unitary group U_a , A is self-adjoint and nonnegative, and Π_0 is exactly the spectral projection of A at the eigenvalue 0. Define the nonzero spatial part

$$q_i := p_i - p_i^0 = (I - \Pi_0)p_i.$$

Equations (8.5) and (8.6) imply, on the nonzero spectral subspace,

$$q_k = D_k A^{-1} (I - \Pi_0) D_i D_j b_{ij}$$

in the standard regularized sense. Explicitly, with

$$q_{k,\varepsilon} := D_k (\varepsilon + A)^{-1} (I - \Pi_0) D_i D_j b_{ij}, \quad \varepsilon > 0,$$

the joint spectral resolution of D_1, D_2, D_3 gives

$$q_{k,\varepsilon} \longrightarrow q_k \quad \text{in } L^2(\mathcal{M}, \mu) \quad \text{as } \varepsilon \downarrow 0.$$

The convergence is just the multiplier convergence

$$\frac{\xi_k \xi_i \xi_j}{\varepsilon + |\xi|^2} \longrightarrow \frac{\xi_k \xi_i \xi_j}{|\xi|^2} \quad (\xi \neq 0),$$

together with $(I - \Pi_0)$, which removes the point $\xi = 0$. The curl-free relation (8.6) says that the nonzero spatial spectral part of p is parallel to ξ , and (8.5) fixes its longitudinal coefficient; there is therefore no additional harmonic component.

For each $\varepsilon > 0$, skew-adjointness of D_k and (8.3) give

$$\begin{aligned} \int_{\mathcal{M}} u_k q_{k,\varepsilon} d\mu &= \int_{\mathcal{M}} u_k D_k (\varepsilon + A)^{-1} (I - \Pi_0) D_i D_j b_{ij} d\mu \\ &= - \int_{\mathcal{M}} (D_k u_k) (\varepsilon + A)^{-1} (I - \Pi_0) D_i D_j b_{ij} d\mu = 0. \end{aligned}$$

Passing to the limit in L^2 yields

$$\int_{\mathcal{M}} u_k q_k d\mu = 0,$$

which is (8.2). Since p_i^0 lies in the range of Π_0 and $u_i - u_i^0$ is orthogonal to that range,

$$\int_{\mathcal{M}} (u_i - u_i^0) p_i^0 d\mu = 0.$$

Combining this with (8.2) proves (8.1). □

Theorem 8.26 (Rigidity of recurrent profile tail cores). *Let $\mathcal{M}$ be a compact tail-minimal recurrent core for (1.2) arising from the normalized Seregin setup in `proof_setup.tex`, with the affine zero mode interpreted by the affine symmetry Theorem 8.13. Equivalently, after applying one fixed affine normalization S_c , the spatially homogeneous part of the core satisfies the bounded mild centered formulation recorded in `proof_setup.tex`. Then every $W \in \mathcal{M}$ is spatially and temporally constant. More precisely, there exists $m \in \mathbb{R}^3$ such that*

$$W(y, \tau) \equiv m \quad \text{for every } W \in \mathcal{M}.$$

After applying the affine symmetry S_m , the core is reduced to the zero solution.

Proof. The group generated by covariant translations and time translations is amenable on the compact invariant set $\mathcal{M}$. Choose an invariant Borel probability measure μ with full support on a minimal component, and write

$$\langle F \rangle := \int_{\mathcal{M}} F(W) d\mu(W)$$

for local observables. Let P_W be a local pressure associated to W . As in Theorem 8.25, set

$$u_i(W) := W_i(0, 0), \quad p_i(W) := \partial_i P_W(0, 0),$$

and let Π_0 be the orthogonal projection onto observables invariant under all spatial covariant translations. Write

$$u_i^0 := \Pi_0 u_i, \quad p_i^0 := \Pi_0 p_i.$$

Invariance gives

$$\langle \partial_\tau F(W)(0, 0) \rangle = 0, \quad \langle \partial_j F(W)(0, 0) \rangle = 0$$

for every smooth local observable F . These identities follow first for difference quotients generated by the group action and then by smoothness.

Evaluate (1.2) at $(0, 0)$. The drift term vanishes at $y = 0$, and we get

$$\partial_\tau W(0, 0) - \Delta W(0, 0) + \frac{1}{2} W(0, 0) + (W(0, 0) \cdot \nabla) W(0, 0) + \nabla P_W(0, 0) = 0.$$

Project this identity onto the spatially invariant subspace. The spatial derivative terms vanish under Π_0 : the Laplacian is $D_j D_j u_i$, and the nonlinear term is $D_j(u_j u_i)$ because $D_j u_j = 0$. Time translations do not literally commute with the covariant spatial translations; by Theorem 8.5 they normalize them:

$$\Theta_s T_a \Theta_{-s} = T_{e^{s/2} a}.$$

Hence the subspace of observables invariant under all T_a is invariant under Θ_s , and the orthogonal projection Π_0 commutes with the time-translation representation. Therefore $\Pi_0 \partial_\tau u_i = \partial_\tau u_i^0$. Thus

$$(8.7) \quad \partial_\tau u_i^0 + \frac{1}{2} u_i^0 + p_i^0 = 0 \quad \text{in } L^2(\mathcal{M}, \mu).$$

Taking the inner product of (8.7) with u_i^0 , summing in i , and using invariance under time translations,

$$\langle u_i^0 \partial_\tau u_i^0 \rangle = \frac{1}{2} \langle \partial_\tau |u^0|^2 \rangle = 0,$$

gives

$$(8.8) \quad \langle u_i^0 p_i^0 \rangle = -\frac{1}{2} \langle |u^0|^2 \rangle.$$

Dot the equation with W , evaluate at $(0, 0)$, and average. The time term is

$$\langle W \cdot \partial_\tau W \rangle = \frac{1}{2} \langle \partial_\tau |W|^2 \rangle = 0.$$

The diffusion term gives

$$-W \cdot \Delta W = |\nabla W|^2 - \operatorname{div}(W \cdot \nabla W),$$

and therefore

$$-\langle W \cdot \Delta W \rangle = \langle |\nabla W|^2 \rangle.$$

The nonlinear term is a divergence:

$$W \cdot (W \cdot \nabla) W = \frac{1}{2} W \cdot \nabla |W|^2 = \frac{1}{2} \operatorname{div}(|W|^2 W),$$

so its average is zero. Decompose

$$p_i = p_i^0 + (p_i - p_i^0).$$

The nonzero spatial part of the pressure gradient does no work in invariant average by [Theorem 8.25](#), and the remaining spatial zero mode is evaluated by (8.8). Hence

$$\langle u_i p_i \rangle = \langle u_i^0 p_i^0 \rangle = -\frac{1}{2} \langle |u^0|^2 \rangle.$$

Thus

$$(8.9) \quad \langle |\nabla W|^2 \rangle + \frac{1}{2} \langle |W|^2 \rangle - \frac{1}{2} \langle |u^0|^2 \rangle = 0.$$

Both terms are nonnegative. The second nonnegative quantity is

$$\langle |u|^2 \rangle - \langle |u^0|^2 \rangle = \langle |u - u^0|^2 \rangle,$$

because Π_0 is an orthogonal projection in $L^2(\mathcal{M}, \mu)$. Therefore

$$\langle |\nabla W|^2 \rangle = 0, \quad \langle |u - u^0|^2 \rangle = 0.$$

Since μ has full support and the observables are continuous in the local smooth topology, $\nabla W(0, 0) = 0$ for every $W \in \mathcal{M}$. Invariance moves $(0, 0)$ to any (y, τ) , so $\nabla W(y, \tau) = 0$ everywhere. Thus every element of $\mathcal{M}$ is spatially constant:

$$W(y, \tau) = a_W(\tau).$$

It remains only to identify the allowed spatially homogeneous zero mode. This is not a pressure-gauge assertion. By the normalization in the setup package quoted in the statement, there is one fixed affine normalization S_c for which the spatially homogeneous representatives satisfy the bounded mild centered formulation recorded in `proof_setup.tex`. For a spatially constant mild centered solution $B(\tau)$, the nonlinear Duhamel term in the mild formulation vanishes, and the Ornstein–Uhlenbeck–Stokes semigroup acts on constants by

$$S(\tau - \sigma)B(\sigma) = e^{-(\tau - \sigma)/2} B(\sigma).$$

Thus

$$B(\tau) = e^{-(\tau - \sigma)/2} B(\sigma) \quad (\sigma < \tau).$$

Letting $\sigma \rightarrow -\infty$ and using boundedness of the ancient trajectory gives $B(\tau) = 0$ for every τ . Undoing the fixed affine normalization gives a single vector $m = c$ and

$$W(y, \tau) \equiv m \quad (W \in \mathcal{M}).$$

The affine symmetry S_m then maps the core to the zero solution. □

9. TERMINAL CONCENTRATION PROFILES AND SEPARATED CENTERS

The terminal part of the proof is local in parabolic cylinders. It does not require a global L^3 bound, a global CKN budget, a finite critical-mass budget, or a global nonlinear profile decomposition for the original residual profile. Everything is formulated on fixed compact terminal cylinders after applying the recentering sequence under consideration.

Definition 9.1 (Parabolic recentering sequence). Let (V_n, P_n) be a terminal extraction sequence on expanding domains $D_n \uparrow \mathbb{R}^3 \times \mathbb{R}$, with local suitability and pressure-gauge control. A parabolic recentering sequence is a sequence

$$\mathbf{c} = (z_n)_{n \geq 1}, \quad z_n = (y_n, \tau_n) \in D_n.$$

Two recentering sequences are equivalent if

$$\sup_n \left(|y_n - y'_n| + |\tau_n - \tau'_n|^{1/2} \right) < \infty,$$

and asymptotically separated if the same parabolic distance tends to infinity. The recentered fields associated to $\mathbf{c}$ are denoted

$$V_n^{\mathbf{c}}(y, s) := V_n(y + y_n, s + \tau_n), \quad P_n^{\mathbf{c}}(y, s) := P_n(y + y_n, s + \tau_n) - a_n^{\mathbf{c}}(s),$$

where $a_n^{\mathbf{c}}$ is an admissible time-dependent local pressure gauge. All convergence assertions for a recentering sequence are made on fixed compact sets in the variables (y, s) .

Definition 9.2 (Combined raw-covariant descendant recentering). Let $g_n = (y_n, \tau_n)$ be a raw parabolic recentering sequence for a terminal extraction (V_n, P_n) , and let $\zeta_n = (\xi_n, \sigma_n)$ be a covariant tail recentering sequence in the centered limit generated by g_n . The combined raw-covariant descendant recentering is denoted

$$d_n = (g_n; \xi_n, \sigma_n)$$

and acts on the original terminal sequence by

$$V_n^d(y, s) := V_n(y + y_n + e^{s/2} \xi_n, s + \tau_n + \sigma_n),$$

with the pressure pulled back by the same formula and then adjusted by an admissible time-dependent gauge. Its relative distance from g_n is

$$|\xi_n| + |\sigma_n|^{1/2}.$$

Thus d_n is separated from g_n exactly when this relative distance tends to infinity. This definition is used only to lift descendants of a centered profile back to the original terminal extraction; ordinary first-generation terminal recenterings remain the raw parabolic recenterings of [Theorem 9.1](#).

Definition 9.3 (Local CKN measure and concentrating recentering sequences). After choosing local pressure gauges $a_n(\tau)$, define

$$d\mu_n := \left(|V_n|^3 + |P_n - a_n(\tau)|^{3/2} \right) dy d\tau, \quad d\nu_n := |\nabla V_n|^2 dy d\tau.$$

The local mass of a recentering sequence $\mathbf{c} = (z_n)$ is

$$m(\mathbf{c}) := \limsup_{n \rightarrow \infty} \mu_n(Q_1(z_n)).$$

For a combined raw-covariant descendant recentering $d_n = (g_n; \xi_n, \sigma_n)$, the same notation means the mass of the pulled back CKN measure in the combined coordinates:

$$m(\mathfrak{d}) := \limsup_{n \rightarrow \infty} \mu_n^{\mathfrak{d}}(Q_1(0, 0)),$$

where $\mu_n^\flat$ is obtained from μ_n by the coordinate change in [Theorem 9.2](#) and by the admissible pressure gauges used in that recentered frame. Thus raw first-generation recentering sequences and lifted covariant descendants use the same local unit-cylinder criterion, expressed in their own local coordinates. For a fixed small-data threshold $\varepsilon_{\text{sd}} > 0$, the recentering sequence is ε_{sd} -concentrating if $m(\mathbf{c}) \geq \varepsilon_{\text{sd}}$, and locally vanishing at unit scale otherwise. It has loss of local compactness if

$$\limsup_{n \rightarrow \infty} \nu_n(Q_2(z_n)) = \infty$$

or if the local pressure gauges fail to be uniformly controlled in $L^{3/2}(Q_2(z_n))$.

Lemma 9.4 (Gauge compatibility and retained-activity semicontinuity). *Let $(U_n, \Pi_n) \rightarrow (U, \Pi)$ in the topology of $\mathfrak{X}_M$ on a fixed compact cylinder containing $Q_1(0, 0)$. After choosing the pressure gauges used in that convergence, the following hold.*

- (i) *The pressure gauges are compatible with centered time shifts, covariant translations, covariant tail recenterings, affine normalization, and diagonal profile-tree extraction: applying any of these operations only changes the gauge by another admissible function of time.*
- (ii) *The local CKN quantity is lower semicontinuous in the direction needed for lifting active limits:*

$$\mathcal{E}_1(U, \Pi; 0) \leq \liminf_{n \rightarrow \infty} \mathcal{E}_1(U_n, \Pi_n; 0).$$

Consequently, if the limit profile has $\mathcal{E}_1(U, \Pi; 0) \geq \eta$, then all sufficiently far approximating profiles have CKN mass at least any prescribed $\eta' < \eta$.

- (iii) *If retained activity is carried by the velocity part, then it is closed under the same convergence:*

$$\liminf_{n \rightarrow \infty} \iint_{Q_1} |U_n|^3 \geq \eta \implies \iint_{Q_1} |U|^3 \geq \eta.$$

In applications where the lower bound is first obtained for velocity alone and then converted into a CKN lower bound, the descendant threshold is chosen below the resulting positive constant as in [Theorem 3.2](#).

Proof. Time shifts and covariant translations compose the pressure with a smooth local change of variables; affine normalization adds the explicit linear pressure $\frac{1}{2}c \cdot y$; and every local pressure is defined only up to a time-dependent function. Thus admissible gauges remain admissible under all operations listed in (i). For (ii), strong local convergence of U_n gives

$$\iint_{Q_1} |U_n|^3 \rightarrow \iint_{Q_1} |U|^3.$$

The pressure terms are convex in the weak $L^{3/2}$ topology after the fixed gauges, hence

$$\iint_{Q_1} |\Pi - c(\tau)|^{3/2} \leq \liminf_{n \rightarrow \infty} \iint_{Q_1} |\Pi_n - c_n(\tau)|^{3/2}.$$

Adding the two inequalities gives the claimed lower semicontinuity direction for $\mathcal{E}_1$. The velocity-only assertion follows directly from strong $L^3(Q_1)$ convergence. The statement is entirely local on Q_1 . $\square$

Lemma 9.5 (Setup compactness excludes retained local compactness failure). *Let (V_n, P_n) be a terminal sequence obtained from the Seregin setup in `proof_setup.tex`, or from a later terminal extraction of such a sequence. If a recentering sequence $\mathbf{c} = (z_n)$ carries retained compact velocity L^3 -mass, then after passing to a subsequence it has no loss of local compactness and its recentered fields converge locally in the Seregin topology to a suitable bounded terminal profile.*

Proof. Fix $R < \infty$. The recentered cylinders $Q_{2R}(z_n)$ are compact terminal cylinders in the variables of the chosen extraction. The setup package gives local suitability, local pressure-gauge compactness, and the inherited Type I bound on every such compact cylinder. Therefore, after choosing the pressure gauges a_n^c , the sequence (V_n^c, P_n^c) has the standard local bounds needed for the Seregin compactness theorem on $Q_R(0, 0)$: strong local L^3 compactness for the velocity, weak local $L^{3/2}$ compactness for the pressure, and the local energy inequality in the limit. A diagonal argument over $R = 1, 2, \dots$ gives a suitable bounded terminal profile. The two failures used in the definition of loss of local compactness are precisely the negations of these local compactness and pressure-gauge conclusions, so a retained recentering sequence arising from the Seregin setup in `proof_setup.tex` cannot remain in the local-compactness-failure alternative. $\square$

Theorem 9.6 (Local active-locus extraction). *Let $Z = \mathbb{R}^3 \times \mathbb{R}$ be the centered covariant observer space, and let $K_m \subseteq Z$, $m \geq 1$, be a compact exhaustion. Let μ_n be the local CKN measures of a terminal sequence, with local finite mass on compact observer windows:*

$$\sup_n \mu_n(K_m^+) < \infty \quad (m \geq 1),$$

where K_m^+ denotes the union of the supports of the transported bumps defined below for $z \in K_m$. Fix $\chi \in C_c^\infty(Q_1(0, 0))$, $0 \leq \chi \leq 1$, with $\chi \equiv 1$ on $Q_{1/2}(0, 0)$. For $z \in Z$, put

$$\chi_z(Y, \tau) := \chi(\mathcal{R}_z^{-1}(Y, \tau)), \quad M_n(z) := \int \chi_z d\mu_n.$$

For $\eta > 0$, define the compact-window active locus

$$A_{n,m}^\eta := \{z \in K_m : M_n(z) \geq \eta\}.$$

After passing to a subsequence, for every m and every rational $\eta > 0$, the compact sets $A_{n,m}^\eta$ converge in the Hausdorff–Fell topology to compact sets $A_m^\eta \subset K_m$. Moreover, for every open neighborhood $O \subset K_m$ of A_m^η ,

$$\limsup_{n \rightarrow \infty} \sup_{z \in K_m \setminus O} M_n(z) \leq \eta.$$

Thus outside every neighborhood of the active locus A_m^η , no η -active covariant unit tube remains in the compact observer window K_m . The active object is the closed locus A_m^η , not a finite or countable enumeration of tubes.

Proof. For fixed n , the map $z \mapsto \chi_z$ is continuous in C_c on K_m , and all supports are contained in the compact enlargement K_m^+ . Since μ_n is a Radon measure finite on K_m^+ , M_n is continuous on K_m . Therefore $A_{n,m}^\eta$ is compact.

The hyperspace of compact subsets of the compact metric space K_m is compact in the Hausdorff–Fell topology. Hence, for each fixed pair (m, η) , a subsequence exists along which $A_{n,m}^\eta$ converges. A diagonal argument over $m \in \mathbb{N}$ and $\eta \in \mathbb{Q}_+$ gives simultaneous convergence.

Let $O \subset K_m$ be an open neighborhood of A_m^η . If the stated residual bound failed, there would exist $\delta > 0$, a subsequence n_k , and points $z_{n_k} \in K_m \setminus O$ such that

$$M_{n_k}(z_{n_k}) \geq \eta + \delta.$$

After replacing O by a slightly smaller open neighborhood of A_m^η , we may assume $K_m \setminus O$ is compact. Passing to a further subsequence,

$$z_{n_k} \rightarrow z_* \in K_m \setminus O.$$

For all large k ,

$$z_{n_k} \in A_{n_k, m}^{\eta + \delta/2}.$$

Hausdorff–Fell convergence gives $z_* \in A_m^{\eta+\delta/2}$. Since $A_m^{\eta+\delta/2} \subset A_m^\eta \subset O$, this contradicts $z_* \in K_m \setminus O$. The residual estimate follows. All bounds used here are local on K_m^+ ; no global CKN budget or countability of active profiles is assumed. $\square$

Lemma 9.7 (Oscillation-active observers are CKN-active). *Let μ_n be the local CKN measure used in Theorem 9.6, so that μ_n dominates the velocity measure $|U_n|^3 dy d\tau$. If a covariant unit observer z satisfies*

$$\text{osc}_3(U_n; \mathcal{Q}_1^{\text{cov}}(z)) \geq \eta,$$

then the same observer is CKN-active at threshold η :

$$\mu_n(\mathcal{Q}_1^{\text{cov}}(z)) \geq \eta.$$

Consequently, after the CKN active loci of Theorem 9.6 are removed at every positive threshold, no positive affine-normalized oscillation-active unit observer remains outside those loci.

Proof. By definition of the affine-normalized oscillation,

$$\text{osc}_3(U_n; \mathcal{Q}_1^{\text{cov}}(z)) = \inf_{c \in \mathbb{R}^3} \iint_{\mathcal{Q}_1^{\text{cov}}(z)} |U_n - c|^3 \leq \iint_{\mathcal{Q}_1^{\text{cov}}(z)} |U_n|^3.$$

The local CKN measure contains the velocity contribution, hence

$$\iint_{\mathcal{Q}_1^{\text{cov}}(z)} |U_n|^3 \leq \mu_n(\mathcal{Q}_1^{\text{cov}}(z)).$$

Thus oscillation activity implies CKN activity with the same threshold. The last statement follows by applying the local active-locus extraction theorem to all rational positive thresholds and then lowering thresholds by harmless fixed cover and cutoff constants when the smoothed bump convention is used. $\square$

Lemma 9.8 (Separation of recentered cylinders). *Let $\mathbf{c}_i = (z_{i,n})$ and $\mathbf{c}_j = (z_{j,n})$ be asymptotically separated recentering sequences. For every fixed $R < \infty$, the cylinder $Q_R(z_{j,n})$, written in the coordinates of $\mathbf{c}_i$, leaves every compact subset of the recentered variables. Consequently no positive compact velocity L^3 -mass assigned to $\mathbf{c}_j$ is visible on any fixed compact cylinder of the $\mathbf{c}_i$ -frame.*

Proof. In the $\mathbf{c}_i$ -coordinates the center of $Q_R(z_{j,n})$ is

$$(y_{j,n} - y_{i,n}, \tau_{j,n} - \tau_{i,n}).$$

The parabolic distance of this point from the origin tends to infinity by asymptotic separation. Hence, for every fixed compact terminal cylinder K , one has

$$K \cap (Q_R(z_{j,n}) - z_{i,n}) = \emptyset$$

for all sufficiently large n . This is a purely local statement about the relative positions of the two recentering sequences. It does not estimate the solution on all of space and does not use a global norm. $\square$

Theorem 9.9 (Paired terminal active-profile decomposition). *Every terminal concentration sequence admits a paired decomposition*

$$\text{active retained profile locus} \quad + \quad \text{residual remainder}.$$

More precisely, after passing to a subsequence, the compact observer-window active loci A_m^η of Theorem 9.6 exist simultaneously for every m and every rational $\eta > 0$. Every observer sequence that remains in these loci at a fixed positive threshold and has no loss of local compactness generates a bounded retained terminal profile after local pressure gauges. Outside every neighborhood of the

active locus in each fixed compact observer window, the residual local CKN mass is below the chosen threshold. The residual remainder is then of exactly one of the following types:

$$\mathcal{R}_{\text{van}}, \quad \mathcal{R}_{\text{diff}}, \quad \mathcal{R}_{\text{noncomp}}.$$

Here $\mathcal{R}_{\text{van}}$ means no critical mass remains on expanding regions, $\mathcal{R}_{\text{diff}}$ means diffuse exterior critical mass remains on expanding regions while every fixed unit cylinder is below threshold, and $\mathcal{R}_{\text{noncomp}}$ means that some escaping residual recentering loses the local Seregin compactness or pressure-gauge control. Thus mixed states such as one active compact profile plus diffuse exterior remainder are part of the decomposition; they are not silently identified with the pure diffuse stratum.

Proof. Apply the local active-locus extraction theorem, [Theorem 9.6](#), on the compact observer exhaustion $\{K_m\}$. It produces closed active loci A_m^η and the residual estimate

$$\limsup_{n \rightarrow \infty} \sup_{z \in K_m \setminus O} M_n(z) \leq \eta$$

outside every neighborhood O of A_m^η . This is the precise replacement for a maximal finite or countable tube family: active centers are recorded by closed loci, and the residual is defined by restricting away from their neighborhoods.

If an observer sequence stays in a positive-threshold active locus and has no loss of the local compactness specified in [Theorem 9.3](#), then [Theorem 9.5](#) and the standard diagonal Seregin compactness argument give a bounded retained terminal profile. If such an observer sequence loses local dissipation or pressure-gauge control, the corresponding remainder is $\mathcal{R}_{\text{noncomp}}$.

It remains to classify what is outside the active loci. If the residual critical CKN mass tends to zero on every expanding terminal region, the remainder is $\mathcal{R}_{\text{van}}$. If a positive amount of critical CKN mass persists on expanding regions while every fixed compact observer window is below every active threshold after removing neighborhoods of the active loci, the remainder is $\mathcal{R}_{\text{diff}}$. These three residual states are mutually exclusive once the active loci are fixed, and they exhaust the residual possibilities. No step asserts that a diffuse or exterior remainder cannot coexist with active compact profiles. $\square$

Corollary 9.10 (Persistent local velocity mass forces retained concentration). *If a terminal sequence satisfies*

$$\liminf_{n \rightarrow \infty} \iint_{Q_1(0,0)} |V_n|^3 dy d\tau \geq \eta_0 > 0$$

with $\eta_0 \geq \varepsilon_{\text{sd}}$, then the terminal decomposition contains a local concentration profile unless the distinguished observer sequence has loss of local compactness. In the generic residual after local vanishing and local compactness failure are excluded, the persistent compact velocity mass is therefore carried by the active retained profile locus of [Theorem 9.9](#); the locus may contain one, finitely many, infinitely many, or continuum many active observer classes, and the residual remainder may still be diffuse until the mixed compact-diffuse stratum is excluded.

Proof. The local CKN measure μ_n dominates the velocity contribution. Hence the origin recentering sequence has μ -mass at least η_0 , and is ε_{sd} -concentrating unless it has loss of local compactness. If it has no loss of local compactness, active-locus extraction places its observer class in a positive-threshold active locus A_m^η of [Theorem 9.9](#). $\square$

Lemma 9.11 (Local vanishing and exterior alternatives do not carry compact concentration). *Let a terminal extraction inherit the positive compact velocity L^3 -mass supplied by the setup package `proof_setup.tex`. Then the retained compact velocity mass cannot be carried solely by local vanishing, exterior escape, or diffuse exterior concentration.*

Proof. The assertion is local. By the local no-escape and concentration input in `proof_setup.tex`, failure of local vanishing at a singular terminal point gives a positive local velocity concentration sequence. The setup package carries this through the Type I Seregina extraction as the compact velocity lower bound (1.5). Cover the fixed compact cylinder K in (1.5) by finitely many unit terminal cylinders. At least one cylinder in the cover carries a positive velocity lower bound, say $\eta_K > 0$. Since μ_n dominates $|V_n|^3 dy d\tau$, choosing $\varepsilon_{sd} \leq \eta_K$ makes that unit cylinder ε_{sd} -concentrating. Thus the distinguished compact concentration cannot be locally vanishing.

Local vanishing after removal of concentrating recentering sequences means that the residual state has local CKN mass below ε_{sd} on each fixed compact terminal cylinder; it therefore cannot contain the fixed compact velocity lower bound. An exterior escaping concentration is visible only after its own recentering; relative to the distinguished compact terminal cylinder it escapes every bounded window. By Theorem 9.8, it cannot be the sole location of a lower bound that remains in the distinguished compact window. Finally, diffuse exterior concentration is defined by persistence through expanding regions while all fixed unit cylinders are locally vanishing. If such a sequence contained the compact lower-bound threshold from `proof_setup.tex`, the positive concentration input in that package would select a unit terminal cylinder with mass at least the retained threshold, contradicting local vanishing. Thus none of these alternatives can be the unique location of the retained compact velocity mass. $\square$

Theorem 9.12 (Local exclusion of non-compact and non-concentrating terminal alternatives). *For terminal sequences generated by the Seregina setup in `proof_setup.tex` theorem, loss of local compactness, local vanishing, exterior escape, and diffuse exterior concentration are not possible sole locations of the retained compact velocity mass.*

Proof. Retained recentering sequences with loss of local compactness are removed by Theorem 9.5. Local vanishing, exterior escape, and diffuse exterior concentration are removed by Theorem 9.11. The proof uses only the positive concentration input and the local compactness package in `proof_setup.tex`, and fixed compact terminal cylinders. $\square$

Theorem 9.13 (Diffuse-defect compactness). *Let μ_n^{osc} be the residual affine-normalized oscillation measures of Theorem 8.8, after removing neighborhoods of all positive-threshold active loci from Theorem 9.6. By Theorem 9.7, this removal also removes all positive-threshold oscillation-active unit observers on compact windows. Assume there are observer regions $E_n \subset Z$, escaping every compact subset of Z , and a number $m_0 > 0$ such that*

$$\mu_n^{\text{osc}}(E_n) \geq m_0,$$

while no unit observer in E_n carries a positive oscillation threshold:

$$\sup_{z \in E_n} \text{osc}_3(U_n; Q_1^{\text{cov}}(z)) \longrightarrow 0.$$

Let $\overline{Z}$ be a compact metrizable observer compactification on which the fixed covariant observer maps extend continuously, and define probability measures

$$\lambda_n := \frac{\mu_n^{\text{osc}} \lfloor E_n}{\mu_n^{\text{osc}}(E_n)} \quad \text{on } \overline{Z}.$$

After passing to a subsequence,

$$\lambda_n \rightharpoonup \lambda$$

weakly as probability measures on $\overline{Z}$. The limit satisfies

$$\lambda(\overline{Z}) = 1, \quad \lambda(B) = 0 \quad \text{for every } B \Subset Z.$$

Thus the limit is a nonzero diffuse-defect state supported at the boundary of the observer compactification, not a local active profile.

Proof. The measures λ_n are probability measures on the compact metric space $\overline{Z}$, so weak compactness gives a convergent subsequence. Testing against the constant function 1 preserves total mass and gives $\lambda(\overline{Z}) = 1$.

Let $B \subseteq Z$. Since E_n escapes every compact subset of Z , one has $B \cap E_n = \emptyset$ for all sufficiently large n . Hence $\lambda_n(B) = 0$ eventually, and the portmanteau theorem gives $\lambda(B) = 0$. The additional unit-scale diffuseness assumption gives the stronger moving-window estimate

$$\sup_{z \in E_n} \lambda_n(B_Z(z, 1)) \leq \frac{C}{m_0} \sup_{\zeta \in E_n} \text{osc}_3(U_n; \mathcal{Q}_1^{\text{cov}}(\zeta)) \longrightarrow 0.$$

Here $B_Z(z, 1)$ is any fixed unit ball for a proper metric on Z , and C is the bounded-overlap constant of the observer cover. The restriction $\lambda_n = \mu_n^{\text{osc}}|_{E_n}/\mu_n^{\text{osc}}(E_n)$ is why only cover centers lying in E_n enter the estimate. Therefore no positive unit-scale local oscillation profile is hidden inside the escaping windows. The proof is local on the chosen observer windows and uses no global critical-mass or energy budget. $\square$

Definition 9.14 (Diffuse-defect state space). Let $\mathfrak{D}_M$ be the compact closure, in the weak topology of probability measures on $\overline{Z}$ together with the local $\mathfrak{X}_M$ -topology for the associated velocity hulls, of all normalized diffuse-defect states produced by [Theorem 9.13](#). The diffuse-defect mass functional is normalized to be identically one on $\mathfrak{D}_M$. This compactness is the product of weak compactness of probability measures on the compact observer space $\overline{Z}$ and the local Seregin compactness of bounded velocity hulls in $\mathfrak{X}_M$, after the admissible pressure gauges. The covariant observer group

$$G := \mathbb{R}^3 \rtimes \mathbb{R}$$

acts on $\mathfrak{D}_M$ by the observer maps generated by T_a and Θ_s .

Theorem 9.15 (Recurrent diffuse-defect measure). *The covariant observer action of G is continuous on $\mathfrak{D}_M$. Every nonzero diffuse-defect state $\lambda \in \mathfrak{D}_M$ generates a G -invariant Borel probability measure $\mathbb{P}$ on $\mathfrak{D}_M$. The normalized diffuse-defect mass remains one under this averaging.*

Proof. Continuity is part of the compactified observer topology in [Theorem 9.14](#): the action is continuous on the compact observer compactification, and the associated velocity hulls transform continuously in the local smooth topology with the admissible pressure gauges. The group $G = \mathbb{R}^3 \rtimes \mathbb{R}$, with the semidirect law from [Theorem 8.5](#), is solvable and hence amenable. Let (F_α) be a left Følner net for G , with Haar measure $|F_\alpha|$, and define probability measures on $\mathfrak{D}_M$ by

$$\mathbb{P}_\alpha := \frac{1}{|F_\alpha|} \int_{F_\alpha} \delta_{g\lambda} dg.$$

Compactness of $\mathfrak{D}_M$ implies weak-* compactness of its probability measures, so a subnet converges weakly to a probability measure $\mathbb{P}$. For every continuous observable F on $\mathfrak{D}_M$ and every fixed $h \in G$,

$$\int F(h\omega) d\mathbb{P}_\alpha(\omega) - \int F(\omega) d\mathbb{P}_\alpha(\omega) = \frac{1}{|F_\alpha|} \int_{F_\alpha} (F(hg\lambda) - F(g\lambda)) dg.$$

The Følner property makes the right-hand side tend to zero. Passing to the weak limit gives G -invariance of $\mathbb{P}$. Since the diffuse-defect mass is identically one on $\mathfrak{D}_M$, its average against $\mathbb{P}$ is also one. $\square$

Theorem 9.16 (Diffuse-defect trichotomy). *Let $(U_n, \Pi_n) \subset \mathfrak{X}_M$ be a normalized terminal residual sequence with $\|U_n\|_{L^\infty} \leq M$. Suppose that, after removing the active loci at every positive affine-normalized threshold, there are escaping observer windows E_n such that*

$$\mu_n^{\text{osc}}(E_n) \geq m_0 > 0.$$

Then, after passing to a subsequence, the residual diffuse defect falls into one of the following ordered alternatives:

D1. Active regeneration. There exist covariant observers z_n and a threshold $\eta > 0$ such that

$$\text{osc}_3(U_n; \mathcal{Q}_1^{\text{cov}}(z_n)) \geq \eta.$$

This produces a retained active descendant.

D2. Affine/parasitic collapse. Every velocity tangent associated with the diffuse-defect hull is locally constant after the admissible affine normalization; the defect belongs to the affine/parasitic lower stratum of [Theorem 8.14](#).

D3. Critical diffuse tail. There is a nonzero recurrent diffuse-defect state supported on $\bar{Z} \setminus Z$, carrying normalized diffuse oscillation mass one, annihilating every fixed unit active observable, and not contained in the affine/parasitic lower stratum.

Equivalently, residual diffuse defects are contained in the union of the active regeneration alternative, the affine/parasitic stratum, and the critical diffuse-tail class $\mathcal{C}_{\text{critdiff}}$.

Proof. If D1 occurs there is nothing to prove. Assume D1 does not occur. Then, for every fixed $\eta > 0$, all sufficiently large n satisfy

$$\sup_{z \in E_n} \text{osc}_3(U_n; \mathcal{Q}_1^{\text{cov}}(z)) < \eta.$$

The hypotheses of [Theorem 9.13](#) apply to $\mu_n^{\text{osc}}[E_n]$, and hence produce a diffuse-defect state λ with total mass one and zero mass on each compact subset of Z . By [Theorem 9.15](#), the orbit closure of λ carries a G -invariant probability measure $\mathbb{P}$.

If every local velocity tangent in the support of $\mathbb{P}$, after the canonical affine normalization, is locally constant, then [Theorem 8.14](#) places the defect in the affine lower stratum. This is D2. If not, the invariant diffuse boundary state has positive normalized diffuse oscillation mass, no unit-scale active observable, and non-affine velocity content. This is precisely the critical diffuse-tail alternative D3. The alternatives are ordered by first extracting all active regeneration and then all affine/parasitic collapse; after those removals, the remaining nonzero diffuse state is $\mathcal{C}_{\text{critdiff}}$. $\square$

Definition 9.17 (Critical-tail compactification). The class $\mathcal{C}_{\text{critdiff}}$ consists of recurrent diffuse-defect states produced by [Theorem 9.16](#) which are supported on the boundary of the covariant observer compactification, carry normalized affine oscillation mass one, annihilate every fixed unit active observable, and are not affine/parasitic.

The critical-tail compactification $\mathfrak{T}_M$ is the compact state space of triples

$$\mathcal{T} = (\bar{W}, \mathcal{Y}, \Lambda)$$

generated by such critical diffuse states after macroscopic blow-down. Here $\bar{W}$ is the barycentric macroscopic velocity, $\mathcal{Y}_{x,s}$ is the Young measure recording unresolved bounded macroscopic oscillation, and Λ is the log-scale diffuse measure recording mass escaping every fixed multiplicative annulus. The coherent subspace $\mathfrak{T}_{\text{coh}}$ is the special case

$$\mathcal{Y}_{x,s} = \delta_{\bar{W}(x,s)}, \quad \Lambda = 0.$$

The purely log-diffuse subspace is denoted $\mathfrak{T}_{\text{log diff}}$, and the oscillatory Young-tail subspace is denoted $\mathfrak{T}_{\text{Young}}$.

For coherent elements, the corresponding macroscopic tangent pair (W, Q) belongs to $\mathcal{C}_{\text{cohcrit}}$ and satisfies on annular cylinders $\{0 < r_0 < |x| < r_1 < \infty\} \times I$

$$\partial_s W + \frac{1}{2} x \cdot \nabla W + \frac{1}{2} W + \nabla Q = 0, \quad \nabla \cdot W = 0$$

in distributions, with nonzero affine-normalized diffuse oscillation and recurrence under the induced logarithmic dilation flow.

9.1. Log-annular virial identities for recurrent critical defects. Throughout this subsection let $\chi \in C_c^\infty((-2, 2))$ satisfy $0 \leq \chi \leq 1$ and $\chi \equiv 1$ on $(-1, 1)$. For $\rho \in \mathbb{R}$ and $A \geq 1$, set

$$\phi_{\rho,A}(y) := \chi\left(\frac{\log|y| - \rho}{A}\right), \quad R := e^\rho.$$

The cutoff is used only on annuli avoiding the origin. When differentiating with respect to ρ , A is kept fixed.

Lemma 9.18 (Log-annular oscillation virial identity). *Let $(U, \Pi) \in \mathfrak{X}_M$ be a smooth bounded solution of (1.2). Fix $A \geq 1$, $\rho \in \mathbb{R}$, and put $\phi = \phi_{\rho,A}$. Define the weighted affine gauge*

$$c_{\rho,A}(\tau) := \frac{\int \phi(y) U(y, \tau) dy}{\int \phi(y) dy}, \quad V_{\rho,A}(y, \tau) := U(y, \tau) - c_{\rho,A}(\tau),$$

so that

$$\int \phi V_{\rho,A} dy = 0.$$

Set

$$\begin{aligned} E_{\rho,A}(\tau) &:= \int \phi |V_{\rho,A}|^2 dy, & \mathcal{V}_{\rho,A}(\tau) &:= R^{-1} E_{\rho,A}(\tau), \\ D_{\rho,A}(\tau) &:= \int \phi |\nabla U|^2 dy, & \mathcal{D}_{\rho,A}(\tau) &:= R^{-1} D_{\rho,A}(\tau). \end{aligned}$$

Then $\mathcal{V}_{\rho,A}$ is locally absolutely continuous in τ , and for a.e. τ , after subtracting an arbitrary pressure gauge $\pi(\tau)$, one has

$$\begin{aligned} (9.1) \quad \partial_\tau \mathcal{V}_{\rho,A} + 2\mathcal{D}_{\rho,A} + \frac{1}{2} \partial_\rho \mathcal{V}_{\rho,A} &= R^{-1} \int (\Delta \phi) |V_{\rho,A}|^2 dy \\ &+ R^{-1} \int |V_{\rho,A}|^2 U \cdot \nabla \phi dy \\ &+ 2R^{-1} \int (\Pi - \pi(\tau)) V_{\rho,A} \cdot \nabla \phi dy. \end{aligned}$$

Moreover

$$(9.2) \quad |\nabla \phi_{\rho,A}(y)| \leq \frac{C_\chi}{A|y|}, \quad |\Delta \phi_{\rho,A}(y)| \leq \frac{C_\chi}{A|y|^2} + \frac{C_\chi}{A^2|y|^2},$$

and

$$y \cdot \nabla \phi_{\rho,A}(y) = \frac{1}{A} \chi' \left(\frac{\log|y| - \rho}{A} \right), \quad \partial_\rho \phi_{\rho,A} = -y \cdot \nabla \phi_{\rho,A}.$$

Proof. Since $c_{\rho,A}(\tau)$ is the weighted least-squares affine gauge,

$$\int \phi V_{\rho,A} dy = 0.$$

Hence the derivative of the gauge drops out:

$$\frac{d}{d\tau} \int \phi |U - c_{\rho,A}|^2 = 2 \int \phi V_{\rho,A} \cdot \partial_\tau U.$$

Substitute the centered equation

$$\partial_\tau U = \Delta U - (U \cdot \nabla) U - \frac{1}{2} y \cdot \nabla U - \frac{1}{2} U - \nabla \Pi.$$

The diffusion term gives

$$2 \int \phi V \cdot \Delta U = -2 \int \phi |\nabla U|^2 + \int (\Delta \phi) |V|^2.$$

The convective term gives, using $U \cdot \nabla U = U \cdot \nabla V$ and $\nabla \cdot U = 0$,

$$-2 \int \phi V \cdot (U \cdot \nabla U) = - \int \phi U \cdot \nabla |V|^2 = \int |V|^2 U \cdot \nabla \phi.$$

The centered drift and linear terms give

$$\begin{aligned} - \int \phi V \cdot (y \cdot \nabla U) - \int \phi V \cdot U &= -\frac{1}{2} \int \phi y \cdot \nabla |V|^2 - \int \phi |V|^2 \\ &= \frac{1}{2} \int y \cdot \nabla \phi |V|^2 + \frac{1}{2} \int \phi |V|^2. \end{aligned}$$

The terms involving $c_{\rho,A}$ vanish because $\int \phi V = 0$. Finally, for any scalar gauge $\pi(\tau)$,

$$-2 \int \phi V \cdot \nabla \Pi = 2 \int (\Pi - \pi(\tau)) V \cdot \nabla \phi,$$

because $\nabla \cdot V = 0$ and

$$\int \pi(\tau) V \cdot \nabla \phi dy = \pi(\tau) \int \nabla \cdot (\phi V) dy = 0.$$

Combining these identities yields

$$\begin{aligned} \partial_\tau E_{\rho,A} + 2D_{\rho,A} &= \int (\Delta \phi) |V|^2 + \int |V|^2 U \cdot \nabla \phi + 2 \int (\Pi - \pi) V \cdot \nabla \phi \\ &\quad + \frac{1}{2} \int \phi |V|^2 + \frac{1}{2} \int y \cdot \nabla \phi |V|^2. \end{aligned}$$

Since

$$\partial_\rho \phi_{\rho,A} = -y \cdot \nabla \phi_{\rho,A}$$

and the weighted least-squares condition also removes the $\partial_\rho c_{\rho,A}$ term,

$$\partial_\rho E_{\rho,A} = \int \partial_\rho \phi_{\rho,A} |V|^2 = - \int y \cdot \nabla \phi_{\rho,A} |V|^2.$$

Therefore

$$\frac{1}{2} R^{-1} \left(\int \phi |V|^2 + \int y \cdot \nabla \phi |V|^2 \right) = -\frac{1}{2} \partial_\rho (R^{-1} E_{\rho,A}) = -\frac{1}{2} \partial_\rho \mathcal{V}_{\rho,A}.$$

Thus, after multiplying the energy identity by R^{-1} , the centered drift and zeroth-order contribution is $-\frac{1}{2} \partial_\rho \mathcal{V}_{\rho,A}$. Moving it to the left gives (9.1). The cutoff bounds follow by direct differentiation in polar coordinates. $\square$

Remark 9.19 (Virial normalization). The normalization $R^{-1} \int \phi |U - c|^2$ is the correct scale for an actual spatial tail of size $|y|^{-1}$. It is not automatically a compact observable for an arbitrary bounded residual profile. In the critical-tail reduction below, one uses it either on selected scale-critical defect windows or as an observable on the compact recurrent defect hull after the required scale-critical virial tightness has been verified. Without that compactness, the virial identity remains valid for smooth annular cutoffs but cannot by itself close the critical-tail alternatives.

Lemma 9.20 (Recurrent virial equality case and routing). *Let $\mathcal{D} \in \mathfrak{T}_M$ be a recurrent critical-tail defect generated by a bounded normalized terminal sequence after active-locus removal and affine normalization. Assume that the log-annular virials in Theorem 9.18 are legitimate compact observables on the recurrent hull of $\mathcal{D}$, in the sense of Theorem 9.19, and let $\mathbb{P}$ be an invariant probability measure on that hull for the joint time-translation and log-translation flow. Assume also that the averaged pressure gauges are chosen by a local annular pressure decomposition, so that the three right-hand side terms in (9.1) have limits as defect observables. Then the following conclusions hold.*

(i) Zero-flux equality case. *If*

$$(9.3) \quad \lim_{A \rightarrow \infty} \int \left| R^{-1} \int (\Delta \phi) |V|^2 + R^{-1} \int |V|^2 U \cdot \nabla \phi + 2R^{-1} \int (\Pi - \pi) V \cdot \nabla \phi \right| d\mathbb{P} = 0,$$

then every local tangent in the support of $\mathbb{P}$ has zero spatial gradient. Consequently

$$\mathcal{D} \in \mathcal{C}_{\text{aff}}.$$

(ii) Tight Dirac flux routing. *Suppose (9.3) fails, and suppose the nonzero limiting flux is neither concentrated on a positive-threshold unit active observer nor lost to logarithmic infinity. Suppose moreover that the annular blow-downs selected by this nonzero flux generate a Dirac Young measure. Then the selected annular blow-downs converge strongly in L^3_{loc} on macroscopic annular cylinders to a nonzero coherent critical tail. Hence*

$$\mathcal{D} \in \mathcal{C}_{\text{cohcrit}}.$$

The first-bad mesoscopic routing theorem proved below supplies the tight-Dirac alternative for realized recurrent defects. Combined with that theorem, the preceding alternatives give the routing

$$\mathcal{D} \in \mathcal{C}_{\text{active}} \cup \mathcal{C}_{\text{aff}} \cup \mathcal{C}_{\text{cohcrit}}.$$

Proof. Integrate (9.1) against the invariant measure $\mathbb{P}$. The averaged $\partial_\tau \mathcal{V}_{\rho,A}$ term vanishes by time-translation invariance. The averaged $\partial_\rho \mathcal{V}_{\rho,A}$ term vanishes by log-translation invariance. Therefore

$$2 \int \mathcal{D}_{\rho,A} d\mathbb{P} = \int \mathcal{F}_{\rho,A} d\mathbb{P},$$

where $\mathcal{F}_{\rho,A}$ denotes the sum of the three right-hand side flux terms in (9.1). Under (9.3) and by nonnegativity of the dissipation,

$$\lim_{A \rightarrow \infty} \int \mathcal{D}_{\rho,A} d\mathbb{P} = 0.$$

Choose a countable exhausting family of ordinary unit covariant cylinders. For each cylinder, choose A large enough that the corresponding cutoff is identically one on a fixed enlargement. The preceding limit then implies that, for $\mathbb{P}$ -a.e. state, the local dissipation $\int_Q |\nabla U|^2$ vanishes on every cylinder in the exhausting family. Hence $\nabla U = 0$ distributionally on every compact cylinder. Thus every local tangent is spatially constant. By the affine normalization dichotomy, this is precisely the affine/parasitic lower stratum $\mathcal{C}_{\text{aff}}$.

We now prove the tight Dirac flux routing. Since the averaged flux does not vanish and is not concentrated on a unit active observer, there are logarithmic annuli with nonzero normalized flux but with no positive-threshold unit oscillation-active subcylinder. Since the flux is not lost to logarithmic infinity, these annuli may be selected with a tight macroscopic scale $R_n \rightarrow \infty$. Passing to the corresponding macroscopic variables gives bounded blown-down fields

$$W_n(x, s) = U_n(y_n + R_n x, \sigma_n + s) - c_n(s)$$

on every fixed annular cylinder. The terminal compactness package gives a weak-* L^∞_{loc} limit and an associated Young measure. By the Dirac hypothesis on the selected flux, the Young

measure is δ_W almost everywhere. Uniform boundedness then upgrades convergence to strong L^3_{loc} convergence on annular cylinders. Passing to the limit in the blown-down equation, the viscous and nonlinear terms vanish by the factors R_n^{-2} and R_n^{-1} , and the pressure gauges converge in annular distributions. The limit therefore satisfies

$$\partial_s W + \frac{1}{2}x \cdot \nabla W + \frac{1}{2}W + \nabla Q = 0, \quad \nabla \cdot W = 0,$$

on macroscopic annular cylinders. The selected flux is nonzero, so the limit has nonzero affine-normalized oscillation. Thus $(W, Q) \in \mathcal{C}_{\text{cohcrit}}$. $\square$

9.2. Logarithmic escape as a hidden-scale obstruction. The logarithmic escape alternative is not discharged below by a separate global tail estimate. The point of the next reduction is narrower and more useful: once tight annular descendants have no hidden-scale variance, a nonzero logarithmic escape component cannot survive. Thus the log-diffuse alternative and the Young alternative are both reduced to the same first-bad mesoscopic routing theorem.

Definition 9.21 (Log-diffuse recurrent critical defect). A realized recurrent critical-tail defect is log-diffuse if, after projecting the normalized diffuse oscillation measures to the logarithmic radial variable $\rho = \log |y|$, their log marginals λ_n satisfy

$$\sup_{\rho_0 \in \mathbb{R}} \lambda_n([\rho_0 - A, \rho_0 + A]) \longrightarrow 0 \quad \text{for every fixed } A < \infty.$$

Equivalently, the critical-tail compactification has a nonzero logarithmic escape component Λ . This alternative is denoted by $\mathcal{C}_{\text{log diff}}$.

Definition 9.22 (Log-window descendant). Let λ_n be the normalized logarithmic defect marginals of a log-diffuse critical defect. A log-window descendant is obtained by choosing numbers $A \geq 1$, centers ρ_n , and intervals

$$I_n = [\rho_n - A, \rho_n + A]$$

with $\lambda_n(I_n) > 0$, restricting the defect measures to I_n , and renormalizing:

$$\tilde{\lambda}_n := \frac{\lambda_n|_{I_n}}{\lambda_n(I_n)}.$$

After applying the logarithmic recentering $\rho \mapsto \rho - \rho_n$, the renormalized measures are tight in the fixed interval $[-A, A]$. Any critical-tail state obtained as a subsequential limit of these normalized restrictions is called a log-window descendant of the original log-diffuse defect.

Lemma 9.23 (Nonzero logarithmic escape has tight log-window descendants). *Let $\mathcal{D} \in \mathcal{C}_{\text{log diff}}$ be a nonzero realized recurrent critical-tail defect after active-locus removal and affine normalization. Then, for some fixed $A \geq 1$, $\mathcal{D}$ has a nonzero log-window descendant. The descendant is a tight macroscopic annular critical-tail realization and retains normalized affine oscillation mass.*

Proof. The logarithmic marginals λ_n are probability measures. Fix any $A \geq 1$. Cover $\mathbb{R}_\rho$ by the intervals

$$I_{k,A} := [2Ak - A, 2Ak + A], \quad k \in \mathbb{Z}.$$

For each n , at least one interval has positive λ_n -mass. Choose one such interval and denote it by $I_n = [\rho_n - A, \rho_n + A]$. Restricting to I_n , renormalizing, and translating ρ_n to the origin gives probability measures supported in the fixed compact interval $[-A, A]$. Compactness of the observer boundary state space and the local Seregín compactness of $\mathfrak{X}_M$ give a subsequential limit. By construction, the limiting state has total normalized defect mass one, hence is nonzero.

The logarithmic recentering fixes the selected multiplicative annulus, so the resulting critical-tail realization is tight in logarithmic scale. The original diffuse defect was measured after

affine normalization by the oscillation defect measure; the normalized restriction preserves total oscillation mass equal to one. Hence the descendant carries nonzero affine-normalized oscillation. No positive lower bound for the unnormalized mass $\lambda_n(I_n)$ is required, because the descendant is a normalized boundary state. $\square$

Lemma 9.24 (Renormalized log-window heredity). *Let $\mathcal{D} \in \mathcal{C}_{\log \text{ diff}}$ be an ancestor-realized log-diffuse defect, and let $\mathcal{F} \subset \mathfrak{T}_M$ be a closed union of alternatives which is closed under retained descendants. Suppose that every nonzero normalized log-window descendant of $\mathcal{D}$ belongs to $\mathcal{F}$. Then the ancestor log-diffuse defect itself has no surviving mass outside $\mathcal{F}$. In particular, if $\mathcal{F}$ has already been discharged by the lower-stratum closure theorems, then $\mathcal{D}$ is impossible.*

Proof. The only point needing clarification is that the selected logarithmic windows may have unnormalized mass tending to zero. This causes no loss of ancestry. Let Λ be the compactified logarithmic defect measure associated with $\mathcal{D}$. If $\Lambda(\mathfrak{T}_M \setminus \mathcal{F}) > 0$, then by regularity of Borel probability measures on the compact defect space there is a compact set

$$K \Subset \mathfrak{T}_M \setminus \mathcal{F}$$

with $\Lambda(K) > 0$. Pulling K back to the approximating observer compactification and to the logarithmic coordinate gives, after passing to a subsequence, positive-measure windows I_n whose normalized restrictions converge to a probability measure supported in K . Equivalently, one obtains a nonzero normalized log-window descendant outside $\mathcal{F}$.

This contradicts the hypothesis that every nonzero log-window descendant lies in $\mathcal{F}$. Hence Λ is supported on $\mathcal{F}$. Since $\mathcal{F}$ is closed under retained descendants and is already discharged, the ancestor defect has no admissible surviving normalized support state. The fact that $\lambda_n(I_n)$ may tend to zero is irrelevant: the normalized restriction records a support state of the ancestor probability measure, and a positive ancestor defect measure cannot be supported entirely on discharged closed alternatives. $\square$

Theorem 9.25 (Logarithmic escape reduces to hidden-scale compactness). *Assume Theorem 9.48. Then every nonzero log-diffuse critical-tail defect generated after active-locus removal and affine normalization has a retained descendant in*

$$\mathcal{C}_{\text{active}} \cup \mathcal{C}_{\text{aff}} \cup \mathcal{C}_{\text{cohcrit}}.$$

Consequently, after the active, affine/parasitic, and coherent critical-tail alternatives have been removed, no log-diffuse alternative remains.

Proof. Let $\mathcal{D} \in \mathcal{C}_{\log \text{ diff}}$ be nonzero. By Theorem 9.23, nonzero tight log-window descendants exist. Let $\tilde{\mathcal{D}}$ be any such descendant. If the descendant regenerates a positive-threshold unit active observer, then it belongs to $\mathcal{C}_{\text{active}}$. If its affine-normalized oscillation vanishes, then Theorem 8.19 places it in $\mathcal{C}_{\text{aff}}$. We may therefore assume that it is neither active nor affine.

Because $\tilde{\mathcal{D}}$ was obtained by restriction to a fixed logarithmic window, its logarithmic escape component is zero. The only remaining possible obstruction in the critical-tail compactification is a non-Dirac annular Young defect. By Theorem 9.48, every tight annular blow-down generated by the descendant has no hidden-scale variance. Hence Theorem 9.28 makes the Young measure Dirac. The coherent extraction Theorem 9.50 then gives a nonzero coherent critical tail; thus $\tilde{\mathcal{D}} \in \mathcal{C}_{\text{cohcrit}}$.

In all cases the log-window descendant lies in the closed discharged set

$$\mathcal{F} := \mathcal{C}_{\text{active}} \cup \mathcal{C}_{\text{aff}} \cup \mathcal{C}_{\text{cohcrit}}.$$

By Theorem 9.24, if every normalized log-window descendant of the original log-diffuse defect lies in $\mathcal{F}$, then the ancestor defect has no surviving mass outside $\mathcal{F}$. Once the active, affine/parasitic,

and coherent critical-tail alternatives are removed, this makes the original log-diffuse defect impossible. $\square$

Corollary 9.26 (Log-diffuse alternative discharge from first bad scale routing). *Inside the retained residual decomposition,*

$$\mathcal{C}_{\log \text{ diff}} = \emptyset.$$

Equivalently, after affine/parasitic lower-stratum removal, logarithmic hidden escape is discharged by the same hidden-scale compactness theorem that removes the Young alternative.

Proof. By Theorem 9.46, Theorem 9.48 applies to all tight annular descendants. Hence Theorem 9.25 reduces every nonzero log-diffuse defect to a descendant in $\mathcal{C}_{\text{active}} \cup \mathcal{C}_{\text{aff}} \cup \mathcal{C}_{\text{cohcrit}}$. The active alternative has been removed by terminal active-locus extraction, the affine stratum by Theorem 9.53, and the coherent critical-tail alternative by Theorem 9.59. Hence no log-diffuse alternative remains. $\square$

9.3. Hidden-scale Young variance.

Definition 9.27 (Hidden-scale variance). Let W_n be a macroscopic critical-tail blow-down on a compact annular cylinder

$$K \Subset \{0 < |x| < \infty\} \times I.$$

For $0 < \delta < 1$, set

$$\mathfrak{H}_K(W_n, \delta) := \sup_{\substack{|h| < \delta \\ |\theta| < \delta}} \|W_n(\cdot + h, \cdot + \theta) - W_n\|_{L^3(K)},$$

where the translated functions are evaluated on a fixed slightly larger annular cylinder containing all $(x + h, s + \theta)$ with $(x, s) \in K$ and $|h| + |\theta| < \delta$. The sequence has no hidden-scale variance on K if

$$\lim_{\delta \downarrow 0} \limsup_{n \rightarrow \infty} \mathfrak{H}_K(W_n, \delta) = 0.$$

It has no hidden-scale variance if this holds on every compact macroscopic annular cylinder.

Lemma 9.28 (Hidden-scale variance is the Young obstruction). *Let W_n be uniformly bounded in L_{loc}^∞ on macroscopic annular cylinders and suppose*

$$W_n \rightharpoonup^* \bar{W} \quad \text{in } L_{\text{loc}}^\infty.$$

On a compact annular cylinder K , the following are equivalent:

- (i) $W_n \rightarrow \bar{W}$ strongly in $L^3(K)$;
- (ii) the generated Young measure is Dirac on K :

$$\mathcal{Y}_{x,s} = \delta_{\bar{W}(x,s)} \quad \text{for a.e. } (x, s) \in K;$$

- (iii) W_n has no hidden-scale variance on K .

Consequently a realized Young critical tail is precisely a failure of the hidden-scale variance condition on some compact annular cylinder.

Proof. The uniform $L^\infty(K)$ bound gives a uniform $L^3(K)$ bound. The Fréchet–Kolmogorov compactness criterion on $L^3(K)$, applied after extending the functions to a fixed slightly larger cylinder, says that this bounded family is relatively compact in $L^3(K)$ if and only if the translation modulus in Theorem 9.27 tends to zero. Thus (iii) is equivalent to relative compactness in $L^3(K)$. Any strong subsequential limit must agree with the weak-* limit $\bar{W}$, so relative compactness is equivalent to strong convergence to $\bar{W}$. This proves (i) $\Leftrightarrow$ (iii).

Strong $L^3(K)$ convergence implies convergence in measure, hence the associated Young measure is $\delta_{\bar{W}}$. Conversely, if the generated Young measure is Dirac and $\{W_n\}$ is uniformly bounded in

$L^\infty(K)$, then $W_n \rightarrow \bar{W}$ in measure on K . Uniform boundedness gives uniform integrability of $|W_n - \bar{W}|^3$, so Vitali convergence yields strong $L^3(K)$ convergence. Hence (i) $\Leftrightarrow$ (ii). $\square$

Definition 9.29 (Multiscale affine oscillation). Let (U_n, Π_n) be a terminal sequence. For a covariant observer $z = (\xi, \sigma)$ and $r \geq 1$, define the critical-tail cylinder

$$\mathcal{Q}_{r,\text{ct}}^{\text{cov}}(z) := \mathcal{R}_z(B_r(0) \times (-1, 1)).$$

Define

$$\mathcal{A}_n(z, r) := r^{-3} \inf_{c \in L^\infty((-1, 1); \mathbb{R}^3)} \int_{-1}^1 \int_{B_r} |(\mathcal{R}_z U_n)(y, s) - c(s)|^3 dy ds.$$

The exponent 3 is the spatial dimension; the centered time interval is kept at fixed size because critical-tail blow-downs are spatial/logarithmic blow-downs, not parabolic rescalings. Allowing the minimizing constant to depend on time removes the harmless spatially homogeneous affine mode. Unit active-locus removal implies that $\sup_z \mathcal{A}_n(z, 1)$ is below the retained oscillation threshold on the residual exterior window, after the fixed cover constants are absorbed into the threshold.

Definition 9.30 (First bad mesoscopic scale). Let $R_n \rightarrow \infty$ be the macroscopic scale of a tight critical-tail blow-down. A first bad mesoscopic scale is a sequence of covariant centers z_n , radii r_n , and a number $\varepsilon_0 > 0$ such that

$$1 \ll r_n \ll R_n, \quad \mathcal{A}_n(z_n, r_n) \geq \varepsilon_0,$$

and, after replacing r_n by a dyadic comparable scale if necessary,

$$\sup_z \sup_{1 \leq r \leq r_n/2} \mathcal{A}_n(z, r) < \varepsilon_0/2$$

on the selected residual exterior window. Thus all smaller retained subscales are affine-flat, while the scale r_n is the first scale at which non-affine oscillation is visible.

Lemma 9.31 (Finite-overlap control of affine oscillation). *Let $I \Subset \mathbb{R}$, let $f \in L_{\text{loc}}^\infty(\mathbb{R}^3 \times I)$ with $\|f\|_{L^\infty} \leq M$, and let*

$$B_1(a_0), B_1(a_1), \dots, B_1(a_N)$$

be a finite chain of unit balls such that

$$|B_1(a_j) \cap B_1(a_{j+1})| \geq \kappa > 0 \quad (j = 0, \dots, N-1).$$

Assume that for each j there exists $c_j \in L^\infty(I; \mathbb{R}^3)$, $|c_j| \leq M$, such that

$$\int_I \int_{B_1(a_j)} |f(y, s) - c_j(s)|^3 dy ds \leq \delta.$$

Then, for $0 \leq j, k \leq N$,

$$(9.4) \quad \int_I |c_j(s) - c_k(s)|^3 ds \leq C(N, \kappa) \delta,$$

and

$$(9.5) \quad \int_I \int_{B_1(a_j)} |f(y, s) - c_k(s)|^3 dy ds \leq C(N, \kappa) \delta.$$

Consequently, if all unit cylinders in a bounded chain have affine-normalized oscillation $o(1)$, then all their local affine gauges agree up to $o(1)$ in $L^3(I)$.

Proof. For neighboring indices put $E_j = B_1(a_j) \cap B_1(a_{j+1})$. Since $|E_j| \geq \kappa$, the pointwise inequality

$$|c_j(s) - c_{j+1}(s)|^3 \leq 4\kappa^{-1} \int_{E_j} (|f(y, s) - c_j(s)|^3 + |f(y, s) - c_{j+1}(s)|^3) dy$$

holds for a.e. s . Integrating in s gives

$$\int_I |c_j - c_{j+1}|^3 ds \leq C_\kappa \delta.$$

Summing along the chain and using

$$\left| \sum_{\ell=j}^{k-1} (c_\ell - c_{\ell+1}) \right|^3 \leq |j - k|^2 \sum_{\ell=j}^{k-1} |c_\ell - c_{\ell+1}|^3$$

proves (9.4). Finally,

$$|f - c_k|^3 \leq 4|f - c_j|^3 + 4|c_j - c_k|^3$$

on $B_1(a_j) \times I$, and (9.5) follows from (9.4). $\square$

Lemma 9.32 (Spacetime finite-overlap translation control). *Let $K \Subset \mathbb{R}^3 \times I$, let $(h_n, \theta_n) \rightarrow 0$, and let f_n be uniformly bounded in L^∞ on a fixed neighborhood of*

$$K \cup (K + (h_n, \theta_n)).$$

Assume that every unit covariant cylinder in a finite spacetime chain covering the segment joining $(x, s) \in K$ to $(x + h_n, s + \theta_n)$ has affine-normalized oscillation $o(1)$. Then there are spatially homogeneous functions $a_n, b_n \in L^\infty(I; \mathbb{R}^3)$ such that

$$\|f_n - a_n\|_{L^3(K)} + \|f_n(\cdot + h_n, \cdot + \theta_n) - b_n\|_{L^3(K)} = o(1),$$

and

$$\|f_n(\cdot + h_n, \cdot + \theta_n) - f_n - (b_n - a_n)\|_{L^3(K)} = o(1).$$

Consequently, if the translated difference has a positive $L^3(K)$ lower bound while no unit active observer is present, then that lower bound is carried by the spatially homogeneous time-dependent mode $b_n - a_n$. In the ordered residual decomposition this descendant belongs to the affine/parasitic stratum $\mathcal{C}_{\text{aff}}$.

Proof. Cover K and $K + (h_n, \theta_n)$ by finitely many unit spacetime cylinders contained in the fixed neighborhood. Refine the cover so that any two successive cylinders in the chain have spatial overlap bounded below and time overlap bounded below by constants depending only on K . On each cylinder choose a near-minimizing spatially homogeneous gauge. Applying Theorem 9.31 on each time-overlap interval and summing along the finite chain shows that all gauges on the chain agree in L^3 on their common time intervals up to $o(1)$. Gluing the finitely many local gauges by this overlap relation gives global-in-space functions $a_n(s)$ on the part of the chain covering K and $b_n(s)$ on the part covering $K + (h_n, \theta_n)$. The first displayed estimate follows from the local small oscillation and finite overlap. Subtracting the two approximations gives the second displayed estimate.

If the translated difference remains bounded below, then $\|b_n - a_n\|_{L^3(\text{proj}_s K)}$ stays bounded below. The corresponding descendant has zero affine-normalized spatial oscillation on every compact cylinder but a nonzero homogeneous time-dependent component. That is the affine/parasitic lower stratum in the ordered state-space decomposition. $\square$

Lemma 9.33 (Translation defect produces activity, affine collapse, or a bad radius). *Let $K \Subset \{0 < |x| < \infty\} \times I$, let*

$$W_n(x, s) = U_n(y_n + R_n x, \sigma_n + s) - c_n(s), \quad R_n \rightarrow \infty,$$

be a bounded macroscopic blow-down, and suppose that for some $(h_n, \theta_n) \rightarrow 0$ and $\eta_0 > 0$,

$$\|W_n(\cdot + h_n, \cdot + \theta_n) - W_n\|_{L^3(K)} \geq \eta_0.$$

Set

$$\ell_n := R_n(|h_n| + |\theta_n|^{1/2}).$$

Then, after passing to a subsequence, exactly one of the following alternatives occurs.

- (i) *A positive-threshold unit active observer is regenerated.*
- (ii) *The defect is carried only by a spatially homogeneous time-dependent affine mode; equivalently the descendant is assigned to $\mathcal{C}_{\text{aff}}$.*
- (iii) *$1 \ll \ell_n \ll R_n$, and there are covariant centers z_n , radii $r_n \sim \ell_n$, and $\varepsilon_0 = \varepsilon_0(\eta_0, K) > 0$ such that*

$$\mathcal{A}_n(z_n, r_n) \geq \varepsilon_0.$$

Proof. Since $(h_n, \theta_n) \rightarrow 0$, $\ell_n/R_n \rightarrow 0$. First assume ℓ_n is bounded. The two points compared in the original centered variables are then connected, on the compact set K , by a uniformly finite chain of covariant unit cylinders. If a unit cylinder in that chain has a positive affine-normalized oscillation lower bound, alternative (i) holds. Otherwise the affine oscillation on every cylinder in the chain tends to zero. By [Theorem 9.32](#), the displayed translation defect can persist only through a spatially homogeneous time-dependent mode. This is alternative (ii). Thus, if neither (i) nor (ii) occurs, $\ell_n \rightarrow \infty$.

In that remaining case $1 \ll \ell_n \ll R_n$. Cover K and its translated copy by finitely many balls of radius comparable to $|h_n| + |\theta_n|^{1/2}$ in macroscopic variables. Pulling this cover back to the centered variables gives covariant cylinders of radius comparable to ℓ_n . If all such cylinders had affine-normalized oscillation below a sufficiently small $\varepsilon_0 = \varepsilon_0(\eta_0, K)$, then [Theorem 9.32](#), after rescaling the finite overlap constants, would again force the translated fields to differ by only an affine/parasitic time mode. This is alternative (ii), contrary to the present case. Therefore one such cylinder has $\mathcal{A}_n(z_n, r_n) \geq \varepsilon_0$ with $r_n \sim \ell_n$, proving (iii). $\square$

Lemma 9.34 (Failure of hidden-scale compactness produces a first bad scale). *Let $W_n(x, s) = U_n(y_n + R_n x, \sigma_n + s) - c_n(s)$ be a tight macroscopic annular blow-down after active-locus removal. If W_n fails the hidden-scale variance condition on some compact annular cylinder K , then either a positive-threshold unit active observer is regenerated, the affine stratum $\mathcal{C}_{\text{aff}}$ is reached by a descendant, or there is a first bad mesoscopic scale in the sense of [Theorem 9.30](#).*

Proof. Failure of hidden-scale variance gives a compact annular cylinder K , a number $\eta_0 > 0$, and translations $(h_n, \theta_n) \rightarrow 0$ such that, after passing to a subsequence,

$$\|W_n(\cdot + h_n, \cdot + \theta_n) - W_n\|_{L^3(K)} \geq \eta_0.$$

By [Theorem 9.33](#), the translation defect either regenerates a unit active observer, reaches $\mathcal{C}_{\text{aff}}$, or gives centers z_n , radii ρ_n with $1 \ll \rho_n \ll R_n$, and $\varepsilon_0 > 0$ such that

$$\mathcal{A}_n(z_n, \rho_n) \geq \varepsilon_0.$$

In the last case, let r_n be the smallest dyadic radius in $[1, C\rho_n]$ for which

$$\sup_z \mathcal{A}_n(z, r_n) \geq \varepsilon_0$$

on the selected residual exterior window. Unit active-locus removal excludes $r_n = O(1)$, so $r_n \rightarrow \infty$. Since $r_n \lesssim \rho_n$ and $\rho_n/R_n \rightarrow 0$, one has $r_n/R_n \rightarrow 0$. By dyadic minimality, all smaller

scales up to $r_n/2$ have oscillation below $\varepsilon_0/2$, after slightly lowering ε_0 . This is a first bad mesoscopic scale. $\square$

Definition 9.35 (First-bad spatial blow-down). Let z_n and r_n be a first bad mesoscopic scale. Write

$$U_n^{z_n} := \mathcal{R}_{z_n} U_n, \quad \Pi_n^{z_n} := \mathcal{R}_{z_n} \Pi_n.$$

Choose measurable velocity gauges $c_n : (-T, T) \rightarrow \mathbb{R}^3$, with $|c_n| \leq M$, which are minimizing, or within $o(1)$ of minimizing, the affine oscillation on each compact time interval. The first-bad spatial blow-down is

$$V_n(x, s) := U_n^{z_n}(r_n x, s), \quad W_n(x, s) := V_n(x, s) - c_n(s).$$

Thus V_n is the ungauged blow-down and W_n is the affine-normalized velocity used to measure non-affine oscillation. The associated scaled pressure is

$$Q_n(x, s) := r_n^{-1} \Pi_n^{z_n}(r_n x, s),$$

modulo the admissible addition of affine functions of x and arbitrary functions of s . Affine pressure terms are always paired with the corresponding spatially homogeneous velocity gauge in the weak formulation; no pointwise time derivative of c_n is required.

Lemma 9.36 (First-bad blow-down equation). *Let V_n, W_n, Q_n, c_n be the first-bad blow-downs of Theorem 9.35. On every compact cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$, V_n and W_n are divergence-free, $\|V_n\|_{L^\infty(K)} \leq M$, $\|W_n\|_{L^\infty(K)} \leq 2M$, and the ungauged field satisfies*

$$(9.6) \quad \partial_s V_n - r_n^{-2} \Delta V_n + \frac{1}{2} x \cdot \nabla V_n + \frac{1}{2} V_n + r_n^{-1} (V_n \cdot \nabla) V_n + \nabla Q_n = 0$$

in distributions. The affine-normalized field $W_n = V_n - c_n(s)$ is used for oscillation and Young-measure compactness. Its equation is understood only in the quotient sense made precise in Theorem 9.37; no pointwise derivative of c_n and no $L_{\text{loc}}^{3/2}$ affine pressure for W_n is assumed. Moreover

$$(9.7) \quad -\Delta Q_n = r_n^{-1} \partial_i \partial_j (V_{n,i} V_{n,j})$$

in distributions, after the same gauges.

Proof. The covariant recentering $\mathcal{R}_{z_n}$ is an exact symmetry of the centered equation, so $(U_n^{z_n}, \Pi_n^{z_n})$ solves (1.2). Since centered time is not rescaled,

$$\begin{aligned} \partial_s V_n &= (\partial_\tau U_n^{z_n})(r_n x, s), & \Delta_y U_n^{z_n} &= r_n^{-2} \Delta_x V_n, \\ y \cdot \nabla_y U_n^{z_n} &= x \cdot \nabla_x V_n, & (U_n^{z_n} \cdot \nabla_y) U_n^{z_n} &= r_n^{-1} (V_n \cdot \nabla_x) V_n. \end{aligned}$$

If $Q_n = r_n^{-1} \Pi_n^{z_n}(r_n x, s)$, then

$$\nabla_x Q_n = (\nabla_y \Pi_n^{z_n})(r_n x, s).$$

Substituting these identities gives the scaled equation for V_n . We do not treat $W_n = V_n - c_n(s)$ as carrying an ordinary pressure unless the gauges have additional time regularity. Instead, all PDE identities below are first written for the ungauged field V_n , and the passage to W_n is made only after quotienting out spatially homogeneous modes. Taking the divergence of the ungauged scaled equation gives (9.7); the linear drift has zero divergence contribution because V_n is divergence-free. The L^∞ bounds follow from $\|U_n\|_{L^\infty} \leq M$ and $|c_n| \leq M$. $\square$

Lemma 9.37 (First-bad quotient formulation). *Let V_n, W_n, c_n, Q_n be as in Theorem 9.36. The first-bad analysis uses the following quotient formulation.*

- (i) *The distributional equation and local energy inequality are always applied to the ungauged field V_n and its pressure Q_n .*

- (ii) The affine-normalized field $W_n = V_n - c_n(s)$ is used only for oscillation functionals and Young measures, which are invariant under spatially homogeneous shifts after the gauge sequence is either compact or routed to $\mathcal{C}_{\text{aff}}$.
- (iii) If c_n is not compact in L^3_{loc} in time, the homogeneous gauge sequence itself generates an affine/parasitic descendant. If $c_n \rightarrow c$ strongly in L^3_{loc} , all Young-measure variances and strong compactness statements pass from V_n to W_n by deterministic translation of the generated Young measures.

Thus the symbol “affine pressure paired with c_n ” means a quotient functional, not an asserted $L^{3/2}_{\text{loc}}$ pressure unless the time regularity of c_n has been separately proved.

Proof. Statement (i) is exactly the construction in [Theorem 9.36](#). The local energy inequality is stable under the first-bad spatial blow-down for V_n , because V_n is obtained from an actual centered solution by an exact covariant recentering and a spatial scaling; no subtraction of c_n is involved.

For (ii) and (iii), observe that subtracting a spatially homogeneous function does not change any affine-normalized oscillation seminorm in which the same class of homogeneous gauges is minimized over. If the sequence c_n is not compact in L^3_{loc} , [Theorem 9.41](#) places the resulting homogeneous descendant in $\mathcal{C}_{\text{aff}}$. If $c_n \rightarrow c$ strongly, then [Theorem 9.42](#) identifies the Young measure of W_n with the deterministic translate of the Young measure of V_n , and the variance densities agree. Hence all compactness and variance conclusions used for W_n follow from identities proved for V_n , without requiring $\partial_s c_n$ to be a function. $\square$

Proposition 9.38 (Normalized pressure representation modulo affine parasitic modes). *Every normalized Seregin, terminal, covariant descendant, first-bad blow-down, or active descendant profile used in the residual state space is represented by a pressure of the form*

$$(9.8) \quad \Pi(\cdot, s) = \mathcal{R}_i \mathcal{R}_j (U_i U_j)(\cdot, s) + \beta(s) \cdot y + \alpha(s),$$

where α is an arbitrary time gauge and $\beta(s) \cdot y$ is exactly the affine pressure paired with the spatially homogeneous parasitic velocity mode. Outside the affine/parasitic stratum $\mathcal{C}_{\text{aff}}$, the representative is taken with $\beta = 0$.

Proof. At the finite-energy Leray level, choose the standard whole-space pressure representative. For the approximating suitable Leray–Hopf solutions one has, modulo a function of time,

$$p_n = \mathcal{R}_i \mathcal{R}_j (u_{n,i} u_{n,j})$$

in $\mathcal{D}'(\mathbb{R}^3)$. This formula is stable under every operation used before the affine quotient: physical Type I scaling, the change to centered variables, centered time translations, spatial translations, covariant translations, and local smooth limits after the admissible time gauges. Indeed each operation either conjugates the Riesz representative by a Euclidean/scaling change of variables or adds only a time-dependent gauge. Therefore no harmonic pressure component with nonzero Hessian can be generated from the finite-energy approximants.

The only operation that changes the velocity by a spatially homogeneous vector is the affine/parasitic normalization. For a smooth homogeneous mode $b(s)$, substitution into the centered equation gives the affine pressure gradient

$$\nabla_y [(-\partial_s b - \tfrac{1}{2}b) \cdot y].$$

For merely bounded measurable homogeneous gauges this expression is interpreted only as the quotient distribution described in [Theorems 8.14](#) and [9.37](#). Thus the only non-Riesz pressure direction admitted in the normalized state space is the affine term $\beta(s) \cdot y + \alpha(s)$.

Conversely, suppose a normalized limit contained a residual harmonic pressure component $h(\cdot, s)$ which was not affine in y . Then $\nabla^2 h \not\equiv 0$ on some compact ball and time interval.

Testing the approximating pressure identities against compactly supported functions with zero spatial mean and passing to the local pressure-gauge limit would force this nonzero Hessian component to arise as a limit of zero Hessian components of the allowed gauges, which is impossible. Equivalently, the difference between the limit pressure and the Riesz representative is harmonic and is produced by the allowed gauges; the only such gauges in the normalized state space are time-dependent constants and affine parasitic terms. Hence (9.8) is the full pressure representation.

Finally, when $\beta \neq 0$ has a retained effect, the associated velocity component is spatially homogeneous and belongs to the affine/parasitic lower stratum $\mathcal{C}_{\text{aff}}$. Outside that stratum the canonical parasitic-free representative has $\beta = 0$. $\square$

Lemma 9.39 (Scaled pressure compactness modulo affine gauges). *Let (U_n, Π_n) be bounded centered solutions with $\|U_n\|_{L^\infty} \leq M$, let z_n be covariant centers, let $r_n \rightarrow \infty$, and set*

$$\tilde{U}_n(y, s) := (\mathcal{R}_{z_n} U_n)(y, s), \quad \tilde{\Pi}_n(y, s) := (\mathcal{R}_{z_n} \Pi_n)(y, s).$$

For

$$\tilde{Q}_n(x, s) := r_n^{-1} \tilde{\Pi}_n(r_n x, s)$$

and every compact cylinder $K = B_R \times I \Subset \mathbb{R}^3 \times \mathbb{R}$, there are affine functions

$$L_n(x, s) = b_n(s) \cdot x + a_n(s)$$

such that

$$(9.9) \quad \tilde{Q}_n - L_n \rightarrow 0 \quad \text{strongly in } L^{3/2}(K).$$

The same statement holds for the first-bad pressures Q_n in the quotient sense of [Theorem 9.37](#). If the affine parts L_n carry a nonzero retained descendant, then that descendant lies in the affine/parasitic stratum $\mathcal{C}_{\text{aff}}$. Otherwise the affine parts may be discarded in the affine-normalized quotient, and the non-affine pressure converges strongly to zero.

Proof. The covariant recentering is an exact symmetry, so it suffices to work with a single bounded centered solution (U, Π) . For a.e. s , the pressure Poisson equation is

$$-\Delta \Pi(\cdot, s) = \partial_i \partial_j (U_i U_j)(\cdot, s)$$

in $\mathbb{R}^3$. Let

$$\Pi^{\text{loc}}(\cdot, s) := \mathcal{R}_i \mathcal{R}_j (U_i U_j)(\cdot, s)$$

be the whole-space Calderón–Zygmund representative. Since $\|U\|_{L^\infty} \leq M$,

$$\|\Pi^{\text{loc}}(\cdot, s)\|_{\text{BMO}(\mathbb{R}^3)} \leq CM^2$$

uniformly in s . Hence, by the John–Nirenberg inequality, for every ball B_ρ ,

$$\inf_{a \in \mathbb{R}} \|\Pi^{\text{loc}}(\cdot, s) - a\|_{L^{3/2}(B_\rho)} \leq CM^2 |B_\rho|^{2/3}.$$

After the scaling $x \mapsto r_n x$, this gives

$$\inf_{a_n(s)} \|r_n^{-1} \Pi^{\text{loc}}(r_n \cdot, s) - a_n(s)\|_{L^{3/2}(B_R)} \leq C_{R,M} r_n^{-1} \rightarrow 0.$$

By [Theorem 9.38](#), the only remaining pressure component in the normalized state space is

$$\beta(s) \cdot y + \alpha(s).$$

After scaling,

$$r_n^{-1} (\beta(s) \cdot r_n x + \alpha(s)) = \beta(s) \cdot x + r_n^{-1} \alpha(s),$$

which is an affine function of x . Taking this function as L_n , the remaining pressure is exactly the scaled Leray representative already estimated above, and it tends to zero strongly in $L^{3/2}(B_R \times I)$.

This proves (9.9). The first-bad pressure is used only up to the affine quotient described in Theorem 9.37; in that quotient the same estimate applies. $\square$

Lemma 9.40 (Pressure compactness at a first bad scale). *Let Q_n be as in Theorem 9.36. For every compact cylinder $K \Subset \mathbb{R}^3 \times \mathbb{R}$, exactly one of the following holds:*

- (i) *the affine pressure/velocity gauges produce a retained descendant in $\mathcal{C}_{\text{aff}}$;*
- (ii) *after discarding those affine gauges in the affine-normalized quotient, there exists a pressure limit Q such that*

$$Q_n \rightarrow Q \quad \text{strongly in } L^{3/2}(K).$$

In fact, in the non-affine quotient one may take $Q = 0$. Consequently, if $W_n \rightharpoonup^ \bar{W}$ in L_{loc}^∞ , then*

$$Q_n W_n \rightharpoonup Q \bar{W} \quad \text{in distributions on } K.$$

Proof. This is Theorem 9.39 applied to the first-bad scaling. If the affine gauges retain nonzero descendant activity, we are in alternative (i). Otherwise the affine gauges are precisely the quotient directions removed by affine normalization, and the remaining pressure converges strongly to 0 in $L^{3/2}(K)$. Strong convergence of the pressure and weak-* convergence of the uniformly bounded velocity give the product convergence in distributions. $\square$

Lemma 9.41 (Homogeneous gauge compactness or affine stratum). *Let $c_n : I \rightarrow \mathbb{R}^3$ be uniformly bounded in $L^\infty(I)$ on every compact interval $I \Subset \mathbb{R}$. Then either c_n is precompact in $L_{\text{loc}}^3(\mathbb{R}; \mathbb{R}^3)$ after passing to a subsequence, or a time recentering of the spatially homogeneous profiles*

$$C_n(y, s) := c_n(s)$$

generates a nonzero affine/parasitic descendant. In the ordered residual decomposition the second alternative belongs to $\mathcal{C}_{\text{aff}}$.

Proof. By the Fréchet–Kolmogorov criterion on compact time intervals, failure of precompactness in L_{loc}^3 is equivalent, after passing to a subsequence, to the existence of an interval I , translations $\theta_n \rightarrow 0$, and $\eta > 0$ such that

$$\|c_n(\cdot + \theta_n) - c_n\|_{L^3(I)} \geq \eta.$$

The associated spatially homogeneous profiles $C_n(y, s) = c_n(s)$ have zero affine-normalized spatial oscillation on every compact cylinder, while the displayed lower bound gives a nonzero homogeneous time-dependent descendant. This is exactly the affine/parasitic lower stratum. If this alternative does not occur, the time translation modulus of c_n vanishes on every compact interval; boundedness then gives L_{loc}^3 precompactness. $\square$

Lemma 9.42 (Variance is invariant under homogeneous velocity gauges). *Let V_n be uniformly bounded in L_{loc}^∞ , let $c_n \rightarrow c$ strongly in $L_{\text{loc}}^3(\mathbb{R})$, and set*

$$W_n(x, s) := V_n(x, s) - c_n(s).$$

If V_n generates a Young measure $\nu_{x,s}$ with barycenter $\bar{V}$, then W_n generates the translated Young measure

$$\tilde{\nu}_{x,s}(A) = \nu_{x,s}(A + c(s))$$

with barycenter $\bar{W} = \bar{V} - c(s)$. The variance densities agree:

$$\langle |\xi - \bar{V}(x, s)|^2 \rangle_{\nu_{x,s}} = \langle |\xi - \bar{W}(x, s)|^2 \rangle_{\tilde{\nu}_{x,s}}.$$

Consequently any distributional variance inequality proved for V_n passes unchanged to the affine-normalized fields W_n .

Proof. After passing to a subsequence, $c_n(s) \rightarrow c(s)$ a.e. and in measure on every compact time interval. For every continuous compactly supported function $F(\xi)$ and every compactly supported test function $\varphi(x, s)$,

$$\iint \varphi F(W_n) dx ds = \iint \varphi F(V_n - c_n(s)) dx ds \rightarrow \iint \varphi \int F(\xi - c(s)) d\nu_{x,s}(\xi) dx ds.$$

This is precisely the translated Young measure. Translation by a deterministic vector preserves centered second moments, giving the variance identity. The distributional inequality involves only this variance density and therefore is unchanged. $\square$

Lemma 9.43 (Barycenter equation and variance inequality). *Let V_n, W_n, Q_n, c_n be a first-bad blow-down. Then either the sequence has a retained affine/parasitic descendant in $\mathcal{C}_{\text{aff}}$, or, after passing to a subsequence, $c_n \rightarrow c$ strongly in L^3_{loc} ,*

$$V_n \rightharpoonup^* \bar{V}, \quad W_n \rightharpoonup^* \bar{W} \quad \text{in } L^\infty_{\text{loc}},$$

and V_n, W_n generate Young measures $\nu_{x,s}$ and $\tilde{\nu}_{x,s}$, respectively. Set

$$E(x, s) := \langle |\xi|^2 \rangle_{\tilde{\nu}_{x,s}}, \quad \mathcal{V}(x, s) := E(x, s) - |\bar{W}(x, s)|^2.$$

Then $\bar{V}$ satisfies the coherent transport equation with pressure Q . The affine-normalized barycenter $\bar{W} = \bar{V} - c(s)$ satisfies the same equation only as a quotient modulo spatially homogeneous modes; after passing to the parasitic-free representative outside $\mathcal{C}_{\text{aff}}$, this quotient equation is represented as

$$(9.10) \quad \partial_s \bar{W} + \frac{1}{2} x \cdot \nabla \bar{W} + \frac{1}{2} \bar{W} + \nabla Q = 0, \quad \nabla \cdot \bar{W} = 0,$$

and the nonnegative variance density $\mathcal{V} \in L^\infty_{\text{loc}}$ satisfies

$$(9.11) \quad \partial_s \mathcal{V} + \frac{1}{2} x \cdot \nabla \mathcal{V} + \mathcal{V} \leq 0$$

in distributions.

Proof. By [Theorem 9.41](#), either the homogeneous gauges generate an affine/parasitic descendant, or $c_n \rightarrow c$ strongly in L^3_{loc} after passing to a subsequence. We work in the second case. The weak-* compactness of V_n, W_n and generation of Young measures are standard consequences of the uniform L^∞_{loc} bound. Testing [\(9.6\)](#) against a compactly supported divergence-free vector field, the viscous term is bounded by

$$Cr_n^{-2} \|W_n\|_{L^\infty},$$

and the nonlinear term by

$$Cr_n^{-1} \|V_n\|_{L^\infty}^2.$$

Both tend to zero. The pressure compactness in [Theorem 9.40](#) gives either a retained descendant in $\mathcal{C}_{\text{aff}}$, in which case we are done, or the pressure gradient limit in the affine-normalized quotient. Thus $\bar{V}$ satisfies the coherent transport equation. Since $\bar{W} = \bar{V} - c(s)$, passing from $\bar{V}$ to $\bar{W}$ changes only the spatially homogeneous quotient component. We do not differentiate c as a function. Instead, [Theorem 9.37](#) says that this is an affine/parasitic quotient direction. If it has retained effect, the profile is in $\mathcal{C}_{\text{aff}}$; otherwise the canonical parasitic-free representative removes it, and the representative satisfies [\(9.10\)](#).

We next pass to the local energy inequality for the ungauged fields V_n . Dot [\(9.6\)](#) with V_n . In distributional local-energy form, using $\nabla \cdot V_n = 0$, one obtains

$$\begin{aligned} \partial_s \frac{|V_n|^2}{2} - r_n^{-2} \Delta \frac{|V_n|^2}{2} + r_n^{-2} |\nabla V_n|^2 + \frac{1}{2} x \cdot \nabla \frac{|V_n|^2}{2} + \frac{1}{2} |V_n|^2 + \text{div}(Q_n V_n) \\ = -r_n^{-1} \text{div} \left(\frac{1}{2} |V_n|^2 V_n \right). \end{aligned}$$

There is no affine pressure work term in this identity; the homogeneous gauge has not been subtracted. When tested against nonnegative compactly supported scalar functions, the viscous diffusion and nonlinear-flux terms vanish in the limit because of the factors r_n^{-2} and r_n^{-1} , while the dissipation term is nonnegative and may be dropped. The strong pressure convergence gives

$$\operatorname{div}(Q_n V_n) \rightarrow \operatorname{div}(Q \bar{V}).$$

Therefore

$$\partial_s \frac{E_V}{2} + \frac{1}{2} x \cdot \nabla \frac{E_V}{2} + \frac{1}{2} E_V + \operatorname{div}(Q \bar{V}) \leq 0$$

in distributions, where

$$E_V(x, s) := \langle |\xi|^2 \rangle_{\nu_{x,s}}$$

is the second moment of the Young measure generated by V_n . On the other hand, dotting the barycenter equation for $\bar{V}$ with $\bar{V}$ gives

$$\partial_s \frac{|\bar{V}|^2}{2} + \frac{1}{2} x \cdot \nabla \frac{|\bar{V}|^2}{2} + \frac{1}{2} |\bar{V}|^2 + \operatorname{div}(Q \bar{V}) = 0.$$

Subtracting gives the variance inequality for the variance of V_n . By [Theorem 9.42](#), the same variance density is $\mathcal{V}$, the variance of W_n . This proves [\(9.11\)](#). $\square$

Lemma 9.44 (Bounded ancient variance Liouville theorem). *Let $F \in L_{\text{loc}}^\infty(\mathbb{R}^3 \times \mathbb{R})$ satisfy*

$$0 \leq F \leq C < \infty$$

and

$$\partial_s F + \frac{1}{2} x \cdot \nabla F + F \leq 0$$

in distributions on $\mathbb{R}^3 \times \mathbb{R}$. Then $F \equiv 0$.

Proof. Mollify F in space-time and apply the standard renormalized characteristic argument; then let the mollification parameter tend to zero. Equivalently, test the distributional inequality against the adjoint transport kernel. Along a backward characteristic

$$x(\sigma) = e^{(\sigma-s)/2} x, \quad \sigma \leq s,$$

the inequality gives, for every $T > 0$,

$$F(x, s) \leq e^{-T} F(e^{-T/2} x, s - T)$$

at Lebesgue points of F . Since $F \leq C$, we obtain

$$F(x, s) \leq C e^{-T}.$$

Letting $T \rightarrow \infty$ gives $F(x, s) = 0$ at every Lebesgue point, hence almost everywhere. The distributional inequality then gives $F \equiv 0$. $\square$

Lemma 9.45 (First-bad blow-downs are strongly compact). *Every first-bad blow-down W_n either has a retained affine/parasitic descendant in $\mathcal{C}_{\text{aff}}$, or has, after passing to a subsequence, a limit $\bar{W}$ such that*

$$W_n \rightarrow \bar{W} \quad \text{strongly in } L_{\text{loc}}^3(\mathbb{R}^3 \times \mathbb{R}).$$

The limit satisfies [\(9.10\)](#).

Proof. By [Theorem 9.43](#), either the affine/parasitic alternative has occurred, or the Young-measure variance $\mathcal{V}$ satisfies the hypotheses of [Theorem 9.44](#). Hence $\mathcal{V} \equiv 0$. Thus

$$\tilde{\nu}_{x,s} = \delta_{\bar{W}(x,s)} \quad \text{for a.e. } (x, s).$$

Uniform L_{loc}^∞ boundedness then upgrades convergence in measure to strong convergence in L_{loc}^3 . The limit equation has already been proved in [Theorem 9.43](#). $\square$

Theorem 9.46 (First bad mesoscopic scale routing). *Every first bad mesoscopic scale generated by a realized recurrent critical-tail extraction after active-locus removal and affine normalization has a descendant in*

$$\mathcal{C}_{\text{active}} \cup \mathcal{C}_{\text{aff}} \cup \mathcal{C}_{\text{cohcrit}}.$$

Proof. Let z_n, r_n be a first bad mesoscopic scale. If, before passing to the first-bad blow-down, some unit covariant observer in the selected residual window carries a positive affine-normalized oscillation threshold, then the active alternative is regenerated and the descendant lies in $\mathcal{C}_{\text{active}}$. We may therefore assume unit activity has been removed at all fixed positive thresholds.

Form the first-bad blow-down W_n of [Theorem 9.35](#). By [Theorem 9.45](#), either the affine/parasitic stratum $\mathcal{C}_{\text{aff}}$ is reached, or, after passing to a subsequence,

$$W_n \rightarrow W \quad \text{strongly in } L_{\text{loc}}^3,$$

and (W, Q) satisfies the coherent critical-tail transport equation

$$\partial_s W + \frac{1}{2} x \cdot \nabla W + \frac{1}{2} W + \nabla Q = 0, \quad \nabla \cdot W = 0.$$

The first-bad lower bound and strong convergence give

$$\inf_{c \in L^\infty((-1,1);\mathbb{R}^3)} \int_{-1}^1 \int_{B_1} |W - c(s)|^3 dx ds \geq \varepsilon_0.$$

Thus the limit is not spatially affine/parasitic on the unit critical-tail window. If this non-affine lower bound were lost in a further descendant, the descendant would be assigned to $\mathcal{C}_{\text{aff}}$ by [Theorem 8.19](#). Otherwise the strongly realized nonzero limit is a coherent critical-tail descendant, hence belongs to $\mathcal{C}_{\text{cohcrit}}$. Therefore every first bad mesoscopic scale routes to $\mathcal{C}_{\text{active}}$, $\mathcal{C}_{\text{aff}}$, or $\mathcal{C}_{\text{cohcrit}}$, as claimed. $\square$

Remark 9.47 (Critical-tail routing theorem). [Theorem 9.46](#) is the critical-tail alternative-routing theorem proved above. It is a local multiscale rigidity statement: oscillation first appearing at an intermediate scale $1 \ll r_n \ll R_n$ must either regenerate an active observer, collapse into the affine/parasitic lower stratum, or produce a coherent critical tail. The log-diffuse alternative has been reduced above to the same hidden-scale compactness statement through tight log-window descendants.

Theorem 9.48 (No hidden-scale variance from first bad scale routing). *Every tight macroscopic annular blow-down generated by a realized recurrent critical-tail extraction after active-locus removal and affine normalization has no hidden-scale variance on compact macroscopic annular cylinders.*

Proof. Suppose that hidden-scale variance fails. By [Theorem 9.34](#), either a unit active observer is regenerated, the affine stratum is reached, or a first bad mesoscopic scale exists. The first alternative contradicts active-locus removal. The affine alternative is removed by [Theorem 9.53](#). In the third alternative, [Theorem 9.46](#) gives a descendant in $\mathcal{C}_{\text{active}} \cup \mathcal{C}_{\text{aff}} \cup \mathcal{C}_{\text{cohcrit}}$. The active case again contradicts active-locus removal; the affine case is removed by [Theorem 9.53](#); and the coherent critical-tail case is closed by [Theorem 9.59](#). Thus all alternatives are impossible, and hidden-scale variance cannot fail. $\square$

Theorem 9.49 (Critical-tail compactification extraction). *Let $(U_n, \Pi_n) \subset \mathfrak{X}_M$ be a bounded normalized terminal residual sequence, $\|U_n\|_{L^\infty} \leq M$. Assume that, after removal of all affine-normalized active covariant observers, there remains a nonzero diffuse exterior oscillation defect. Then, after passing to a subsequence, the sequence generates a nonzero element*

$$\mathcal{T} = (\bar{W}, \mathcal{Y}, \Lambda) \in \mathfrak{T}_M.$$

Moreover, the barycentric velocity satisfies the linear critical-tail transport law

$$\partial_s \bar{W} + \frac{1}{2} x \cdot \nabla \bar{W} + \frac{1}{2} \bar{W} + \nabla Q = 0, \quad \nabla \cdot \bar{W} = 0$$

in distributions on macroscopic annular cylinders, and the full defect $(\mathcal{Y}, \Lambda)$ is transported by the induced log-dilation flow.

Proof. Let μ_n^{osc} be the affine-normalized oscillation defect measure after active-locus removal. By the mixed-diffuse hypothesis there are escaping observer windows E_n and $m_0 > 0$ such that

$$\mu_n^{\text{osc}}(E_n) \geq m_0.$$

Normalize

$$\lambda_n := \frac{\mu_n^{\text{osc}} \lfloor E_n}{\mu_n^{\text{osc}}(E_n)}.$$

By [Theorem 9.13](#), after passing to a subsequence these probability measures converge to a boundary diffuse-defect state. Because all positive-threshold active observers have been removed, the limit gives no mass to ordinary unit active observers in Z .

We now compactify the critical boundary without imposing annular tightness. Apply Prokhorov compactness first in the observer compactification and then in the logarithmic radial compactification. If the normalized defect is tight in a fixed multiplicative annulus, select macroscopic scales $R_n \rightarrow \infty$, observers z_n , affine gauges $c_n(s)$, and pressure gauges. On fixed annular cylinders set, schematically,

$$W_n(x, s) = U_n(y_n + R_n x, \sigma_n + s) - c_n(s).$$

The uniform bound $\|U_n\|_{L^\infty} \leq M$ gives weak-* compactness of W_n in L_{loc}^∞ , and the fundamental theorem of Young measures gives, after passing to a further subsequence, a barycenter $\bar{W}$ and Young measure $\mathcal{Y}_{x,s}$. If the sequence is strongly compact in L_{loc}^3 , then $\mathcal{Y}_{x,s} = \delta_{\bar{W}(x,s)}$; otherwise $\mathcal{Y}$ records the unresolved oscillation defect. If the defect is not tight in any fixed logarithmic annulus, the escaping logarithmic mass is recorded by Λ . These alternatives are exhaustive and define an element of $\mathfrak{T}_M$. Since the normalized defect mass is one, $\mathcal{T} \neq 0$.

It remains to identify the equation satisfied by the barycenter. After the change $y = y_n + R_n x$ and the admissible affine/pressure normalizations, the centered equation becomes, in distributions on fixed macroscopic annuli,

$$\partial_s W_n - R_n^{-2} \Delta_x W_n + R_n^{-1} W_n \cdot \nabla_x W_n + \frac{1}{2} x \cdot \nabla_x W_n + \frac{1}{2} W_n + \nabla_x Q_n = o(1).$$

For every compactly supported test function, the viscous and nonlinear contributions are bounded respectively by

$$C R_n^{-2} \|W_n\|_{L^\infty}, \quad C R_n^{-1} \|W_n\|_{L^\infty}^2,$$

and hence both tend to zero. The pressure gauges give distributional compactness of the remaining gradients on annuli. Passing to the weak-* barycentric limit yields the displayed linear critical-tail transport equation. Applying the same weak formulation to transported test functions shows that $(\mathcal{Y}, \Lambda)$ is transported by the induced log-dilation flow. This construction is local on macroscopic annuli and uses no global critical norm, CKN budget, or energy estimate. $\square$

Corollary 9.50 (Coherent extraction under annular tightness and strong compactness). *If the critical-tail defect $\mathcal{T} \in \mathfrak{T}_M$ produced by [Theorem 9.49](#) satisfies $\Lambda = 0$ and $\mathcal{Y}_{x,s} = \delta_{W(x,s)}$ for a.e. (x, s) , then there exist macroscopic scales, covariant observers, affine gauges, and pressure gauges for which the blown-down fields converge strongly in L_{loc}^3 and weak-* in L_{loc}^∞ to a nonzero coherent critical tail $(W, Q) \in \mathcal{C}_{\text{cohcrit}}$.*

Proof. This is the coherent subcase of [Theorem 9.49](#). The condition $\Lambda = 0$ rules out loss of mass across logarithmic scale, and the Dirac-valued Young measure gives strong L^3_{loc} convergence by uniform L^∞ boundedness. Nontriviality follows from the normalized affine oscillation mass. $\square$

Theorem 9.51 (Coherent critical-tail classification). *Let $(W, Q) \in \mathcal{C}_{\text{cohcrit}}$ be a coherent critical tail. In logarithmic radial variables*

$$x = e^\rho \omega, \quad Z(\rho, \omega, s) = e^\rho W(e^\rho \omega, s),$$

its compact log-translation/time-translation hull belongs to exactly one of

$$\mathcal{C}_{\text{affine}}, \quad \mathcal{C}_{\text{homcrit}}, \quad \mathcal{C}_{\text{logper}}, \quad \mathcal{C}_{\text{ap}}.$$

Here $\mathcal{C}_{\text{homcrit}}$ is the homogeneous (-1) -tail class, $\mathcal{C}_{\text{logper}}$ is the nonzero log-periodic tail class, and $\mathcal{C}_{\text{ap}}$ is the remaining aperiodic minimal log-translation hull class.

Proof. Let

$$\mathcal{H}(Z) := \overline{\{Z(\rho + a, \omega, s + b) : a, b \in \mathbb{R}\}}$$

be the compact logarithmic translation hull. If the affine-normalized oscillation vanishes on every compact log-cylinder, then the tail is affine/parasitic. Otherwise, if every element of $\mathcal{H}(Z)$ is independent of ρ , then W is (-1) -homogeneous and belongs to $\mathcal{C}_{\text{homcrit}}$. If the log-translation orbit has a nonzero period, then W is discretely self-similar at infinity and belongs to $\mathcal{C}_{\text{logper}}$. If neither alternative holds, compact-flow decomposition gives a minimal compact invariant subsystem with no nonzero log period; this is the aperiodic log-translation class $\mathcal{C}_{\text{ap}}$. These alternatives exhaust compact log-translation hulls. $\square$

Theorem 9.52 (Realized critical-tail rigidity by stratification). *Let $\mathcal{T} = (\bar{W}, \mathcal{Y}, \Lambda) \in \mathfrak{T}_M$ be a nonzero realized critical-tail state generated by a bounded normalized residual Navier–Stokes terminal sequence after terminal active-locus removal. Then exactly one of the following ordered state-space alternatives occurs:*

- (a) $\mathcal{T}$ regenerates affine-normalized local activity;
- (b) $\mathcal{T}$ is affine/parasitic;
- (c) $\mathcal{T}$ is log-diffuse;
- (d) $\mathcal{T}$ is a Young critical tail;
- (e) $\mathcal{T}$ is a coherent homogeneous critical tail;
- (f) $\mathcal{T}$ is a coherent log-periodic critical tail;
- (g) $\mathcal{T}$ is a coherent aperiodic critical tail.

In the ordered refined decomposition, the log-diffuse, Young, homogeneous, log-periodic, and aperiodic critical-tail alternatives are lower structured strata. Thus every nonzero realized critical-tail state which remains after these lower strata have been removed either regenerates local activity or collapses to the affine/parasitic lower stratum.

Proof. If the covariant observer orbit closure of $\mathcal{T}$ intersects an affine-normalized active set $\{U : \text{osc}_3(U; Q_1^{\text{cov}}) \geq \eta\}$ for some $\eta > 0$, then alternative (a) holds. We remove this case first.

If every affine-normalized oscillation observable vanishes on the state, then [Theorems 8.14](#) and [8.19](#) place the state in the affine/parasitic lower stratum. This is alternative (b). We therefore assume from now on that the state has nonzero non-affine diffuse content and does not regenerate unit-scale activity.

By [Theorem 9.49](#), the residual normalized diffuse mass is represented by the critical-tail compactification. Thus the boundary mass has only two mechanisms: mass escaping every bounded logarithmic annulus, recorded by Λ , and annular blow-down mass, recorded by $(\bar{W}, \mathcal{Y})$. If $\Lambda \neq 0$, the state is log-diffuse, giving (c). If $\Lambda = 0$ but $\mathcal{Y}_{x,s}$ is not a Dirac mass on a set of positive macroscopic measure, the state is a Young critical tail, giving (d).

The remaining case is coherent: $\Lambda = 0$ and $\mathcal{Y}_{x,s} = \delta_{\bar{W}(x,s)}$ a.e. Then [Theorem 9.50](#) gives a coherent critical tail $(\bar{W}, Q) \in \mathcal{C}_{\text{cohcrit}}$. Applying [Theorem 9.51](#) gives exactly one of the homogeneous, log-periodic, or aperiodic coherent alternatives, namely (e), (f), or (g), unless the affine/parasitic case already occurred. The alternatives are mutually exclusive after the priority order above: active regeneration is removed first, then affine collapse, then log-diffuse mass, then Young oscillation, and finally the coherent log-translation classification. The final assertion follows from this ordered classification. $\square$

9.4. Complete closure of lower alternatives. The critical-tail classification above is a classification theorem, not by itself an emptiness theorem. The complete residual decomposition is discharged by local lower-stratum closures together with the first-bad mesoscopic routing theorem [Theorem 9.46](#). That theorem first gives hidden-scale compactness through [Theorem 9.48](#). The log-diffuse alternative is reduced to hidden-scale compactness by the log-window descendant selection of [Theorem 9.25](#); the Young alternative is reduced to it by the Fréchet–Kolmogorov/Young-measure equivalence in [Theorem 9.28](#). The critical-tail routing theorem used in both reductions is [Theorem 9.46](#). These are local state-space statements about realized residual descendants; none is a global $L_t^\infty L_x^3$, finite-energy-tail, or global CKN-budget estimate.

Theorem 9.53 (Oscillatory entry normalization). *Let (U, Π) be a retained residual state after the earlier lower classes recorded in the setup package `proof_setup.tex` have been removed. Then exactly one of the following holds:*

- (i) *after an admissible affine symmetry S_c , the profile retains positive non-affine oscillatory activity:*

$$\Phi(S_c U) > 0;$$

- (ii) *the profile belongs to the affine/parasitic stratum $\mathcal{C}_{\text{aff}}$.*

Consequently the retained residual decomposition has

$$\mathcal{C}_{\text{aff}} = \emptyset.$$

Proof. The dichotomy is exactly [Theorem 8.19](#). If $\Phi(S_c U) = 0$, then $S_c U$ is globally constant and the profile is in the affine/parasitic lower stratum $\mathcal{C}_{\text{aff}}$. If the residual state is retained after lower-stratum removal, this alternative is not available. Hence a positive affine-normalized oscillatory activity threshold survives. A state in $\mathcal{C}_{\text{aff}}$ therefore cannot carry retained residual concentration, so $\mathcal{C}_{\text{aff}} = \emptyset$ inside the retained residual decomposition. $\square$

Corollary 9.54 (Young alternative discharge from first bad scale routing).

$$\mathcal{C}_{\text{Young}} = \emptyset.$$

Proof. Let a retained residual state generate a nonzero Young critical-tail alternative. By [Theorem 9.26](#), the log-diffuse alternative is already empty, so the defect has a tight macroscopic annular realization. Let W_n be the corresponding annular blow-down and $\mathcal{Y}_{x,s}$ its Young measure. By [Theorems 9.46](#) and [9.48](#), W_n has no hidden-scale variance on compact annular cylinders. Hence [Theorem 9.28](#) gives

$$\mathcal{Y}_{x,s} = \delta_{\bar{W}(x,s)} \quad \text{for a.e. } (x, s),$$

contradicting the definition of $\mathcal{C}_{\text{Young}}$. Therefore $\mathcal{C}_{\text{Young}} = \emptyset$. $\square$

Theorem 9.55 (Critical-tail rigidity from first bad scale routing). *Every nonzero recurrent critical-tail defect generated by a realized residual terminal sequence belongs to*

$$\mathcal{C}_{\text{active}} \cup \mathcal{C}_{\text{aff}} \cup \mathcal{C}_{\text{cohcrit}}.$$

Equivalently, after active regeneration, affine/parasitic collapse, and coherent critical tails have been removed, no recurrent critical-tail defect remains.

Proof. Let $\mathcal{D} \in \mathfrak{T}_M$ be a nonzero recurrent critical-tail defect. By [Theorem 9.52](#), exactly one ordered alternative occurs: active regeneration, affine/parasitic collapse, log-diffuse defect, Young critical tail, or coherent critical tail. The first two alternatives are precisely $\mathcal{C}_{\text{active}}$ and $\mathcal{C}_{\text{aff}}$. The log-diffuse alternative is empty by [Theorem 9.26](#). The Young alternative is empty by [Theorem 9.54](#). The only remaining nonempty critical-tail alternative is therefore the coherent alternative $\mathcal{C}_{\text{cohcrit}}$. This proves the inclusion. Once $\mathcal{C}_{\text{active}}$, $\mathcal{C}_{\text{aff}}$, and $\mathcal{C}_{\text{cohcrit}}$ are removed, the inclusion leaves no recurrent critical-tail defect. $\square$

Lemma 9.56 (Bounded velocity gauges for realized blow-downs). *Let $f : E \rightarrow \mathbb{R}^3$ be measurable on a finite positive-measure set E , and assume $|f| \leq M$ a.e. Then the minimization problem*

$$\inf_{c \in \mathbb{R}^3} \int_E |f - c|^3$$

has a minimizer in $\overline{B_M(0)}$. Consequently every affine-normalized blow-down of a sequence satisfying $\|U_n\|_{L^\infty} \leq M$ may be represented, without increasing any local oscillation functional, by velocity gauges c_n with $|c_n| \leq M$. If the normalization is performed time-slice by time-slice, the same conclusion holds for a measurable choice $c_n(s)$, with $|c_n(s)| \leq M$ for a.e. s .

Proof. The functional is continuous and coercive in c , so a minimizer exists. Let $\pi : \mathbb{R}^3 \rightarrow \overline{B_M(0)}$ be the nearest-point projection. Since $\overline{B_M(0)}$ is closed and convex, for every $v \in \overline{B_M(0)}$ and every $c \in \mathbb{R}^3$,

$$|v - \pi(c)| \leq |v - c|.$$

Taking $v = f(y)$ and integrating gives

$$\int_E |f - \pi(c)|^3 \leq \int_E |f - c|^3.$$

Thus any minimizing sequence may be projected into $\overline{B_M(0)}$, and a minimizer can be chosen there. Applying this to the local velocity fields used to define the affine-normalized observer oscillations gives the asserted bounded representatives. The time-slice version is the same argument on each slice, with measurability obtained by the usual measurable-selection theorem for a continuous coercive finite-dimensional minimization problem. The projection step is pointwise in s , so the bound $|c_n(s)| \leq M$ is preserved. $\square$

Theorem 9.57 (Bounded-origin realization of coherent critical tails). *Every coherent critical tail generated by a bounded normalized residual Navier–Stokes terminal sequence is realized, after subsequence extraction and admissible gauges, as a weak-* L^∞_{loc} limit on full macroscopic compact cylinders, including cylinders meeting the macroscopic origin. In particular, every coherent critical tail has a representative which is locally bounded near $x = 0$ on compact macroscopic time intervals.*

Proof. Let $(W, Q) \in \mathcal{C}_{\text{cohcrit}}$ be a coherent critical tail generated by a bounded terminal sequence $(U_n, \Pi_n) \subset \mathfrak{X}_M$, with $\|U_n\|_{L^\infty} \leq M$. By coherent extraction, after passing to a subsequence there are macroscopic scales $R_n \rightarrow \infty$, covariant observer centers (y_n, σ_n) , affine velocity gauges, and pressure gauges such that on every compact annular cylinder

$$A_{r_0, r_1, I} := \{r_0 < |x| < r_1\} \times I, \quad 0 < r_0 < r_1 < \infty,$$

the fields

$$W_n(x, s) := U_n(y_n + R_n x, \sigma_n + s) - c_n(s)$$

converge to W strongly in $L^3(A_{r_0, r_1, I})$ and weak-* in $L^\infty(A_{r_0, r_1, I})$.

We first check that the same gauges which realize the annular coherent tail are bounded on compact time intervals. Fix a compact interval I and use the annulus $1 < |x| < 2$. The weak- $*$ L^∞ convergence on $\{1 < |x| < 2\} \times I$ implies, after discarding finitely many n , that

$$\|W_n\|_{L^\infty(\{1 < |x| < 2\} \times I)} \leq C_I.$$

For a.e. $s \in I$, if $|c_n(s)| > M + C_I$, then for every x with $1 < |x| < 2$,

$$|U_n(y_n + R_n x, \sigma_n + s) - c_n(s)| \geq |c_n(s)| - M > C_I,$$

contradicting the preceding L^∞ bound. Hence

$$|c_n(s)| \leq M + C_I \quad \text{for a.e. } s \in I.$$

This argument keeps the annular representative fixed. The bounded-gauge lemma above shows, in addition, that the affine-normalized construction may always be started with bounded canonical gauges.

Therefore, on every full macroscopic cylinder

$$K_{R,I} := B_R(0) \times I,$$

one has

$$|W_n(x, s)| \leq |U_n(y_n + R_n x, \sigma_n + s)| + |c_n(s)| \leq 2M + C_I$$

for a.e. $(x, s) \in K_{R,I}$. The terminal sequence is defined on the whole ancient space, or on expanding domains containing $y_n + R_n B_R$ for all large n ; hence W_n is defined on $K_{R,I}$ for all sufficiently large n .

Choose an exhaustion of $\mathbb{R}^3 \times \mathbb{R}$ by full compact macroscopic cylinders $K_j = B_j(0) \times [-j, j]$. Banach–Alaoglu gives, by a diagonal extraction that does not change notation, a field

$$W^{\text{full}} \in L_{\text{loc}}^\infty(\mathbb{R}^3 \times \mathbb{R}),$$

such that

$$W_n \rightharpoonup^* W^{\text{full}} \quad \text{in } L^\infty(K_j)$$

for every j . The subsequence still has the annular strong L_{loc}^3 convergence to the coherent tail W . Therefore, for every annular cylinder $A_{r_0, r_1, I}$,

$$W^{\text{full}} = W \quad \text{a.e. on } A_{r_0, r_1, I}.$$

Since the annuli exhaust $(\mathbb{R}^3 \setminus \{0\}) \times \mathbb{R}$, the annular coherent tail is the restriction of the full weak- $*$ limit W^{full} away from $x = 0$.

We take W^{full} as the full-cylinder representative of the coherent tail. It is locally bounded on every compact macroscopic cylinder, including cylinders meeting $x = 0$, and it realizes the same annular tail because it agrees with W on all punctured compact cylinders. Pressure gauges are not needed for the boundedness conclusion; if one records the pressure as well, the local pressure-gauge compactness in the terminal state space gives distributional pressure-gradient limits on the same full cylinders after the usual time-dependent gauges. The construction uses only the uniform local boundedness inherited from the Type I terminal state space and finite-dimensional affine normalization; it uses no global critical norm or global tail budget. $\square$

Lemma 9.58 (Minimal coherent hulls inherit bounded-origin vanishing). *Let $(W, Q) \in \mathcal{C}_{\text{coherit}}$ be realized by the bounded-origin theorem, and write*

$$Z(\rho, \omega, s) = e^\rho W(e^\rho \omega, s).$$

Let $\mathcal{M}$ be a compact minimal invariant subset of the logarithmic translation hull generated by Z . Then every $Z_ \in \mathcal{M}$ is itself realized by a full locally bounded macroscopic tangent near the origin, and*

$$Z_*(\rho, \cdot, \cdot) \rightarrow 0 \quad \text{as } \rho \rightarrow -\infty$$

in the local logarithmic topology. Consequently $0 \in \mathcal{H}(Z_*)$. If $\mathcal{M} = \mathcal{H}(Z_*)$ is minimal, then $\mathcal{M} = \{0\}$.

Proof. Elements of the logarithmic hull are limits of logarithmic translations of the same ancestor-realized coherent tail. Logarithmic translation corresponds to choosing another macroscopic scale in the bounded-origin realization. The bounded-origin theorem gives L_{loc}^∞ bounds on every full compact cylinder, including neighborhoods of the macroscopic origin, with constants that are stable under passing to local hull limits. Hence each hull element Z_* corresponds to a velocity

$$W_*(r\omega, s) = r^{-1}Z_*(\log r, \omega, s)$$

which is locally bounded near $r = 0$.

For $r = e^\rho \downarrow 0$, local boundedness gives

$$|Z_*(\rho, \omega, s)| = r |W_*(r\omega, s)| \leq C_K r$$

on every compact angular-time set K . Thus $Z_*(\rho, \cdot, \cdot) \rightarrow 0$ locally as $\rho \rightarrow -\infty$, and the zero state belongs to the orbit closure $\mathcal{H}(Z_*)$. If Z_* lies in a minimal compact invariant set $\mathcal{M}$, then $\mathcal{H}(Z_*) = \mathcal{M}$. Since $\{0\}$ is closed and invariant, minimality forces $\mathcal{M} = \{0\}$. $\square$

Corollary 9.59 (Coherent critical-tail alternative closure). *One has*

$$\mathcal{C}_{\text{homcrit}} = \emptyset, \quad \mathcal{C}_{\text{logper}} = \emptyset, \quad \mathcal{C}_{\text{ap}} = \emptyset.$$

Proof. Let (W, Q) be a coherent critical tail and use the bounded-origin realization theorem to view W as locally bounded near $x = 0$ on compact time intervals.

If $W \in \mathcal{C}_{\text{homcrit}}$, then

$$W(r\omega, s) = r^{-1}A(\omega, s).$$

Local boundedness near $r = 0$ gives

$$|A(\omega, s)| = r |W(r\omega, s)| \leq Mr$$

for small r . Letting $r \downarrow 0$ gives $A = 0$, hence $W \equiv 0$, contradicting the nonzero affine-normalized oscillation of a realized coherent tail. Thus $\mathcal{C}_{\text{homcrit}} = \emptyset$.

If $W \in \mathcal{C}_{\text{logper}}$, set

$$Z(\rho, \omega, s) = e^\rho W(e^\rho \omega, s).$$

Bounded-origin realization gives $Z(\rho, \cdot, \cdot) \rightarrow 0$ as $\rho \rightarrow -\infty$ in the local logarithmic topology. If Z has nonzero period L , then

$$Z(\rho, \omega, s) = Z(\rho - kL, \omega, s)$$

for every integer k . Sending $k \rightarrow +\infty$ gives $Z(\rho, \omega, s) = 0$ for every ρ , hence $W \equiv 0$, again contradicting nonzero critical tail mass. Thus $\mathcal{C}_{\text{logper}} = \emptyset$.

If $W \in \mathcal{C}_{\text{ap}}$, let $\mathcal{M}$ be the nonzero minimal aperiodic logarithmic subsystem used to define this alternative and choose $Z_* \in \mathcal{M}$. By [Theorem 9.58](#), $0 \in \mathcal{H}(Z_*)$. Since $\mathcal{M}$ is minimal, $\mathcal{H}(Z_*) = \mathcal{M}$; hence the closed invariant set $\{0\}$ contained in $\mathcal{M}$ forces $\mathcal{M} = \{0\}$. This contradicts the nonzero affine-normalized oscillation required in $\mathcal{C}_{\text{ap}}$. Therefore $\mathcal{C}_{\text{ap}} = \emptyset$. $\square$

Corollary 9.60 (Critical-tail discharge, complete form). *No nonzero realized critical-tail state remains after terminal active-locus removal and affine/parasitic lower-stratum removal.*

Proof. Let $\mathcal{T} \in \mathfrak{T}_M$ be a remaining nonzero realized critical-tail state. By [Theorem 9.52](#), exactly one ordered alternative occurs: active regeneration, affine/parasitic collapse, log-diffuse defect, Young critical tail, coherent homogeneous tail, coherent log-periodic tail, or coherent aperiodic tail.

Active regeneration contradicts terminal active-locus removal. The affine/parasitic alternative is impossible by [Theorem 9.53](#). The log-diffuse alternative is impossible by [Theorem 9.26](#), and the

Young alternative is impossible by [Theorem 9.54](#). The three coherent alternatives are impossible by [Theorem 9.59](#). All alternatives are therefore impossible, so no such $\mathcal{T}$ remains. $\square$

Theorem 9.61 (No mixed compact–diffuse residual state). *Let (U, Π) be a retained terminal profile in the refined residual class, after the affine normalization of [Theorem 8.14](#). Then U cannot have a diffuse exterior oscillation remainder, nor an escaping retained noncompact recentering remainder, coexisting with its compact retained core. Equivalently, the paired terminal states*

$$\text{one active compact core} + \mathcal{R}_{\text{diff}}, \quad \text{one active compact core} + \mathcal{R}_{\text{noncomp}},$$

and their finite, infinite, or continuum active-locus analogues do not occur in the refined residual class.

Proof. The noncompact escaping case is local. In this theorem the noncompact remainder is retained in the velocity or affine-oscillation sense. If an escaping recentering with retained mass loses the local dissipation or pressure-gauge bounds in [Theorem 9.3](#), then after its own local normalization it is a retained compact-mass recentering sequence. This contradicts [Theorem 9.5](#). Thus it remains to exclude a compact core plus diffuse exterior remainder.

Assume that such a diffuse exterior remainder exists. By the paired active-locus decomposition [Theorem 9.9](#), remove neighborhoods of the active locus at every positive affine-normalized activity threshold. The mixed assumption says that a nonzero diffuse oscillation residual remains. Apply [Theorem 9.16](#).

If active regeneration occurs, then a positive oscillation-active observer survives after all positive-threshold active loci were removed, contradicting the definition of the diffuse residual. If affine/parasitic collapse occurs, then [Theorems 8.14](#) and [10.11](#) places the relevant descendant hull in an already closed lower stratum. This cannot occur inside $\mathcal{C}_{\text{gen}}$.

The only remaining alternative is a critical diffuse tail. By [Theorem 9.49](#), it generates a nonzero $\mathcal{T} \in \mathfrak{T}_M$. This contradicts [Theorem 9.60](#). Hence no mixed compact–diffuse residual profile exists. $\square$

Definition 9.62 (Finite separated family of retained concentration profiles). A finite separated family of retained concentration profiles is a finite family of recentering sequences. For the original terminal extraction these are the parabolic recenterings of [Theorem 9.1](#); for descendants of a centered profile they are the covariant tail recenterings of [Theorem 8.4](#). We write them uniformly as $g_{1,n}, \dots, g_{J,n}$, $J \geq 2$, such that:

- (i) the recentering sequences are mutually separated;
- (ii) each recentered sequence converges locally smoothly for velocity and locally in $L^{3/2}$ for pressure modulo gauges to a terminal profile (U_j, Π_j) ;
- (iii) each limiting profile has retained local concentration:

$$\mathcal{E}_1(U_j, \Pi_j; 0) \geq \varepsilon_* \quad (j = 1, \dots, J).$$

This is a finite local family, not a statement that all concentration profiles in the whole space have been counted. The separated-profile proof below treats the possible continuation of escaping recenterings by local concentration alternatives, not by a global counting estimate.

10. SEPARATED CONCENTRATION PROFILES, INDECOMPOSABILITY, AND SEQUENCE- L^3

We now close the terminal alternatives carrying retained local concentration. The proof uses no global critical-mass budget. Instead, finite separated concentration families are treated through local recentering and compactness. Iterating retained tail limits from such a finite family has only local state-space outcomes: an indecomposable retained concentration profile, a mixed compact–diffuse or compact–noncompact successor, or an active descendant chain. The mixed

successors are excluded by [Theorem 9.61](#); infinite active descendant chains and finite loops are excluded by the compact active descendant theorem.

Definition 10.1 (Tail limits and terminal indecomposability). Let (U, Π) be a terminal profile with retained local concentration. A retained concentrating tail limit is a limit of covariant tail recenterings

$$\mathcal{T}_{(x_n, \tau_n)}(U, \Pi), \quad |x_n| + |\tau_n|^{1/2} \rightarrow \infty,$$

in the local smooth velocity topology and the local $L^{3/2}$ pressure topology modulo gauges, with retained local concentration at the origin. A raw fixed spatial shift is not used in this definition; by [Theorems 8.3](#) and [8.4](#), it would introduce an unbounded spurious drift for escaping centers. Diffuse exterior concentration means that there are $\tau_n \rightarrow -\infty$, $R_n \rightarrow \infty$, and $\eta > 0$, such that

$$\int_{|y| > R_n} |U(y, \tau_n)|^3 dy \geq \eta,$$

while every unit-scale exterior covariant recentering sequence based in the parabolic exterior windows

$$\{|y| > R_n\} \times (\tau_n - 1, \tau_n + 1)$$

has pressure-normalized parabolic CKN mass below the retained-concentration threshold, unless that recentering sequence has loss of local compactness. Thus the word “diffuse” is always parabolic: localization is tested by unit CKN cylinders, not only by spatial balls on the distinguished time slices. An escaping non-compact recentering sequence is a covariant tail recentering sequence for which the local compactness, suitability, or pressure control needed to extract a terminal profile fails. A retained concentration profile is terminally indecomposable if it has no retained concentrating tail limit, no diffuse exterior concentration, and no escaping non-compact recentering sequence.

Lemma 10.2 (Time-slice activity gives parabolic CKN activity). *Let (U, Π) be a bounded smooth solution of (1.2) with $\|U\|_{L^\infty} \leq M$. If*

$$\int_{B_1(0)} |U(y, 0)|^3 dy \geq \varepsilon_0 > 0,$$

then there are $\delta = \delta(M, \varepsilon_0) > 0$ and $c_0 = c_0(M, \varepsilon_0) > 0$ such that, after the standard local pressure gauge,

$$\iint_{B_1(0) \times (-\delta^2, 0)} |U|^3 dy d\tau \geq c_0, \quad \text{and hence} \quad \mathcal{E}_1(U, \Pi; 0) \geq c_0.$$

The same constants apply to every covariant tail recentering $\mathcal{T}_{(x, \sigma)}(U, \Pi)$.

Proof. Apply interior parabolic regularity on $B_2 \times [-2\delta_0, \delta_0]$, with $\delta_0 > 0$ universal and small. On this cylinder the drift $-\frac{1}{2}y \cdot \nabla U$ has bounded smooth coefficients, and the nonlinear forcing is bounded by M^2 . After choosing local pressure gauges, the standard interior estimates give

$$\|\partial_\tau U\|_{L^\infty(B_1 \times [-\delta_0, 0])} + \|\nabla U\|_{L^\infty(B_1 \times [-\delta_0, 0])} \leq C(M)$$

where $C(M)$ is independent of the particular profile. Hence the map $\tau \mapsto \int_{B_1} |U(y, \tau)|^3 dy$ is uniformly continuous with a modulus depending only on M and $|B_1|$. Choose $\delta \leq \delta_0$ so that

$$\int_{B_1} |U(y, \tau)|^3 dy \geq \frac{1}{2}\varepsilon_0 \quad -\delta^2 < \tau < 0.$$

Integrating in time gives a positive lower bound for the velocity part of $\mathcal{E}_1(U, \Pi; 0)$; the pressure contribution is nonnegative after any admissible gauge. The constants are unchanged under $\mathcal{T}_{(x, \sigma)}$, because that recentering is an exact composition of the centered symmetries in [Theorem 8.4](#). $\square$

Lemma 10.3 (One retained tail limit gives two separated concentration profiles). *Let U be a retained concentration terminal profile. If U has a retained concentrating tail limit, then the corresponding terminal extraction contains a finite separated family of retained concentration profiles.*

Proof. The origin recentering sequence carries retained concentration:

$$\mathcal{E}_1(U, \Pi; 0) \geq \varepsilon_*.$$

Let W be a retained concentrating tail limit. By definition there are covariant relative tail recentering parameters $\zeta_n = (x_n, \tau_n)$, with $|x_n| + |\tau_n|^{1/2} \rightarrow \infty$, such that $\mathcal{T}_{\zeta_n}(U, \Pi) \rightarrow (W, \Pi_W)$ in the terminal topology and

$$\mathcal{E}_1(W, \Pi_W; 0) \geq \varepsilon_*.$$

By local convergence of velocity and pressure after gauges,

$$\mathcal{E}_1(\mathcal{T}_{\zeta_n} U, \mathcal{T}_{\zeta_n} \Pi; 0) \geq \frac{1}{2} \varepsilon_*$$

for all sufficiently large n . Thus, for large n , the origin recentering sequence and the covariant tail sequence ζ_n are separated and have retained local concentration. They form a finite separated two-profile concentration family. $\square$

Proposition 10.4 (No finite separated family and local exclusions imply indecomposability). *Every single retained concentration terminal profile arising from a refined residual candidate is terminally indecomposable.*

Proof. Let U be a single retained concentration terminal profile. If U had a retained concentrating tail limit, [Theorem 10.3](#) would produce a finite two-profile separated family with retained concentration, contradicting [Theorem 10.23](#). If U had diffuse exterior concentration or an escaping non-compact recentering sequence, the corresponding terminal extraction would be a mixed state consisting of the retained compact core U plus a diffuse or noncompact tail remainder. This is excluded by [Theorem 9.61](#). Hence no retained concentrating tail limit, diffuse exterior concentration, or escaping non-compact recentering sequence exists, and U is terminally indecomposable. $\square$

Lemma 10.5 (Indecomposable sequence- L^3 completeness). *Let (U, Π) be a bounded smooth ancient solution of (1.2). Suppose U has retained local concentration and is terminally indecomposable in the sense of [Theorem 10.1](#). Then there exists a sequence $\tau_k \rightarrow -\infty$ such that*

$$\sup_k \|U(\cdot, \tau_k)\|_{L^3(\mathbb{R}^3)} < \infty.$$

Proof. We prove the contrapositive. Define

$$A(\tau) := \int_{\mathbb{R}^3} |U(y, \tau)|^3 dy \in [0, \infty].$$

If the conclusion fails, then

$$\liminf_{\tau \rightarrow -\infty} A(\tau) = \infty.$$

Choose $\tau_n \rightarrow -\infty$ such that $A(\tau_n) \rightarrow \infty$, allowing the value $+\infty$. Since U is bounded by some M , for each finite R ,

$$\int_{|y| \leq R} |U(y, \tau_n)|^3 dy \leq M^3 |B_R|.$$

Choose $R_n \rightarrow \infty$ so slowly that, after discarding finitely many n ,

$$\int_{|y| > R_n} |U(y, \tau_n)|^3 dy \geq 1.$$

If $A(\tau_n) = +\infty$, take $R_n = n$; otherwise choose R_n with $M^3|B_{R_n}| \leq A(\tau_n)/2$.

Define the parabolic exterior activity level

$$m_n^{\text{par}} := \sup_{\substack{|x| > R_n - 2 \\ |s - \tau_n| \leq 1}} \mathcal{E}_1(\mathcal{T}_{(x,s)}(U, \Pi); 0),$$

where $\mathcal{E}_1$ is computed after the admissible local pressure gauge in the recentered variables. This is the correct quantity for the terminal decomposition: it tests unit parabolic CKN cylinders, not merely spatial time-slices.

First suppose $\limsup_{n \rightarrow \infty} m_n^{\text{par}} \geq \varepsilon_*$. Passing to a subsequence, choose (x_n, s_n) , with

$$|x_n| > R_n - 2, \quad |s_n - \tau_n| \leq 1,$$

such that

$$\mathcal{E}_1(\mathcal{T}_{(x_n, s_n)}(U, \Pi); 0) \geq \frac{1}{2}\varepsilon_*.$$

Since $R_n \rightarrow \infty$ and $\tau_n \rightarrow -\infty$, these are escaping covariant tail recenterings. If they have loss of local compactness, terminal indecomposability is contradicted. Otherwise local compactness gives a subsequential terminal limit (W, Π_W) , and lower semicontinuity of the pressure-normalized CKN quantity gives

$$\mathcal{E}_1(W, \Pi_W; 0) \geq c\varepsilon_*$$

after lowering the retained threshold by a fixed harmless factor. Hence W is a retained concentrating covariant tail limit, contradicting terminal indecomposability.

It remains to consider the case in which $m_n^{\text{par}} < \varepsilon_*$ for all large n , after lowering the threshold if necessary. If some unit exterior covariant recentering sequence in the windows

$$\{|y| > R_n - 2\} \times (\tau_n - 1, \tau_n + 1)$$

has loss of local compactness, terminal indecomposability is contradicted by the definition. If no such recentering loses local compactness, then the exterior time-slice lower bound

$$\int_{|y| > R_n} |U(y, \tau_n)|^3 dy \geq 1$$

together with the parabolic smallness $m_n^{\text{par}} < \varepsilon_*$ is exactly the parabolic diffuse exterior concentration alternative in [Theorem 10.1](#). This again contradicts terminal indecomposability. Both parabolic alternatives are impossible, so the desired sequence exists. $\square$

Proposition 10.6 (Stability of the bounded mild formulation). *Let (U_m, Π_m) be normalized parasitic-free centered profiles whose physical pullbacks are bounded mild Navier–Stokes solutions on every finite terminal interval. Assume $U_m \rightarrow U$ locally smoothly in centered variables and that the U_m are uniformly bounded on compact centered time intervals. Then the physical pullback of U is also bounded mild on every finite terminal interval. More precisely, for every $T < 0$, after setting*

$$u^T(x, s) := (-T - s)^{-1/2} U\left(\frac{x}{\sqrt{-T - s}}, -\log(-T - s)\right), \quad s < 0,$$

and translating the interval so that its terminal time is 0, one has, for all $-\infty < s < t < 0$,

$$(10.1) \quad u^T(t) = e^{(t-s)\Delta} u^T(s) - \int_s^t e^{(t-\sigma)\Delta} \mathbb{P} \nabla \cdot (u^T \otimes u^T)(\sigma) d\sigma$$

in the bounded mild duality sense used in the setup package. The same conclusion is preserved by centered time translations, covariant translations, retained tail limits, active successors, recurrent-core limits, and affine normalization outside $\mathcal{C}_{\text{aff}}$.

Proof. For each m , the normalized setup representative satisfies the Duhamel formula on every finite physical interval. Fix $T < 0$ and $-\infty < s < t < 0$. Pull the centered convergence back to the compact physical time interval $[s, t]$ after the finite terminal shift. The velocities converge locally smoothly and are uniformly bounded; hence

$$u_m^T \otimes u_m^T \rightarrow u^T \otimes u^T$$

locally in every finite L^q , and in particular in distributions. Testing the Duhamel identity for u_m^T against a compactly supported smooth vector field and using the heat-kernel/Leray-projection duality gives convergence of the linear heat term and of the bilinear Duhamel term. The bounded mild duality formulation used in the setup package then extends the identity from compactly supported tests to the standard bounded mild class, yielding (10.1) for u^T .

Centered time translations and covariant translations are exact symmetries of (1.2); in physical variables they are compositions of translations, scalings, and finite time shifts, which preserve the Duhamel identity. Retained tail limits, active successors, and recurrent-core limits are locally smooth limits of such normalized representatives, so the preceding limit argument applies. The affine normalization either removes a spatially homogeneous parasitic component or assigns the profile to $\mathcal{C}_{\text{aff}}$; outside $\mathcal{C}_{\text{aff}}$ no homogeneous parasitic forcing remains, and the same Duhamel identity holds for the canonical representative. $\square$

Theorem 10.7 (Parasitic-free mildness inheritance from the normalized package). *Let (U, Π) be a retained terminal profile, retained covariant tail descendant, active descendant limit, or recurrent-core limit represented in the normalized residual state space generated by the Seregin extraction in the setup package. Then exactly one of the following alternatives applies.*

- (i) (U, Π) belongs to the affine/parasitic lower stratum of Theorem 8.14;
- (ii) after the canonical affine normalization already built into the residual representative, the physical pullback

$$u(x, t) = (-t)^{-1/2} U\left(\frac{x}{\sqrt{-t}}, -\log(-t)\right), \quad t < 0,$$

is parasitic-free and has the following property: for every $T < 0$, the finite-terminal shift

$$u^T(x, s) := u(x, T + s), \quad s < 0,$$

is a bounded mild ancient Navier–Stokes solution in the Albritton–Barker sense on $\mathbb{R}^3 \times (-\infty, 0)$.

Proof. Affine-constant centered profiles are precisely the homogeneous parasitic profiles removed by Theorem 8.14; they are assigned to alternative (i). We therefore consider a representative outside that lower stratum.

The setup package constructs normalized Seregin limits as centered bounded mild ancient representatives, after removal of the spatially homogeneous parasitic mode. The stability statement Theorem 10.6 shows that the state-space operations used in this paper preserve the bounded mild formulation: exact centered symmetries preserve the Duhamel identity, local limits preserve it by bounded mild duality, and affine normalization either removes the homogeneous parasitic mode or places the profile in $\mathcal{C}_{\text{aff}}$. Thus retained terminal profiles, covariant tail descendants, active descendant limits, and recurrent-core limits remain parasitic-free mild unless they have fallen into the affine/parasitic lower stratum.

Finally fix $T < 0$. On $\mathbb{R}^3 \times (-\infty, T]$, the physical pullback

$$u(x, t) = (-t)^{-1/2} U\left(\frac{x}{\sqrt{-t}}, -\log(-t)\right)$$

is smooth and bounded by $(-T)^{-1/2}\|U\|_{L^\infty}$, and it solves the physical Navier–Stokes equations by the direct self-similar change of variables. By the inherited parasitic-free mild formulation, for $-\infty < s < t \leq T$,

$$u(t) = e^{(t-s)\Delta}u(s) - \int_s^t e^{(t-\sigma)\Delta}\mathbb{P}\nabla \cdot (u \otimes u)(\sigma) d\sigma.$$

After translating the finite terminal time T to 0, this is the mild ancient formulation for u^T on every finite forward interval $(-\infty, 0)$. The bound $\|u^T\|_{L^\infty(\mathbb{R}^3 \times (-\infty, 0))} \leq (-T)^{-1/2}\|U\|_{L^\infty}$ is finite. Thus u^T is a genuine bounded mild ancient solution in the sense required by the Albritton–Barker endpoint theorem. $\square$

Theorem 10.8 (Albritton–Barker endpoint Liouville theorem). *Let u be a mild ancient solution of the three-dimensional Navier–Stokes equations. If there exists $t_k \downarrow -\infty$ such that*

$$\sup_k \|u(\cdot, t_k)\|_{L^3(\mathbb{R}^3)} < \infty,$$

then $u \equiv 0$.

Proof. This is the endpoint ancient Liouville theorem in the sequence- L^3 form proved by Albritton and Barker [1]. $\square$

Lemma 10.9 (No retained indecomposable concentration profile). *No retained terminally indecomposable concentration profile exists.*

Proof. Let U have retained local concentration and be terminally indecomposable. By [Theorem 10.5](#), choose $\tau_k \rightarrow -\infty$ such that

$$\sup_k \|U(\cdot, \tau_k)\|_{L^3} < \infty.$$

Let u be the physical pullback and set $t_k = -e^{-\tau_k}$. Since $\tau_k \rightarrow -\infty$, the sequence satisfies $t_k \downarrow -\infty$, and critical scaling gives

$$\|u(\cdot, t_k)\|_{L^3(\mathbb{R}^3)} = \|U(\cdot, \tau_k)\|_{L^3(\mathbb{R}^3)}.$$

By [Theorem 10.7](#), either U is in the affine/parasitic lower stratum, which is not part of the normalized residual class, or else for each $T < 0$

$$u^T(x, s) := u(x, T + s), \quad s < 0,$$

is a bounded mild ancient solution. Choose $T < 0$ and discard finitely many indices so that $t_k < T$. Then $s_k := t_k - T \downarrow -\infty$, and

$$\|u^T(\cdot, s_k)\|_{L^3(\mathbb{R}^3)} = \|u(\cdot, t_k)\|_{L^3(\mathbb{R}^3)} \leq \sup_j \|U(\cdot, \tau_j)\|_{L^3(\mathbb{R}^3)} < \infty.$$

[Theorem 10.8](#) gives $u^T \equiv 0$. Since $T < 0$ was arbitrary, the physical pullback vanishes on $\mathbb{R}^3 \times (-\infty, 0)$, and therefore $U \equiv 0$. This contradicts retained local concentration. $\square$

Lemma 10.10 (Diagonal lifting of retained concentrating tail limits). *Let $A = \{\mathfrak{s}_1, \dots, \mathfrak{s}_J\}$, $J \geq 2$, be a finite retained concentration separated family for a refined residual candidate, represented by separated recentering sequences $g_{j,n} = (y_{j,n}, \tau_{j,n})$. Let (U_j, Π_j) be the terminal profile obtained in the j -th recentering. If (U_j, Π_j) has a retained concentrating tail limit (W, Π_W) , then there is a combined raw-covariant descendant recentering sequence $\mathfrak{d} = (d_n)$, in the sense of [Theorem 9.2](#), for the original terminal sequence such that:*

- (i) $\mathfrak{d}$ is asymptotically separated from g_j ;
- (ii) the recentered original sequence at $\mathfrak{d}$ converges locally to (W, Π_W) , after admissible pressure gauges;

(iii) $\mathfrak{d}$ has retained local concentration.

Proof. By definition of retained concentrating tail limit, there are relative covariant tail recenterings $\zeta_m = (\xi_m, \sigma_m)$, with

$$|\xi_m| + |\sigma_m|^{1/2} \rightarrow \infty,$$

such that

$$\mathcal{T}_{\zeta_m}(U_j, \Pi_j) \rightarrow (W, \Pi_W)$$

locally in the terminal topology, and

$$\mathcal{E}_1(W, \Pi_W; 0) \geq \varepsilon_*.$$

Fix m . Since $V_n^{g_j} \rightarrow U_j$ locally smoothly and $(P_n^{g_j}) \rightarrow \Pi_j$ locally in $L^{3/2}$ modulo gauges, the convergence also holds on the fixed covariant tube

$$\{(y + e^{s/2}\xi_m, s + \sigma_m) : (y, s) \in Q_m(0, 0)\}.$$

Therefore one can choose $n(m)$ so large that the covariantly recentered original fields

$$V_{n(m)}(y + y_{j,n(m)} + e^{s/2}\xi_m, s + \tau_{j,n(m)} + \sigma_m)$$

and the corresponding pressure gauges are within m^{-1} of the $\mathcal{T}_{\zeta_m}$ -recentered fields of (U_j, Π_j) on $Q_m(0, 0)$.

Set

$$d_m := (g_{j,n(m)}; \xi_m, \sigma_m).$$

Relabeling the subsequence $n(m)$ as the terminal index gives a combined raw-covariant descendant recentering sequence $\mathfrak{d} = (d_m)$ for the original terminal sequence. Since the covariant parabolic distance from d_m to $g_{j,n(m)}$ is $|\xi_m| + |\sigma_m|^{1/2}$, the sequence $\mathfrak{d}$ is separated from the j -th recentering sequence. The diagonal choice gives local convergence of the original sequence at $\mathfrak{d}$ to (W, Π_W) . Let $\mu_{n(m)}^{\mathfrak{d}}$ denote the local CKN measure of the original terminal sequence pulled back by this combined raw-covariant recentering and by the pressure gauges used in the diagonal convergence. The local CKN quantity is continuous under the strong local velocity convergence and weak lower semicontinuity of the pressure term after these fixed gauges, so

$$\liminf_{m \rightarrow \infty} \mu_{n(m)}^{\mathfrak{d}}(Q_1(0, 0)) \geq \mathcal{E}_1(W, \Pi_W; 0) \geq \varepsilon_*.$$

Thus $\mathfrak{d}$ has retained local concentration. $\square$

Theorem 10.11 (Heredity of retained descendants). *Let (U, Π) be a normalized retained residual profile in $\mathfrak{X}_M$, and let (W, Π_W) be a retained covariant tail descendant of (U, Π) . Then $(W, \Pi_W) \in \mathfrak{X}_M$, it carries retained local concentration at the descendant threshold fixed in [Theorem 3.2](#), and it has the same parasitic-free normalization status as a terminal profile generated directly by the setup package extraction. Consequently exactly one of the following ordered outcomes occurs:*

- (i) (W, Π_W) belongs to one of the previously closed lower strata, in which case that descendant is already discharged;
- (ii) (W, Π_W) belongs to the normalized residual terminal state space, so the active descendant and terminal-indivisibility arguments may be iterated from W .

Proof. The descendant is obtained as a limit of covariant tail recenterings $\mathcal{T}_{\zeta_n}(U, \Pi)$. By [Theorem 8.9](#), the recentered sequence is locally precompact in $\mathfrak{X}_M$, and the limit remains a bounded smooth locally suitable centered solution with the same L^∞ bound and compatible pressure gauges. Retained local concentration is part of the definition of retained descendant; when the descendant is produced through a limit passage, [Theorem 9.4](#) fixes the threshold loss once and for all through [Theorem 3.2](#). The affine/parasitic normalization is canonical for the residual representative and is preserved by T_a , Θ_s , and their composition $\mathcal{T}_{\zeta_n}$; if a limit is

affine-parasitic, it is assigned to the affine-constant lower stratum of [Theorem 8.14](#). Otherwise it is represented in the same normalized residual state space. The ordered residual decomposition then places the descendant either in a previously closed lower class or in the generic residual state space. This is a local state-space reduction and does not use a global critical norm or global tail budget. $\square$

Lemma 10.12 (Escaping recenterings in a separated family). *Let $A = \{\mathfrak{s}_1, \dots, \mathfrak{s}_J\}$, $J \geq 2$, be a finite retained concentration separated family for a refined residual candidate. For each component $\mathfrak{s}_j$, the following ordered alternatives exhaust the possible successor states:*

- (i) *the profile (U_j, Π_j) is terminally indecomposable;*
- (ii) *(U_j, Π_j) has diffuse exterior concentration or an escaping recentering sequence with loss of local compactness;*
- (iii) *(U_j, Π_j) has a retained concentrating tail limit which lifts to a recentering sequence of the original terminal sequence with retained local concentration.*

Proof. This is the definition of terminal indecomposability plus the diagonal lifting lemma, with no disjointness assertion. If (U_j, Π_j) is terminally indecomposable, (i) holds. If it is not terminally indecomposable, at least one of the three excluded alternatives in [Theorem 10.1](#) occurs: a retained concentrating tail limit, diffuse exterior concentration, or an escaping recentering sequence with loss of local compactness. The diffuse exterior and local-compactness-failure cases give (ii). In the retained concentrating tail-limit case, [Theorem 10.10](#) lifts the limit to a recentering sequence of the original terminal sequence with retained local concentration, and [Theorem 10.11](#) places the descendant either in a closed lower stratum or back in the normalized residual state space, giving (iii) in the iterative residual case. If several successor states coexist, they are all recorded; the next lemma removes the nonactive ones. $\square$

Lemma 10.13 (No diffuse or non-compact escaping recentering from a retained family). *In the setting of [Theorem 10.12](#), alternative (ii) cannot occur.*

Proof. Suppose that (U_j, Π_j) has diffuse exterior concentration or an escaping recentering sequence with loss of local compactness. By the same diagonal procedure used in [Theorem 10.10](#), this tail state is realized by a terminal extraction of the original refined residual candidate, after passing to a subsequence and choosing the local pressure gauges supplied by the setup package in `proof_setup.tex`. It is not enough to say that the diffuse or noncompact state is not the sole carrier of the original compact concentration: the compact core (U_j, Π_j) may coexist with that tail state. The paired exclusion [Theorem 9.61](#) rules out exactly this coexistence in the refined residual class. Therefore alternative (ii) cannot occur for a refined residual family with retained local concentration. $\square$

Definition 10.14 (Closed active successor relation). For a fixed threshold $\eta > 0$, let

$$\mathfrak{A}_\eta := \mathfrak{A}_\eta^\# = \{U \in \mathfrak{X}_M^\# : \Phi(U) \geq \eta\}$$

be the compact affine-normalized activity set of [Theorem 8.18](#). We write

$$U \mathcal{R}_\eta V$$

if $U, V \in \mathfrak{A}_\eta$ and V is a retained active covariant tail descendant of U .

Lemma 10.15 (Closedness of the active successor relation). *The relation*

$$\mathcal{R}_\eta \subset \mathfrak{A}_\eta \times \mathfrak{A}_\eta$$

is closed in the local $\mathfrak{X}_M$ topology.

Proof. Let $U_m \rightarrow U$, $V_m \rightarrow V$ in $\mathfrak{X}_M^\#$, with $U_m \mathcal{R}_\eta V_m$. For each m , choose a retained covariant tail observer sequence $\zeta_{m,\ell}$ for U_m whose recentered fields converge to V_m on $Q_m(0,0)$ to accuracy m^{-1} , after admissible pressure gauges. A diagonal choice in (m, ℓ) , using [Theorems 8.9](#) and [9.4](#), gives a single escaping covariant tail sequence for U converging locally to V . Closedness of $\mathfrak{A}_\eta$ follows from [Theorem 8.18](#), and lower semicontinuity of the retained local CKN quantity after the fixed gauges preserves the descendant threshold. Hence $U \mathcal{R}_\eta V$. $\square$

Theorem 10.16 (Active path-space recurrence). *Assume there exists an infinite active descendant chain*

$$U_0 \mathcal{R}_\eta U_1 \mathcal{R}_\eta U_2 \mathcal{R}_\eta \cdots$$

Then the path space

$$\mathcal{P}_\eta := \left\{ (W_j)_{j \geq 0} \in \mathfrak{A}_\eta^\mathbb{N} : W_j \mathcal{R}_\eta W_{j+1} \text{ for all } j \right\}$$

is nonempty and compact, is invariant under the left shift

$$S(W_0, W_1, W_2, \dots) = (W_1, W_2, \dots),$$

and carries an S -invariant Borel probability measure $\mathbb{P}$. Its projected stationary state measure $\nu = (\pi_0)_\# \mathbb{P}$ is supported entirely on $\mathfrak{A}_\eta$.

Proof. The set $\mathfrak{A}_\eta = \mathfrak{A}_\eta^\#$ is compact by [Theorem 8.18](#). Hence $\mathfrak{A}_\eta^\mathbb{N}$ is compact and metrizable. By [Theorem 10.15](#),

$$\mathcal{P}_\eta = \bigcap_{j=0}^{\infty} \{ (W_k)_{k \geq 0} : (W_j, W_{j+1}) \in \mathcal{R}_\eta \}$$

is closed, hence compact. The given infinite chain is an element of $\mathcal{P}_\eta$, so the path space is nonempty, and it is plainly invariant under the left shift.

For the path $p = (U_0, U_1, \dots)$, define empirical measures

$$\mathbb{P}_N := \frac{1}{N} \sum_{k=0}^{N-1} \delta_{S^k p}.$$

Weak-* compactness of probability measures on compact $\mathcal{P}_\eta$ gives a convergent subsequence $\mathbb{P}_{N_\ell} \rightharpoonup \mathbb{P}$. For each continuous F on $\mathcal{P}_\eta$,

$$\int F \circ S d\mathbb{P}_N - \int F d\mathbb{P}_N = \frac{F(S^N p) - F(p)}{N} \rightarrow 0.$$

Thus $\mathbb{P}$ is shift-invariant. Since every coordinate of every path in $\mathcal{P}_\eta$ belongs to $\mathfrak{A}_\eta$, the projected measure $\nu = (\pi_0)_\# \mathbb{P}$ is supported in $\mathfrak{A}_\eta$. $\square$

Lemma 10.17 (Concatenation of retained active descendants). *Let*

$$U_0 \mathcal{R}_\eta U_1 \mathcal{R}_\eta \cdots \mathcal{R}_\eta U_N$$

be a finite active descendant chain. Then U_N is obtained from U_0 by a single diagonal covariant recentering sequence, after admissible pressure gauges, and it remains in $\mathfrak{A}_\eta$. The pressure gauges in the intermediate limits can be chosen compatibly with the final diagonal limit.

Proof. For $N = 1$ this is the definition of $\mathcal{R}_\eta$. Assume it holds up to $N - 1$. The relation $U_{N-1} \mathcal{R}_\eta U_N$ gives escaping covariant observers for U_{N-1} which converge locally to U_N . Compose these observers with the diagonal observers which realize U_{N-1} from U_0 . The semidirect covariance identity in [Theorem 8.5](#) shows that the composition is again a covariant observer sequence for U_0 . A diagonal choice on $Q_m(0,0)$, together with the pressure-gauge compatibility in [Theorem 9.4](#), gives local convergence to U_N . Since each link lies in $\mathfrak{A}_\eta$, the endpoint does too. $\square$

Lemma 10.18 (Recurrent path produces a compact retained core). *Let $\mathbb{P}$ be a shift-invariant probability measure on the active path space $\mathcal{P}_\eta$. For $\mathbb{P}$ -a.e. recurrent path $(W_j)_{j \geq 0}$, the orbit closure of its coordinates, together with all covariant tail limits generated by the returning blocks, contains a nonempty compact invariant set $\mathcal{M}_\eta \subset \mathfrak{A}_\eta$. Every element of $\mathcal{M}_\eta$ is an ancestor-realized retained descendant of the original profile.*

Proof. Poincaré recurrence for the left shift gives a full-measure set of paths which return arbitrarily close to themselves in the product topology. Fix such a path. For each returning block, use [Theorem 10.17](#) to compose the finitely many active successor observers into one covariant recentering of the initial state. The endpoints of the returning blocks remain in the compact set $\mathfrak{A}_\eta$. Taking the closure of all such endpoints and their covariant tail limits inside the compact normalized state space $\mathfrak{X}_M^\#$ gives a nonempty compact set. Closedness of $\mathcal{R}_\eta$ and compactness of $\mathfrak{A}_\eta$ keep the closure inside $\mathfrak{A}_\eta$. Intersecting over all closed invariant subsets of this compact closure and applying Zorn's lemma gives a compact invariant minimal core $\mathcal{M}_\eta$. The diagonal construction in the concatenation lemma preserves ancestry and retained concentration, so every element of $\mathcal{M}_\eta$ is ancestor-realized. $\square$

Lemma 10.19 (Retention and gauges on the recurrent active core). *The compact core $\mathcal{M}_\eta$ of [Theorem 10.18](#) carries the same retained activity threshold:*

$$\Phi(U) \geq \eta \quad (U, \Pi) \in \mathcal{M}_\eta,$$

and its pressure gauges are compatible with the invariant averaging used in [Theorem 8.26](#).

Proof. The inequality $\Phi \geq \eta$ is closed under the local topology by [Theorem 8.18](#), and all approximating states lie in $\mathfrak{A}_\eta$. Hence every element of the compact core lies in $\mathfrak{A}_\eta$. Gauge compatibility follows from [Theorem 9.4](#): on each compact cylinder the pressure gauges are chosen by a fixed normalization, and diagonal limits preserve the normalized local $L^{3/2}$ pressure topology. Therefore the invariant measures on $\mathcal{M}_\eta$ used in the recurrent-core rigidity theorem average genuine normalized states with retained activity, not untracked pressure representatives. $\square$

Theorem 10.20 (No infinite active descendant chain). *No infinite retained active descendant chain can occur in the refined residual class.*

Proof. Suppose an infinite retained active descendant chain exists. If $\limsup_j \Phi(U_j) = 0$, local compactness gives a subsequential limit $U_{j_k} \rightarrow U_\infty$ with $\Phi(U_\infty) = 0$. The diagonal lifting used in [Theorem 10.10](#) shows that this limit is still a retained tail descendant of the original residual profile. By [Theorem 8.19](#), U_∞ belongs to the affine/parasitic lower stratum, contradicting descendant heredity in the refined residual class. Hence $\limsup_j \Phi(U_j) > 0$. Choose $\eta > 0$ and an infinite ordered subchain contained in $\mathfrak{A}_\eta$. By transitivity of retained tail descendants through [Theorem 10.10](#), this ordered subchain is an $\mathcal{R}_\eta$ -chain.

[Theorem 10.16](#) gives a compact shift-invariant active path space and a stationary path measure supported on $\mathfrak{A}_\eta$. [Theorems 10.17](#) to [10.19](#) turn a recurrent path into a compact tail-minimal recurrent core for (1.2) whose states remain in $\mathfrak{A}_\eta$ with compatible pressure gauges. The recurrent-core rigidity theorem [Theorem 8.26](#) forces this core to be affine-constant after the canonical affine normalization. But affine/parasitic profiles have $\Phi = 0$, whereas $\mathfrak{A}_\eta = \{\Phi \geq \eta\}$. This contradiction rules out the infinite active descendant chain. $\square$

Lemma 10.21 (Finite separated families produce indecomposable profiles or active descendant chains). *Let $A = \{\mathfrak{s}_1, \dots, \mathfrak{s}_J\}$, $J \geq 2$, be a finite retained concentration separated family for a refined residual candidate. Then either one of the terminal profiles (U_j, Π_j) is terminally indecomposable, or the family generates an infinite active descendant chain. A finite recurrent loop is counted as an infinite chain by periodic repetition.*

Proof. Assume no (U_j, Π_j) is terminally indecomposable. By [Theorems 10.12](#) and [10.13](#), each (U_j, Π_j) has a retained concentrating tail limit which lifts to a recentering sequence of the original terminal sequence with retained local concentration.

If a lifted concentrating recentering sequence is not equivalent to one of the sequences already in the finite family, enlarge the family by adding this new sequence. This is still a finite retained concentration separated family because only one separated recentering sequence with retained local concentration has been added. Repeat the construction. At each finite stage there are two possibilities. Either an indecomposable profile appears, proving the first alternative, or every existing decomposable profile has a retained concentrating tail limit.

If the process returns to a previously occurring concentrating recentering sequence, the directed recentering relation contains a finite loop

$$\mathfrak{s}_{j_0} \rightarrow \mathfrak{s}_{j_1} \rightarrow \cdots \rightarrow \mathfrak{s}_{j_q} = \mathfrak{s}_{j_0}.$$

Repeating the loop periodically gives an infinite active descendant chain. If the process never returns to a previously occurring concentrating class, the successively lifted retained tail limits themselves form an infinite active descendant chain. The nonactive successor alternatives at each stage are excluded by [Theorem 10.13](#), while coexistence of a compact core with a diffuse or noncompact remainder is handled separately by [Theorem 9.61](#). $\square$

Lemma 10.22 (Finite recurrent loops of retained tail limits are impossible). *No refined residual terminal extraction contains a finite recurrent loop of retained concentrating tail limits.*

Proof. Let

$$\mathfrak{s}_{j_0} \rightarrow \mathfrak{s}_{j_1} \rightarrow \cdots \rightarrow \mathfrak{s}_{j_q} = \mathfrak{s}_{j_0}$$

be such a loop, and write (U_ℓ, Π_ℓ) for the corresponding retained concentration terminal profiles. Each arrow means that $U_{\ell+1}$ is a retained concentrating tail limit of U_ℓ . Thus, for each ℓ , there are escaping tail recentering sequences whose recentered fields converge locally to $U_{\ell+1}$.

Let $\mathcal{K}$ be the closure, in the local smooth velocity topology and local $L^{3/2}$ pressure topology modulo gauges, of all profiles obtained from the finite set $\{U_0, \dots, U_{q-1}\}$ by time translations and covariant terminal translations. The boundedness inherited from the Type I Seregín extraction theorem and the local pressure compactness from `proof_setup.tex` give compactness of $\mathcal{K}$ on every fixed terminal cylinder; a diagonal argument gives compactness of $\mathcal{K}$. The set is invariant under the same time and covariant translations by construction. The recurrent-loop relations imply that each U_ℓ returns to $\mathcal{K}$ through tail limits, so the closed invariant subset of $\mathcal{K}$ generated by the loop is nonempty, compact, invariant, and contains the retained concentration profiles.

Repeating the loop periodically gives an infinite active descendant chain. This contradicts [Theorem 10.20](#). Therefore no finite recurrent loop of retained concentrating tail limits exists. $\square$

Theorem 10.23 (Finite separated retained concentration profile families are impossible). *No refined residual candidate admits a finite separated family of retained concentration profiles.*

Proof. Suppose, for contradiction, that a refined residual candidate admits a finite retained concentration separated family $A = \{\mathfrak{s}_1, \dots, \mathfrak{s}_J\}$, $J \geq 2$. By [Theorem 10.21](#), either some component of the family is terminally indecomposable or the family generates an infinite active descendant chain. The first alternative contradicts [Theorem 10.9](#). The second contradicts [Theorem 10.20](#). Hence the finite retained concentration separated family cannot exist. $\square$

Theorem 10.24 (Terminal local concentration theorem). *Every bounded centered Type I Seregín profile with retained local concentration outside the previously closed lower classes admits the local terminal concentration decomposition of [Theorem 9.9](#) as an active retained profile locus plus a residual remainder. Pure local vanishing, exterior escape, and noncompact loss cannot be the sole locations of the distinguished compact mass by [Theorem 9.12](#); mixed compact–diffuse and*

compact–noncompact states are excluded by Theorem 9.61; infinite active descendant chains are excluded by Theorem 10.20; finite separated retained concentration profile families are excluded by Theorem 10.23; every remaining single retained concentration terminal profile is terminally indecomposable by Theorem 10.4; retained indecomposable profiles have a backward sequence of finite critical norm by Theorem 10.5; and retained concentration terminal profiles are parasitic-free mild, unless they are already in the affine lower stratum, by Theorem 10.7.

Proof. This is the assembly of the local results and explicit state-space arguments in Sections 9 and 10. The local concentration decomposition is the paired active-locus/remainder decomposition Theorem 9.9. The old pure radiative alternatives are only sole-carrier exclusions, and are used only in that form through Theorem 9.12. Coexistence of a compact active core with diffuse or noncompact tail remainder is separately ruled out by Theorem 9.61, using realized critical-tail rigidity Theorems 9.52 and 9.60. Infinite active trees and finite recurrent loops are ruled out by the active path-space recurrence theorem Theorem 10.20, while finite separated active families are ruled out by Theorem 10.23. Therefore any retained concentration terminal profile that survives the ordered lower-class removals is single and has no retained active descendant or diffuse/noncompact tail remainder. By Theorem 10.4 it is terminally indecomposable, and then Theorems 10.5 and 10.7 give the sequence- L^3 and parasitic-free finite-shift mildness conclusions. All estimates used in this assembly are localized to fixed terminal cylinders, except for the final sequence L^3 conclusion, which is derived by contradiction from terminal indecomposability and is not assumed as a global bound. $\square$

11. CLOSURE OF THE GENERIC TERMINAL RESIDUAL

Definition 11.1 (Generic residual profile). A generic residual profile is a smooth bounded normalized Seregian ancient solution of (1.2) which carries persistent compact velocity L^3 -mass and belongs to none of the previously closed alternatives, none of $\mathcal{C}_{\text{ax}}, \mathcal{C}_{\text{rot}}, \mathcal{C}_{\text{stat}}^{\text{ret}}$, and none of the structured or tight classes removed earlier, including the ordered critical-tail lower strata $\mathcal{C}_{\text{log diff}}, \mathcal{C}_{\text{Young}}, \mathcal{C}_{\text{homcrit}}, \mathcal{C}_{\text{logper}}, \mathcal{C}_{\text{ap}}$.

Theorem 11.2 (Closure of the generic residual class by local concentration analysis). *One has*

$$\mathcal{C}_{\text{gen}} = \emptyset.$$

Proof. Suppose, for contradiction, that $V \in \mathcal{C}_{\text{gen}}$. By the tail geometry developed above, constant tails may be removed by the exact affine symmetry and nonconstant recurrent profile tail cores collapse to affine constants. Thus the remaining representative may be taken to be tail-normalized without leaving the centered equation or the residual decomposition. If this normalization eliminates non-affine activity, then Theorem 8.19 places the profile in the affine/parasitic lower stratum already removed from $\mathcal{C}_{\text{gen}}$. Thus, for the residual representative still under consideration, a positive non-affine activity threshold survives.

Apply the paired terminal local concentration decomposition to a terminal extraction of this tail-normalized candidate. The distinguished compact velocity mass from the setup package still forces the terminal active locus to be nonempty by Theorem 9.10; after affine normalization the surviving residual activity is measured by Φ . Pure radiative alternatives cannot be the sole location of the compact mass by Theorem 9.12, and mixed compact–diffuse or compact–noncompact remainders are excluded by Theorem 9.61. Infinite active descendant chains are excluded by Theorem 10.20. Thus the active locus reduces, in the refined residual class, either to a finite separated family or to a single retained profile with no surviving tail remainder.

The finite separated-family case contradicts Theorem 10.23. In the single-profile case, Theorem 10.4 makes the terminal profile terminally indecomposable. Then Theorem 10.9 excludes it. All paired state-space possibilities are impossible. Hence no generic residual profile exists. $\square$

12. FINAL ASSEMBLY

The class closures above form the following complete dependency chain. The rotational relative-equilibrium arrow is the localized Coriolis-flux argument; the terminal active-locus compactness, no finite separated-family, and indecomposability arrows are proved in this paper from local estimates:

$$\begin{aligned}
\mathcal{C}_{\text{ax}} &\xrightarrow{\Gamma\text{-Liouville}} \emptyset, \\
\mathcal{C}_{\text{rot}} &\xrightarrow{\text{local flux}} \emptyset, \\
\mathcal{C}_{\text{stat}}^{\text{ret}} &\xrightarrow{\text{terminal concentration}} \emptyset, \\
\mathcal{C}_{\text{aff}} &\xrightarrow{\text{oscillatory entry}} \emptyset, \\
\mathcal{C}_{\text{log diff}} &\xrightarrow{\text{log-window descendants} + \text{hidden-scale closure}} \emptyset, \\
\mathcal{C}_{\text{Young}} &\xrightarrow{\text{first bad mesoscopic routing}} \emptyset, \\
\mathcal{C}_{\text{homcrit}} \cup \mathcal{C}_{\text{logper}} \cup \mathcal{C}_{\text{ap}} &\xrightarrow{\text{bounded-origin realization}} \emptyset, \\
\mathcal{C}_{\text{gen}} &\xrightarrow{\text{critical-tail closure, finite separation, sequence-}L^3} \emptyset.
\end{aligned}$$

The positive local velocity mass inherited by every normalized Seregin limit, after affine normalization through the non-affine activity functional Φ , is the common contradiction target in the complete ordered decomposition.

Equivalently, the final exclusion is the composition of six certificates:

- (1) *Entry certificate*: The setup package supplies a positive local velocity concentration sequence; the endpoint weak Serrin local Type I package [Theorem 1.19](#) gives the terminal local energy/no-escape estimate in [Theorem 1.20](#). Thus the sequence is admissible in the sense of [Theorem 1.12](#), and [Theorem 1.3](#) turns it into a normalized Seregin profile with retained compact velocity concentration.
- (2) *Stratification certificate*: the ordered residual decomposition places every remaining retained profile in the complete list of alternatives

$$\mathcal{C}_{\text{ax}} \cup \mathcal{C}_{\text{rot}} \cup \mathcal{C}_{\text{stat}}^{\text{ret}} \cup \mathcal{C}_{\text{aff}} \cup \mathcal{C}_{\text{log diff}} \cup \mathcal{C}_{\text{Young}} \cup \mathcal{C}_{\text{homcrit}} \cup \mathcal{C}_{\text{logper}} \cup \mathcal{C}_{\text{ap}} \cup \mathcal{C}_{\text{gen}}.$$

- (3) *Closed-strata certificate*: the small, stationary L^3 , tight, closed structured/decay, axisymmetric, rotational, and stationary-hull strata are discharged by the results cited above.
- (4) *Lower-stratum certificate*: $\mathcal{C}_{\text{aff}}$ is discharged by oscillatory entry normalization. The alternative $\mathcal{C}_{\text{log diff}}$ is discharged by log-window descendant selection plus hidden-scale compactness; see [Theorem 9.26](#). The alternative $\mathcal{C}_{\text{Young}}$ is discharged by the same hidden-scale compactness; see [Theorem 9.54](#). The alternatives $\mathcal{C}_{\text{homcrit}}, \mathcal{C}_{\text{logper}}, \mathcal{C}_{\text{ap}}$ are discharged by bounded-origin realization of coherent critical tails.
- (5) *Residual terminal certificate*: the generic residual is reduced by the paired active-locus decomposition, diffuse-defect compactness, diffuse-defect trichotomy, critical-tail compactification, realized critical-tail rigidity, descendant heredity, active path-space recurrence, finite separated-family exclusion, and terminal indecomposability.
- (6) *Endpoint Liouville certificate*: a terminally indecomposable retained profile is either affine/parasitic lower-stratum or has a parasitic-free bounded mild physical pullback with a backward L^3 -bounded sequence, and is therefore zero by Albritton–Barker.

Proof of Theorem 1.9, revisited. The detailed reductions have now been given in the class sections. The axisymmetric bounded-circulation class is empty unconditionally by Theorem 4.3. The rotational relative-equilibrium class is empty by the local Coriolis-flux and recentered concentration analysis in Theorem 5.8. The degenerate stationary-hull class with retained local concentration is empty by the terminal local concentration theorem, through Theorem 7.6. The affine stratum is empty by Theorem 9.53. The log-diffuse and Young critical-tail alternatives are empty by Theorems 9.26 and 9.54. The coherent critical-tail alternatives are empty by Theorem 9.59. The generic terminal residual is empty by the finite separated-family and indecomposability argument in Theorem 11.2. Since the complete ordered decomposition (1.6) exhausts the residual class by definition, these local exclusions imply that the residual class is empty. $\square$

Remark 12.1 (What is imported and what is proved here). The paper inherits the base setup, including local concentration, the local Type I Type II dichotomy, the Type I Seregina extraction framework, and the initial structural decomposition, from the unified setup package in `proof_setup.tex`. All remaining branch-closure steps are proved in this paper using local concentration and terminal dynamics:

- terminal active-locus extraction;
- compactness of descendant/tail families and pressure controls;
- active path-space recurrence;
- finite separated-family exclusion;
- oscillatory-entry normalization;
- hidden-scale compactness routing for critical tails;
- bounded-origin realization and critical-tail branch exclusion;
- and the compactness and branching exclusions culminating in Theorem 11.2.

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